SingleRAN

IPv4 Transmission Feature Parameter Description

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
  Changes in functions and their corresponding parameters
- Editorial changes
  Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-04-27).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for inter-board Ethernet link aggregation by the BSC6960. For details, see: • **6.4.2 Application on the Base Station Controller (GSM/UMTS)** • **7.10 Active and Backup Link ARP Probe (BSC6960)** • **7.12 LLDP**	Added the **ETHTRUNK**.***INTERBOARDFLAG*** parameter.	BSC6960

Change Description	Parameter Change	Base Station Model
Added support for IPv4 transmission by the UMPThb. For details, see: • 4.2.1 Ethernet Port • 4.3.4 MAC • 5.3.3 uX2 Interface Networking • 5.4 LTE Interface Networking • 5.5 NR Interface Networking • 6.4.3 Application on the Base Station • 7.11.1 Introduction	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for backup power saving data transmission from the base station to the microwave device. For details, see 7.12 LLDP.	Modified parameter: Added the **BKP_PSAV_EVALUATE_DATA_SW** option to the **LLDPGLOBAL**.***BTSMWCOSW (LTE eNodeB, 5G gNodeB)*** parameter.	3900 and 5900 series base stations (macro base stations)
Added support for precise reporting of X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn user-plane fault alarms. For details, see 7.9 GTP-U Echo.	Modified parameter: Added the **INTER_UP_ALM_PRECISE_SW** option to the **TRANSALGEXTPARA**.***TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)*** parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Added descriptions of the feature dependency of the BSC6960. For details, see 8.25 Source-based Routing on the Base Station Side.

Changed the related MOs that cannot be configured for sub-port IP addresses. For details, see 4.4.1.2 IP Address.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR. This feature works the same way with each of these RATs.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
GSM	GBFD-118601	Abis over IP	5.2.1 Abis Interface Networking
GSM	GBFD-118602	A over IP	5.2.2 A Interface Networking
GSM	GBFD-118603	Gb over IP	5.2.3 Gb Interface Networking
GSM	GBFD-118611	Abis IP over E1/T1	4.2.2 E1/T1 Port
GSM	GBFD-118622	A IP over E1/T1	4.2.2 E1/T1 Port
UMTS	WRFD-021404	Single IP Address for NodeB	4.6.5 Protocol Stacks for Other Transport Interfaces
UMTS	WRFD-050402	IP Transmission Introduction on Iub Interface	5.3.1 Iub Interface Networking
UMTS	WRFD-050409	IP Transmission Introduction on Iu Interface	5.3.2 Iu/Iur Interface Networking
UMTS	WRFD-050410	IP Transmission Introduction on Iur Interface	5.3.2 Iu/Iur Interface Networking
UMTS	WRFD-050411	Fractional IP Function on Iub Interface	4.3.3 Fractional IP
GSM UMTS	MRFD-210103	Link Aggregation	6.4 Ethernet Link Aggregation
LTE FDD	LBFD-00202103	SCTP Multi-homing	6.6 Control-Plane SCTP Multi-homing
LTE TDD	TDLBFD-00202103	SCTP Multi-homing	6.6 Control-Plane SCTP Multi-homing
NB-IoT	MLBFD-12000419	SCTP Multi-homing	6.6 Control-Plane SCTP Multi-homing
LTE TDD	TDLOFD-003005	OM Channel Backup	6.8 OM Channel Backup
LTE FDD	LOFD-003005	OM Channel Backup	6.8 OM Channel Backup

RAT	Feature ID	Feature Name	Chapter/Section
NB-IoT	MLOFD-003005	OM Channel Backup	**6.8 OM Channel Backup**
LTE FDD	LBFD-003007	IP Route Backup	**6.5 IP Route Backup**
LTE TDD	TDLBFD-003007	IP Route Backup	**6.5 IP Route Backup**
NB-IoT	MLBFD-12000308	IP Route Backup	**6.5 IP Route Backup**
NR	FBFD-010022	Active/Standby IP Routes	**6.5 IP Route Backup**
LTE TDD	TDLBFD-003003	VLAN Support (IEEE 802.1p/q)	**4.3.5 VLAN**
LTE FDD	LBFD-003003	VLAN Support (IEEE 802.1p/q)	**4.3.5 VLAN**
NB-IoT	MLBFD-12000303	VLAN Support (IEEE 802.1p/q)	**4.3.5 VLAN**
NR	FBFD-010019	VLAN Support (IEEE802.1p/q)	**4.3.5 VLAN**
LTE FDD	LOFD-003007	Bidirectional Forwarding Detection	**7.4 BFD**
LTE TDD	TDLOFD-003007	Bidirectional Forwarding Detection	**7.4 BFD**
NB-IoT	MLOFD-003007	Bidirectional Forwarding Detection	**7.4 BFD**
LTE FDD	LOFD-003008	Ethernet Link Aggregation (IEEE 802.3ad)	**6.4 Ethernet Link Aggregation**
NB-IoT	MLOFD-003008	Ethernet Link Aggregation (IEEE 802.3ad)	**6.4 Ethernet Link Aggregation**
LTE TDD	TDLOFD-003008	Ethernet Link Aggregation (IEEE 802.3ad)	**6.4 Ethernet Link Aggregation**
NR	FBFD-010016	Transmission Networking	**5 Interface Networking**
NR	FOFD-010060	Transmission Network Detection and Reliability Improvement	**6 Transmission Reliability**
			7 Transmission Maintenance and Detection

3 Overview

3.1 IP Transmission Interfaces

IP transmission interfaces include:

- Abis, A, Gb, and Ater interfaces of GSM
- Iub, Iur, Iu, and uX2 interfaces of UMTS

  NOTE

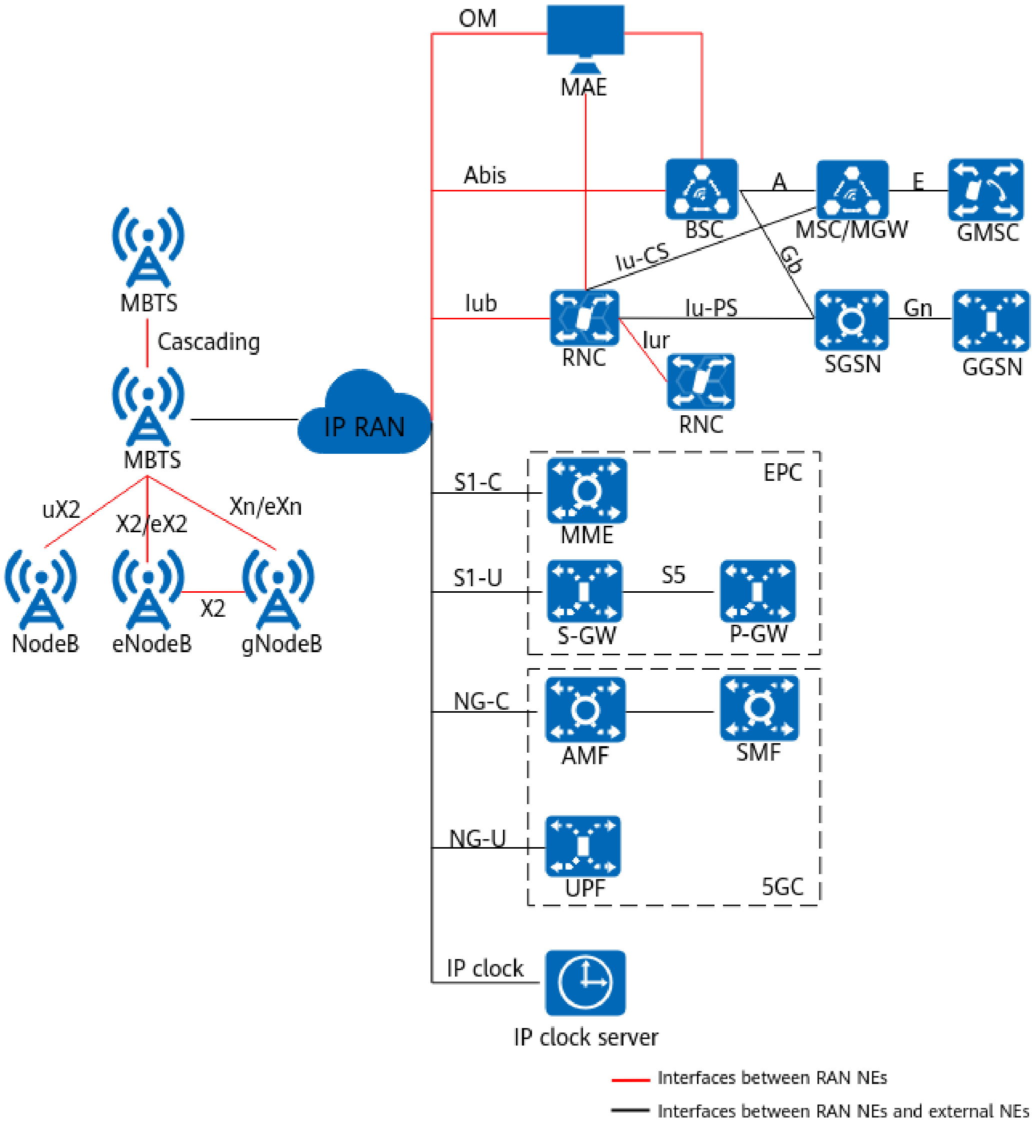
  A uX2 interface connects two NodeBs and implements coordination between them.
- S1, X2, and eX2 interfaces of LTE
- S1, X2, and eXn interfaces of non-standalone (NSA) NR
- NG, Xn, and eXn interfaces of standalone (SA) NR

  NOTE

  An eXn interface connects two gNodeBs and implements coordination between them.
- OM and IP clock interfaces at the RAN side

The following figure shows the transmission interfaces on a network.

Figure 3-1 Transmission interfaces at the G/U/L/T/NR RAN side



3.2 Transmission Configuration Model

As network traffic volume increases, network transmission scenarios become increasingly complicated. For example, transmission ports or boards need to be adjusted to support inter-board functions. The traditional transmission network configuration is complicated and struggles to meet the requirement for fast deployment on the live network. Therefore, a new transmission configuration model is introduced to decouple transmission configurations from physical devices to simplify operations.

A transmission configuration model can either be new or old, which is controlled by the **GTRANSPARA**.***TRANSCFGMODE (LTE eNodeB, 5G gNodeB)*** parameter.

- When this parameter is set to **OLD**, the old transmission configuration model (old model for short) is used. In the old model, location information such as the cabinet number, subrack number, and slot number are configured and the transmission configuration is bound to physical devices.

- When this parameter is set to **NEW**, the new transmission configuration model (new model for short) is used. In the new model, the transmission configuration is decoupled from physical devices. That is, the transmission configuration does not contain the cabinet number, subrack number, or slot number, and IPv4 and IPv6 configuration objects above the IP layer are combined. This model facilitates the expansion of new transmission functions and requires fewer transmission configuration parameters.

Compared with the old model, the new model incorporates an **INTERFACE** MO to isolate the upper layer from the physical layer. In this way, transmission configuration objects are decoupled from physical devices.

In addition, with the new model, when a transmission link is configured or a transmission board or port is modified, the information such as the cabinet, subrack, and slot numbers as well as the subboard type only needs to be configured for the physical layer and data link layer.

The MAE-Deployment function integrated into the MAE supports the conversion from the old transmission configuration model to the new model.

The new model is applicable to base stations excluding GBTSs, pico base stations, and the BBU3910C.

4 Transport Protocols

4.1 Introduction

This chapter describes the basic protocols for IP transmission at the physical, data link, network, and transport layers, as well as the protocol stacks of IP transmission interfaces.

4.2 Physical Layer

Physical layer is the lowest layer of the generic transport protocol stack. It transmits signals and processes protocol data at the physical layer, and provides reliable data transmission for the data link layer. The main physical layer ports to implement IP transmission at the RAN side are as follows:

- Ethernet port
- E1/T1 port
- Synchronous digital hierarchy (SDH) port

4.2.1 Ethernet Port

Base stations and base station controllers support standard Ethernet ports, which can be electrical or optical, providing a data rate level of 10M/100M/GE/10GE/25GE/50GE. The work mode and rate mode of an Ethernet port can be automatically negotiated or manually configured. The Ethernet port transmits signals using CSMA or CD:

- Carrier sense multiple access (CSMA)

  Data is transmitted only after carrier sense verifies that there is no ongoing data transmission on the link. This reduces the possibility of data transmission conflicts. With multiple access, data sent from one site can be received by multiple sites.
- Collision detection (CD)

  Two sites transmit signals at the same time. After the signals are superimposed, the fluctuation of the voltage on the line is one time larger than the normal value. In this case, collision occurs. Detection is performed during signal transmission. If a conflict is detected, the transmission is suspended, and then the signal transmission resumes after a random delay.

The port mode (data rate, duplex mode, and FEC mode) adaption function is supported by the optical Ethernet port that carries traffic through the remote maintenance channel or service channel. If the Ethernet link is interrupted due to inconsistent port modes between the base station and the peer device, the base station uses the Ethernet port mode adaptation function to recover the Ethernet link and reports ALM-26245 Configuration Data Inconsistency. This function is controlled by the **TRANSALGEXTPARA**.***ETHADAPTSW (5G gNodeB, LTE eNodeB)*** parameter.

The recovery of the Ethernet link cannot be ensured because the Ethernet port mode adaptation function depends on the peer device behavior. You are advised to set the same Ethernet port mode for both the base station and the peer device as planned.

4.2.2 E1/T1 Port

Huawei base stations and base station controllers support standard E1/T1 electrical ports.

T1 and E1 systems are two basic PCM communications systems in the plesiochronous digital hierarchy (PDH).

The work modes of the E1/T1 port include E1 unbalanced, E1 balanced, and T1. The E1/T1 port supports encapsulation using data-link-layer protocols time division multiplexing (TDM), asynchronous transfer mode (ATM), and PPP. The E1/T1 port supports standard physical layer alarms and fault detection.

T1 supports a data rate of 1544 kbit/s. It is mainly used in North America and is recommended by the ANSI. J1, which is similar to T1, is used in Japan. The T1 coding types are B8ZS and AMI. T1 frame formats include T1 super frame and T1 extended super frame.

E1 supports a data rate of 2048 kbit/s. It is mainly used in Europe and China and is recommended by the ITU-T. The E1 coding types are HDB3 and AMI. E1 frames are odd-and-even double frames, known as the dual-frame format. The formats of E1 multiframes are CAS, CRC-4, and hybrid. CAS multiframes are no longer used. CRC-4 multiframes are recommended by ITU-T G.704.

4.2.3 SDH Port

Synchronous digital hierarchy (SDH) is a standardized hierarchical structure, used for synchronous transmission, multiplexing, and cross connection of digital signals. The synchronous transfer mode (STM) indicates the data rate level of SDH. STM-1 is the first data rate level of SDH, and an STM-1 port provides a data rate of 155.520 Mbit/s. Huawei base station controllers support the STM-1 optical port. The SDH port supports encapsulation using the data-link-layer protocols TDM, ATM, or PPP.

The SDH port supports both channelized and unchannelized configuration modes. A channelized SDH port supports encapsulation using TDM, ATM, or PPP. An unchannelized SDH port supports only ATM encapsulation.

The SDH port also supports standard SDH physical layer alarms, fault detection, and protection switching.

4.3 Data Link Layer

The data link layer is the second layer of the Transmission Control Protocol/Internet Protocol (TCP/IP) protocol stack. It receives and processes data from the physical layer and provides reliable data transmission for the network layer.

Protocols for IP transmission at the data link layer vary with the port type used at the physical layer, as follows:

- PPP and MLPPP are used if the physical layer uses the E1/T1 or STM-1 port.
- Ethernet and VLAN are used if the physical layer uses the Ethernet port.

4.3.1 PPP

PPP is a data link layer protocol, commonly used in a WAN.

PPP is designed to encapsulate and transmit datagrams of various network layer protocols over point-to-point links. These network layer protocols include IP, Internetwork Packet Exchange (IPX), and AppleTalk.

PPP includes Link Control Protocol (LCP) and Network Control Protocol (NCP).

- LCP: used for creating, maintaining, and removing links. LCP allows the two communication parties to negotiate about parameters, including maximum receive unit (MRU), verification algorithm, protocol field compression (PFC), and address and control field compression (ACFC).
- NCP: used for negotiating about network-layer protocol parameters. NCP includes IPCP and IPXCP. IPCP is used to negotiate about the IP parameters (such as IP address and DNS address) of the PPP link.

The PPP frame format is defined in RFC1661, as shown in **Figure 4-1**.

Figure 4-1 PPP frame format

Flag	Address	Control	Protocol	Information	FCS	F
1 byte	1 byte	1 byte	2 bytes	Variable	2 bytes	1 byte

The fields in **Figure 4-1** are described as follows:

- Flag: specifies the start or end of a frame. This field is set to a fixed value for a PPP frame.
- Address: specifies an address. This field is set to a fixed value for a PPP frame.
- Control: specifies the control field. This field is set to a fixed value for a PPP frame.
- Protocol: specifies the protocol type of the data encapsulated by the Information field. For example, 0x0021 indicates that an IP datagram is encapsulated.
- Information (also known as payload): contains the upper-layer protocol datagram encapsulated by a PPP frame. The size of the Information field

varies and the maximum size is specified by the MRU. The MRU is negotiable during the PPP link establishment. If the MRU is not negotiated, the default size, 1500 bytes, is used.

- Frame check sequence (FCS): specifies the frame check sequence field.

The NR mode does not support the PPP protocol. The following interfaces support PPP:

- Abis, A, and Ater interfaces of the BSC6900, Abis and A interfaces of the BSC6910, and Abis interface of the GBTS or eGBTS
- Iub, Iur, and Iu interfaces of the RNC and the Iub interface of the NodeB
- S1 and X2 interfaces of the eNodeB

4.3.2 MLPPP

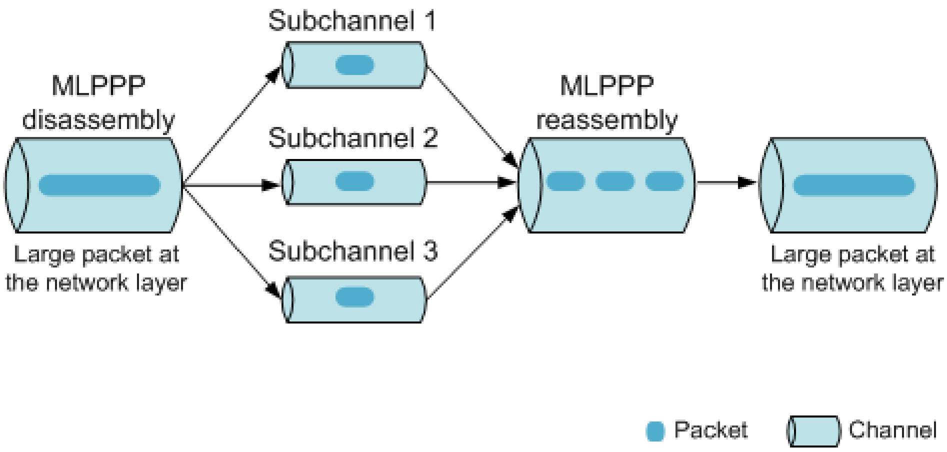
MLPPP (also called MP) is a data link layer protocol that exists between the PPP layer and the network layer. It is an extension of the PPP protocol through which multiple PPP physical links (also called MLPPP sublinks) are combined into one logical link.

Data Transmission Mechanism

MLPPP fragments network layer packets and then transmits the fragmented packets over MLPPP sublinks. This reduces the transmission delay and increases the bandwidth, data transmission rate, and throughput.

Figure 4-2 shows the data transmission mechanism of MLPPP.

Figure 4-2 Data transmission mechanism of MLPPP



Frame Format

The MLPPP frame format is defined in RFC 1990. The MLPPP frame has two formats: with a long sequence number and with a short sequence number, as shown in **Figure 4-3** and **Figure 4-4**, respectively.

Figure 4-3 MLPPP frame format with a long sequence number

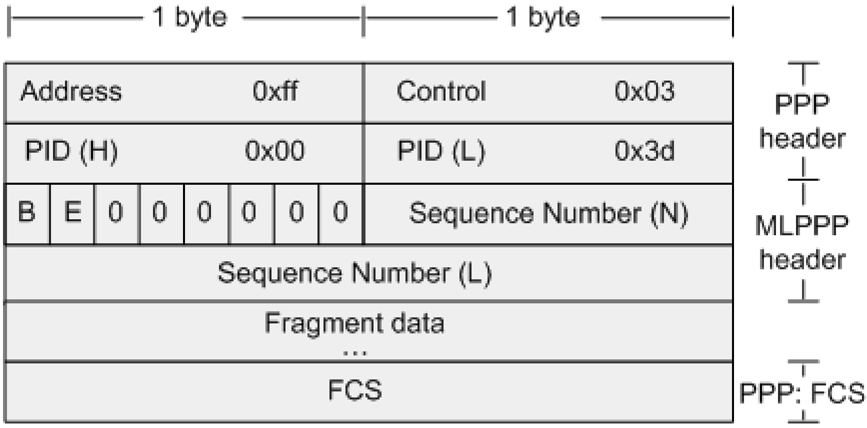
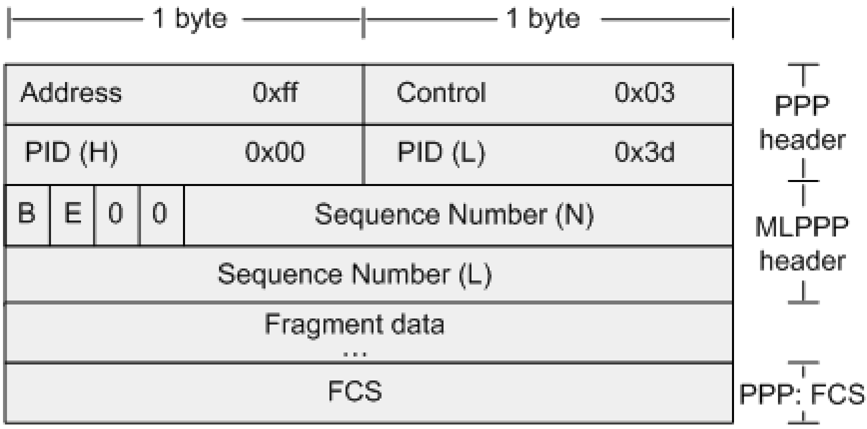


Figure 4-4 MLPPP frame format with a short sequence number



The fields of the MLPPP frame header are described as follows:

- PID: protocol ID for MLPPP encapsulation (2 bytes). The value is 0x00-3d.
- B: beginning fragment bit (one bit). This field specifies whether an MLPPP frame is the first fragment of a PPP frame.
- E: ending fragment bit (one bit). This field specifies whether an MLPPP frame is the last fragment of a PPP frame.
- Sequence Number: unique identifier of a frame (24 or 12 bits). The number increases by one after a fragment is transmitted.

The NR mode does not support the MLPPP protocol. The following interfaces support MLPPP:

- Abis, A, and Ater interfaces of the BSC6900, Abis and A interfaces of the BSC6910, and Abis interface of the GBTS or eGBTS
- Iub, Iur, and Iu interfaces of the RNC and the Iub interface of the NodeB
- S1 and X2 interfaces of the eNodeB

MC-PPP

MC-PPP (also called MC) is an extension of MLPPP. Unlike with MLPPP, MC-PPP provides services of different priorities. High-priority packets can preempt low-priority packets. The MC-PPP frame format is defined in RFC 2686.

4.3.3 Fractional IP

In fractional IP, IP packets are transmitted in partial E1/T1 timeslots. The timeslots not used for IP packet transmission can be used for transmitting other data. The peer end reassembles the original IP packets from these timeslots. This can reduce transmission costs.

One PPP link or MLPPP link can be carried on only one E1/T1 link. Some or all of timeslots 1 to 31 of an E1 link or timeslots 1 to 24 of a T1 link can be used for IP transmission, with the bandwidth 64 kbit/s per timeslot. After passing through a timeslot cross connection device (such as a DXX device) on the intermediate transport network, one PPP/MLPPP link can be carried on only one E1/T1 link.

The BSC6910 does not support fractional IP.

4.3.4 MAC

A Media Access Control (MAC) address, also called a physical address, hardware address, or link address, is pre-written into a device to uniquely identify the device at the MAC layer. During network communication, the physical layer uses the MAC address to identify a host.

Two MAC addresses are required for communication between two devices:

- Destination MAC address of the physical device that receives frames.
- Source MAC address of the physical device that sends frames.

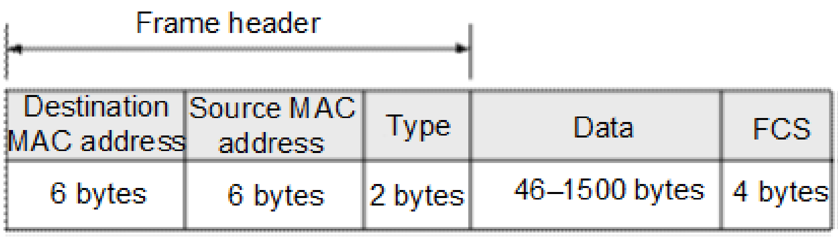
Ethernet ports can be configured to connect to other network elements (NEs). Information, such as port attribute, speed, work mode, and maximum transmission unit (MTU), can be configured for an Ethernet port.

The MTU is the size of the largest packet (excluding the MAC header and VLAN tag) that can be sent through an Ethernet port. An IP packet with a size larger than the MTU can be sent through the port only after the packet is fragmented.

Ethernet has multiple MAC frame formats. The Ethernet II frame format, as shown in **Figure 4-5**, is defined in RFC 894.

Huawei supports the transmission of frames in Ethernet II format.

Figure 4-5 Ethernet II frame format



The fields in **Figure 4-5** are described as follows:

- Destination MAC address specifies the MAC address of the physical device that receives the frame.
- Source MAC address specifies the MAC address of the physical device that transmits the frame.
- Type indicates the type of upper-layer protocol encapsulated by an Ethernet II frame. For example, 0x0800 signifies IP.
- Data contains the upper-layer protocol data encapsulated by the Ethernet frame. The upper-layer protocol data is usually an IP packet. The minimum size of the Data field is 46 bytes. If the size of an IP packet is less than 46 bytes, bits are padded to the Data field. If the size of the upper-layer protocol packet exceeds the size of the MTU, the packet is fragmented. The fragments are then reassembled by the base station or base station controller in the receive direction. The maximum length of a jumbo frame supported by the base station is specified by the **ETHPORT**.***MTU (5G gNodeB, LTE eNodeB)*** (old model)/**INTERFACE**.***MTU4 (5G gNodeB, LTE eNodeB)*** (new model) parameter. The maximum length of a jumbo frame supported by the BSC6910/BSC6900 is specified by the **ETHPORT**.***MTU*** parameter. The maximum length of a jumbo frame supported by the BSC6960 is specified by the **INTERFACE**.***MTU4*** parameter.
- FCS specifies the frame check sequence field. The cyclic redundancy check (CRC) is used. If the receive end detects an error during the frame check, the frame is discarded.

NOTE

According to the hardware type, the maximum size of a data packet that can be received over an Ethernet port is as follows:

- BBU:

  BBU5900C: 9320 bytes
- Main control boards:

  UMDU: 2000 bytes

  UMPTb: 2000 bytes

  UMPTe: 2024 bytes

  UMPTg/UMPTga: 9320 bytes

  UMPTha/UMPThb: 9320 bytes
- Transmission extension board:

  UTRPc: 2000 bytes

4.3.5 VLAN

A VLAN covers a broadcast area and generally a network segment is planned as a VLAN. VLANs provide two functions:

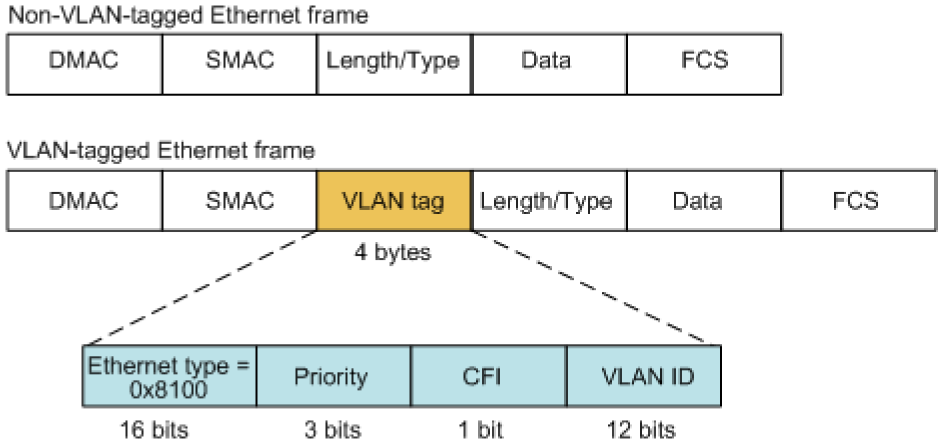
- Prevent broadcast storms and improve network security by communicating with each other only through routes.
- Assign different VLAN priorities to frames forwarded by the data link layer.

Figure 4-6 shows the VLAN-tagged and non-VLAN-tagged Ethernet frame formats.

In a VLAN-tagged frame, the VLAN tag field is between the SMAC field and the Length/Type field, where:

- Ethernet type 0x8100 occupies 16 bits.
- Priority occupies 3 bits. The priority value ranges from 0 to 7. A larger value indicates a higher priority.
- CFI (short for control format indicator) occupies 1 bit.
- VLAN ID occupies 12 bits.

Figure 4-6 Ethernet frame formats



NOTE

CFI: canonical format indicator

DMAC: destination MAC address

FCS: frame check sequence

SMAC: source MAC address

Based on network deployment, both a radio device and a switch can attach VLAN tags to packets. However, a radio device is recommended for two reasons:

- A radio device can identify differentiated services code point (DSCP) values at the network layer. In general, a radio device maps traffic types to DSCP values at the network layer and then maps these DSCP values to VLAN priorities at the data link layer. However, a switch cannot identify DSCP values. Therefore, a switch cannot map DSCP values to VLAN priorities.
- A radio device can attach VLAN tags to traffic flows based on the traffic type. However, a switch attaches VLAN tags based only on the ingress port and cannot attach different VLAN tags to traffic flows sent by the same radio device.

4.3.5.1 VLAN on the Base Station Side

When the old transmission configuration model is used, there are two VLAN modes for a base station: single VLAN and VLAN group. The VLAN mode is specified by the **VLANMAP**.***VLANMODE (LTE eNodeB, 5G gNodeB)*** parameter. You can configure a single VLAN or a VLAN group. The single VLAN mode is recommended for VLAN tagging.

- Single VLAN mode

  In this mode, VLAN tags are attached to traffic flows based only on next-hop IP addresses (**VLANMAP**.***NEXTHOPIP (LTE eNodeB, 5G gNodeB)***).

- VLAN group mode

  In this mode, VLAN tags are attached to traffic flows based on next-hop IP addresses (**VLANMAP.*NEXTHOPIP (LTE eNodeB, 5G gNodeB)***) and traffic types (**VLANCLASS**.***TRAFFIC (LTE eNodeB, 5G gNodeB)***). You are advised to use the **DSCPMAP** MO to configure the mapping between DSCP values and VLAN priorities. You can also use the **VLANCLASS** MO to configure the mapping between traffic types and VLAN priorities. If the two MOs **DSCPMAP** and **VLANCLASS** are both configured, the settings in the **VLANCLASS** MO are used.

NOTE

- Before configuring the mapping between service priorities and VLAN priorities, you need to first use the **DIFPRI**, **UDT**, and **UDTPARAGRP** MOs to configure the mapping between traffic types and DSCP values to differentiate VLAN priorities for different traffic types.
- If the mapping between DSCP values and VLAN priorities has not been configured, the base station will map the DSCP values to VLAN priorities based on the default mapping. For details about the default mapping between DSCP values and VLAN priorities, see **Table 4-4**.

When the new transmission configuration model is used, the interface VLAN mode is also supported in addition to the single VLAN and VLAN group modes. In the new model, the interface VLAN mode is recommended for VLAN tagging.

In interface VLAN mode, the base station attaches a VLAN tag based on the interface (defined by the **INTERFACE** MO). Each VLAN tag includes a VLAN ID (**INTERFACE**.***VLANID (5G gNodeB, LTE eNodeB)***) and a DSCP-to-PCP mapping ID (**INTERFACE**.***DSCP2PCPMAPID (5G gNodeB, LTE eNodeB)***).

NOTE

The interface VLAN mode is mutually exclusive with the single VLAN and VLAN group modes.

The UBBPe/UTRP board does not support the interface VLAN mode.

The base station, whether using a new or old transmission configuration model, does not support the QinQ VLAN mode defined in IEEE802.1ad and cannot process packets with more than one tag.

Single VLAN Mode

In single VLAN mode, the base station searches for a VLAN tag based on the next-hop IP address (**VLANMAP.*NEXTHOPIP (5G gNodeB, LTE eNodeB)***) and the next-hop subnet mask (**VLANMAP.*MASK (5G gNodeB, LTE eNodeB)***). Each VLAN tag includes a VLAN ID (**VLANMAP.*VLANID (5G gNodeB, LTE eNodeB)***) and a VLAN priority (**VLANMAP.*VLANPRIO (5G gNodeB, LTE eNodeB)***). The next-hop IP address and the corresponding subnet mask collectively specify a network segment. This network segment is usually planned as a VLAN and corresponds to a VLAN ID. The base station attaches VLAN tags to traffic flows before sending the traffic flows to the network segment.

- If all traffic flows in a VLAN require the same VLAN priority, set the **VLANMAP.*SETPRIO (5G gNodeB, LTE eNodeB)*** parameter to

ENABLE(Enable). Then set the **VLANMAP.*VLANPRIO (5G gNodeB, LTE eNodeB)*** parameter to the same value for traffic flows with the same next-hop IP address.

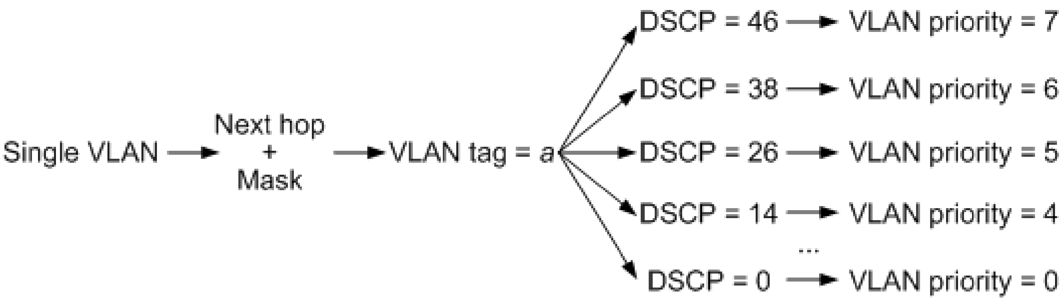
- If different traffic flows in a VLAN require different VLAN priorities as shown in **Figure 4-7**, set the **VLANMAP.*SETPRIO (5G gNodeB, LTE eNodeB)*** parameter to **DISABLE(Disable)**. Then configure the **DSCPMAP** MO (old model) or the **DSCP2PCPMAP** MO (new model) to map different DSCP values to different VLAN priorities.

NOTE

If **VLANMAP**.***SETPRIO (5G gNodeB, LTE eNodeB)*** is set to **DISABLE(Disable)** and no **DSCPMAP** MO (old model) or the **DSCP2PCPMAP** MO (new model) is configured, the base station uses the default mapping between DSCP values and VLAN priorities. For details about the default mapping between DSCP values and VLAN priorities, see **Table 4-4**.

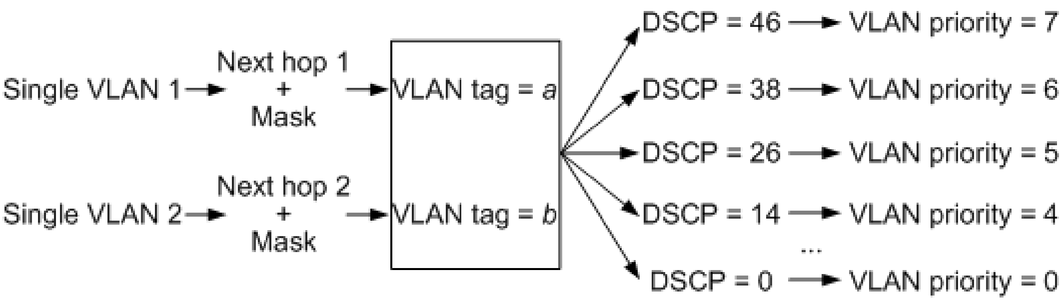
The GBTS identifies a network segment by using the result of the AND operation between its next-hop IP address (**BTSVLANMAP**.***NEXTHOPIP***) and its mask of the device IP address (**BTSDEVIP**.***MASK***).

Figure 4-7 Different traffic flows entering the same VLAN



In single VLAN mode, each network segment corresponds to only one VLAN and traffic flows in a VLAN can be assigned different VLAN priorities. In single VLAN mode, if an operator requires that the signaling, user data, and OM data should be distributed to different VLANs, the interface IP addresses of different network segments must be configured for the signaling, user data, and OM data. **Figure 4-8** shows an example of distributing signaling, user data, and OM data to different VLANs.

Figure 4-8 Different traffic flows entering different VLANs



The base station determines whether to attach VLAN tags and which VLAN tags to attach based on the packet type, as described in **Table 4-1**.

Table 4-1 Mapping between packet types and VLAN tags in single VLAN mode

Packet Type	VLAN Tagging
IP	• The VLAN ID is generated based on the mapping between VLAN IDs and next-hop IP addresses/subnet masks. The mapping is specified in the **VLANMAP** MO. • The VLAN priority is generated based on the mapping between VLAN priorities and DSCP values. The mapping is specified by the **DSCPMAP** MO (old model) or the **DSCP2PCPMAP** MO (new model).
ARP	• The VLAN ID is generated based on the destination IP address contained in the packet as well as the mapping (between next-hop IP addresses/subnet masks and VLAN IDs) specified in the **VLANMAP** MO. • The VLAN priority is always 7.
IEEE 802.1ag and Y.1731	The VLAN ID and VLAN priority of 802.1ag and Y.1731 packets are configured in the **CFMMA** MO.
LACP and 802.3ah	No VLAN tag is attached.
802.1x	No VLAN tag is attached.
LLDP	The VLAN ID and VLAN priority are configured using the **LLDPPORT** (old model)/**LLDP** (new model) MO.

VLAN Group Mode

In VLAN group mode, VLAN tags are attached to packets based on next-hop IP addresses and traffic types.

- If the traffic type is **USERDATA**, user data priorities are differentiated and VLAN tags are attached to packets based on DSCP values in addition to next-hop IP addresses and traffic types.
- If the traffic type is not **USERDATA**, VLAN tags are attached to packets based on next-hop IP addresses and traffic types. Non-USERDATA services also have DSCP values, which are set to the default values by the system. Therefore, you do not need to set the DSCP values for these services.

In the VLAN group mode, the procedure for attaching VLAN tags to packets is as follows:

Step 1 Determine the packet type. The base station determines the traffic type of a packet in a VLAN group according to the mapping shown in **Table 4-2**. Traffic types include **SIG**, **OM_HIGH**, **OM_LOW**, **USERDATA**, and **OTHER**. **Table 4-2** does not apply to the GBTS. **Table 4-3** describes the mapping between the VLAN group traffic types and packet types for the GBTS.

Table 4-2 Mapping between VLAN group traffic types and packet types

<table>
<tr><th>Traffic Type</th><th>Packet Type</th></tr>
<tr><td>SIG</td><td>All types of SCTP packets</td></tr>
<tr><td>OM_HIGH</td><td>● UDP packets, including OM SWITCH, NTP, and DHCP Relay packets
● TCP packets, including all TCP packets except OM_LOW packets, such as packets containing MML commands, alarms, events, tracing, and real-time monitoring files</td></tr>
<tr><td>OM_LOW</td><td>TCP packets used for software download and activation, log upload and download, upload of files related to performance, tracing, and real-time monitoring</td></tr>
<tr><td>USERDATA</td><td>● UDP packets
User data packets and GTP-U echo packets, excluding UDP packets of the OM_HIGH type, BFD packets, IKE packets, and UDP packets of the PTP (1588v2) type.
● ICMP ping request packets
If the DSCP value of an ICMP packet is the same as that of a USERDATA-type service, the ICMP packet is a USERDATA-type packet.
● ICMP ping response packets
The VLAN ID to be included in such a packet is the same as that included in the ICMP ping request packet.
● TraceRT response packets
The VLAN ID to be included in such a packet is the same as that included in the TraceRT request packet.
NOTE
Before configuring the VLAN information for GTP-U echo packets, query the DSCP values of GTP-U echo packets by running the LST GTPU command. Then, check whether there is a VLANCLASS MO whose traffic type is USERDATA and priority equals the DSCP values of GTP-U echo packets. If such an MO does not exist, add it by running the ADD VLANCLASS command.
If the DSCP value of an ICMP packet is different from that of any USERDATA-type packet, the ICMP packet is an OTHER-type packet.</td></tr>
<tr><td>OTHER</td><td>● Packets other than SCTP, UDP, TCP, and ICMP packets
● Packets other than Layer 2 packets such as ETH OAM and LACP packets
NOTE
Traffic types of BFD, IKE, and PTP packets can be determined as follows:
If VLAN information is configured for the OTHER type in the VLANCLASS MO, BFD, IKE, and PTP packets are considered OTHER-type packets.
If VLAN information is not configured for the OTHER type in the VLANCLASS MO, BFD, IKE, and PTP packets are considered USERDATA-type packets.</td></tr>
</table>

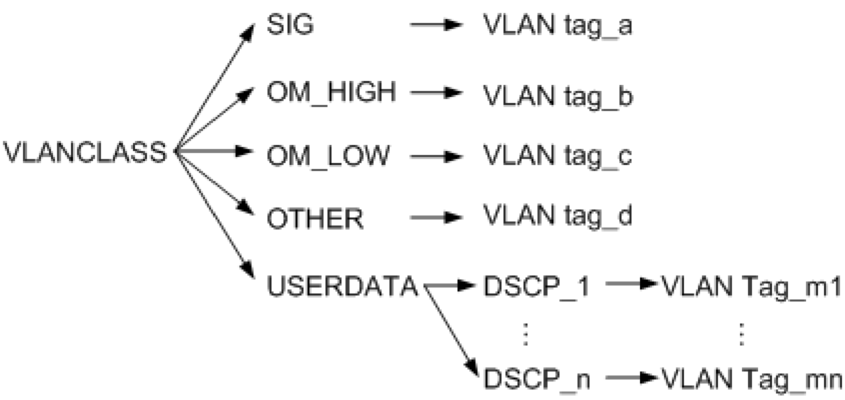
Traffic Type	Packet Type
Undefined	The following types of packets do not fall into the preceding traffic types. These packets do not contain VLAN information or their VLAN information is not configured in the **VLANCLASS** MO. ● Ethernet OAM (IEEE 802.1ag) VLAN information is configured in the **CFMMA** MO. ● Ethernet OAM (IEEE 802.3ah) No VLAN information is carried. ● LACP No VLAN information is carried. ● LLDP VLAN information is configured in the **LLDPPORT** (old model)/**LLDP** (new model) MO. ● 802.1x No VLAN information is carried.

Table 4-3 Mapping between VLAN group traffic types and packet types for the GBTS

Traffic Type	Packet Type
USERDATA	Operation and maintenance link (OML), redundant server link (RSL), extended signaling link (ESL), EML, CS voice, CS data, PS high-priority data, PS low-priority data
OTHERDATA	ARP, ICMP, IPCLK handshake, DHCP, BFD, and 1588V2 negotiation

Step 2 If the traffic type is not USERDATA, skip this step and go to **Step 3**. If the traffic type is USERDATA, attach VLAN tags to packets based on DSCP values, as shown in **Figure 4-9**.

Figure 4-9 Mapping between traffic types and VLAN tags



In VLAN group mode, the same VLAN tag can be attached to different types of traffic flows with the same next-hop address. In a VLAN group, different types of traffic flows can be mapped to different VLAN priorities based on DSCP values by using the **VLANCLASS** MO. The default mapping is configured based on **Table 4-4** or defined by users.

Table 4-4 Default mapping between DSCP values and VLAN priorities

DSCP Value Range	VLAN Priority
56 to 63	7
48 to 55	6
40 to 47	5
32 to 39	4
24 to 31	3
16 to 23	2
8 to 15	1
0 to 7	0

Step 3 Attach VLAN tags to packets based on the mapping between traffic types and VLAN tags, as shown in **Figure 4-9**.

NOTE

ARP request packets are of the OTHER type, not SIG type or USERDATA type. To ensure that ARP request packets carry correct VLAN tags, VLAN information about the OTHER type must be configured. Otherwise, Ethernet ports cannot communicate properly with each other.

----End

If packets are encapsulated in tunnel mode in IPsec scenarios, an encrypted packet contains two IP headers: an inner IP header (encrypted) and an outer IP header (unencrypted).

- The outer IP header uses the interface IP address of the base station and the IP address of the peer end (usually SeGW) as the source and destination IP addresses, respectively.
- The inner IP header uses the service IP address or OM IP address of the base station as the source IP address and the IP address of the MME/S-GW/MAE as the destination IP address.

The base station adds a VLAN tag according to the inner IP header. The process is as follows:

Step 1 The base station searches **VLANCLASS** and **VLANMAP** MO configurations for entries that correspond to the next-hop IP address of the route to the destination IP address in the inner IP header and the traffic type.

Step 2 After service packets are encrypted using IPsec, the base station searches **VLANCLASS** and **VLANMAP** MO configurations that match the next-hop IP address of the route to the destination IP address in the outer IP header, USERDATA type, and DSCP value.

Step 3 If the base station finds the corresponding **VLANCLASS** and **VLANMAP** MOs after IPsec encryption, the base station attaches VLAN tags to packets based on the **VLANCLASS** and **VLANMAP** MOs. If these MOs do not exist, the base station proceeds based on the search result of step 1. If the MOs specified in step 1 exist, the base station attaches VLAN tags to service packets based on the settings in the **VLANCLASS** and **VLANMAP** MOs recorded in step 1. If no matching **VLANCLASS** and **VLANMAP** MOs are found in step 1, the base station does not attach any VLAN tags to service packets.

----End

In IPsec scenarios, the routes to the destination IP addresses in outer and inner IP headers usually have the same next-hop IP address.

In IPsec dual-tunnel scenarios, the route to the destination IP address in the inner IP header has the same next-hop IP address as the route to the destination IP address of the primary IPsec tunnel. The destination IP address in the outer header of an encrypted IP packet may be the SeGW IP address of either the primary or secondary IPsec tunnel. The base station attaches a VLAN tag to the packet based on the USERDATA type with the same DSCP value. Therefore, **VLANCLASS** MOs must be configured for the secondary IPsec tunnel. In these MOs, IP packets of the USERDATA type have the same DSCP values as IP packets of the SIG, OM_HIGH, or OM_LOW type sent through the primary IPsec tunnel. This prevents VLAN tagging failures after a switchover from the primary to secondary IPsec tunnel. For example, if the DSCP value for an IP packet of the SIG type sent through the primary IPsec tunnel is 50, a **VLANCASS** MO in which the DSCP value is **50** and traffic type is **USERDATA** must be configured for the secondary IPsec tunnel.

NOTE

In addition to attaching VLAN tags to packets based on next-hop IP addresses, and based on next-hop IP addresses and traffic types, the base station can also attach VLAN tags to packets based on source IP addresses (subnets), which requires that the **SUBNETVLAN** MO be configured. In this case, the MAE-Deployment does not support export of the deployment list. Therefore, the base station does not support PnP-based deployment. This VLAN tagging method is implemented by the base station and the first-hop router and applies only to special scenarios. Its principles and configurations are not described in this document.

Interface VLAN Mode

In interface VLAN mode, the base station attaches a VLAN tag based on the interface (defined by the **INTERFACE** MO). Each VLAN tag includes a VLAN ID (**INTERFACE**.***VLANID (5G gNodeB, LTE eNodeB)***) and a DSCP-to-PCP mapping ID (**INTERFACE**.***DSCP2PCPMAPID (5G gNodeB, LTE eNodeB)***).

A DSCP-to-PCP mapping ID (specified by **INTERFACE**.***DSCP2PCPMAPID (5G gNodeB, LTE eNodeB)***) identifies a DSCP-to-PCP mapping table (corresponding to the **DSCP2PCPMAP** MO). This table contains the mapping between DSCP values and VLAN priorities. The mapping information is configured using the **DSCP** and **PCP** parameters in the **ADD DSCP2PCPREF** command. A VLAN interface is planned as a VLAN and corresponds to a VLAN ID. The base station attaches VLAN tags to traffic flows before sending them to the interface.

If the port on the peer device connected to the base station is of the access or hybrid link type, the packets sent from the peer device to the base station do not need to carry VLAN tags. In this case, an interface with **INTERFACE**.***ITFTYPE (5G gNodeB, LTE eNodeB)*** set to **NORMAL** must be configured on the base station.

The VLAN ID of the ARP request packet sent by the NE through a VLAN subinterface is the VLAN ID of the interface, and the VLAN priority of the packet is always 7.

Application restrictions of the interface VLAN mode are as follows:

- The UTRP/UBBPe board does not support the interface VLAN mode.
- The interface VLAN, **VLANMAP**, and **SUBNETVLAN** configurations are mutually exclusive.
- The VLAN IDs in the transmit and receive directions of the base station must be the same. If the VLAN IDs are different before the reconstruction from the **VLANMAP** MO to the interface VLAN, the base station may be disconnected or services may be interrupted after the reconstruction.

  When processing the **VLANMAP** MO, the base station controls only the VLAN ID in the transmit direction, but does not control the VLAN ID in the receive direction. When processing the interface VLAN, the base station requires that the VLAN IDs in the transmit and receive directions be the same. Therefore, when the port of the switch connected to the base station is set to the access or hybrid mode, the base station uses the port VLAN ID (PVID) of the access or hybrid port in the transmit direction. The packets sent from the interconnected switch to the base station do not contain VLAN tags. As a result, the base station discards the packets, causing site disconnection or service interruption. In VLAN reconstruction scenarios, if the base station is connected to a switch, the port on the switch is configured to work in access or hybrid mode, and the PVID of the hybrid port is configured on the base station, delete the **VLANMAP** MO corresponding to the PVID on the base station or do not perform VLAN reconstruction.

4.3.5.2 VLAN on the Base Station Controller Side

The BSC6900 supports VLAN tagging based on the next-hop IP address or data flows, and the BSC6910 only supports VLAN tagging based on the next-hop IP address. VLAN tagging based on the next-hop IP address is recommended.

- VLAN tagging based on the next-hop IP address

  VLAN tagging based on the next-hop IP address on the base station controller side is the same as that on the base station side. The VLAN ID is determined based on the preconfigured mapping between next-hop ID addresses and VLAN IDs. The related parameters are **IPADDR** and **VLANID**.
- VLAN tagging based on the data flows

  VLAN tagging based on the data flows allows different VLAN tags to be attached to the SCTP links and IP paths. The VLAN ID is determined based on the preconfigured mapping between SCTP links, IP paths, and VLAN IDs.

  Parameters related to the SCTP link are **SCTPLNKN**, **VLANID1**, and **VLANID2**.

  Parameters related to the IP path are **PATHID** and **VLANID**.

The BSC6960 supports VLAN tagging based on the next-hop IP address and the interface VLAN mode.

- VLAN tagging based on the next-hop IP address

VLAN tagging based on the next-hop IP address on the base station controller side is the same as that on the base station side. The VLAN ID is determined based on the preconfigured mapping between next-hop ID addresses and VLAN IDs. The related parameters are **NEXTHOPIP** and **VLANID**.

- Interface VLAN mode

  In interface VLAN mode, the BSC6960 attaches a VLAN tag based on the interface (defined by the **INTERFACE** MO). Each VLAN tag includes a VLAN ID (indicated by the **INTERFACE.*VLANID*** parameter).

4.3.6 ARP

The Address Resolution Protocol (ARP) defines the mapping between the IP addresses and MAC addresses. The ARP is also used to detect IP address conflicts and link status.

Mapping from IP Addresses to MAC Addresses

Before sending a packet to the next-hop router, the base station or base station controller checks whether the ARP table contains the mapping between the IP address and MAC address of the next hop. If the mapping does not exist, the base station or base station controller establishes the mapping between the IP address and the MAC address. The process is as follows:

1. The base station or base station controller sends an ARP request.
2. After receiving the ARP request, the next-hop router sends an ARP response.
3. The base station or base station controller obtains the MAC address corresponding to the IP address of the next hop from the ARP response packet, and records the mapping between the IP address and the MAC address of the next hop in the ARP table.

If the MAC address corresponding to the next-hop IP address is changed during network adjustment, the mapping recorded in the ARP table may be incorrect. To prevent this problem, ARP aging is supported. After an ARP entry is aged, an ARP request is sent automatically to update the ARP entry. The ARP aging time is specified by the **GTRANSPARA.*ARPAGINGTIME*** parameter. The ARP aging time must be set according to the ARP aging time of the switches and routers on the network. If the ARP aging time of the base station is longer than that of network devices, network problems such as broadcast storms may occur.

IP Address Conflict Detection

After obtaining an IP address, the base station or base station controller sends gratuitous ARP packets to check whether this IP address has been used by another host. If this IP address has been used by another host, an IP address conflict alarm is reported.

For information about link status detection, see **7.2 ARP Probe (GSM/UMTS)**.

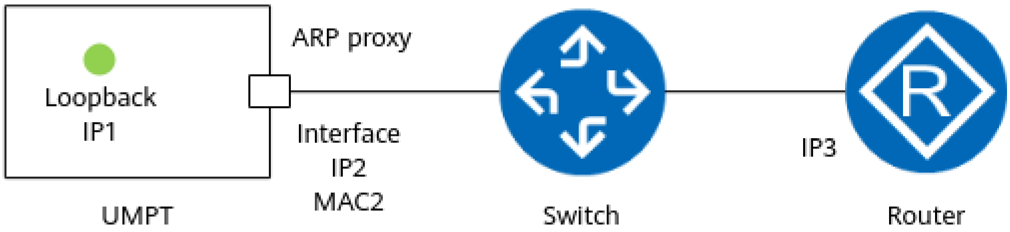
ARP Proxy

The base station supports the ARP proxy function on the Ethernet port for the logical IP address of the local board and the IP address of the base station of a different mode in backplane-based co-transmission scenarios. The Ethernet port acts as a proxy and uses its own MAC address to establish the mapping between the proxy IP address and MAC address with the router.

Single-mode Logical IP Proxy

As shown in **Figure 4-10**, the Ethernet port on the main control and transmission board is configured with IP2 and enabled with the ARP proxy function. This board is also configured with logical IP1. If IP1 is included in the network segment of IP2, the Ethernet port can perform ARP proxy for IP1, and the router can directly obtain the MAC address, MAC2, mapped from IP1.

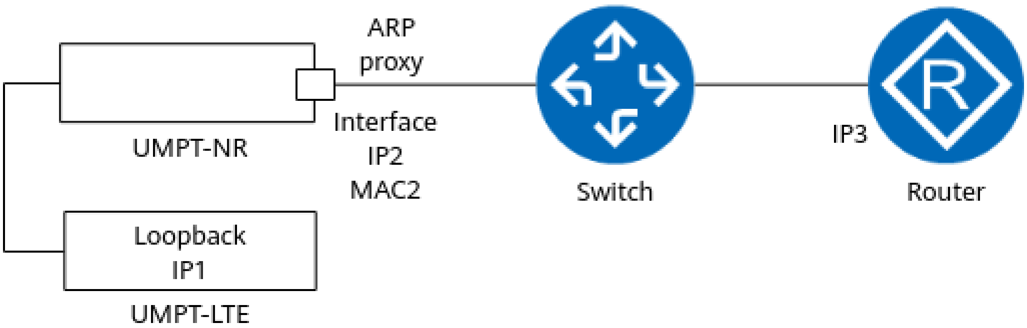
Figure 4-10 Logical IP proxy of a single main control board



Logical IP Proxy in Multimode Backplane-based Co-transmission

As shown in **Figure 4-11**, the main control and transmission board of NR provides co-transmission through backplane interconnection with the main control and transmission board of LTE. IP2 is configured and ARP proxy is enabled on the Ethernet port of NR. Logical IP1 is configured on the LTE side. If IP1 is included in the network segment of IP2 and a destination-based route to IP1 is configured on the NR side, the Ethernet port can function as an ARP proxy for IP1. In this case, the router can directly obtain the MAC address, MAC2, mapped from IP1.

Figure 4-11 Inter-mode IP Proxy in Backplane-based Co-transmission



4.4 Network Layer

The network layer uses IP as the principal protocol and is also referred to as the IP layer.

4.4.1 IP

IP is a primary network layer protocol suite. Using a universal packet format, IP conceals differences at data link layers and enables network interconnection.

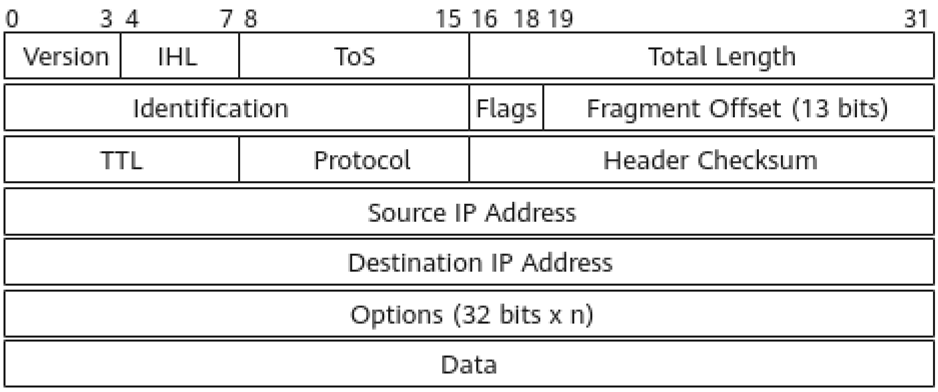
The IP protocol is unreliable and connectionless, providing functions such as IP addressing, routing, IP packet fragmentation, and reassembly.

4.4.1.1 Packet Format

At the transmit end, the IP layer receives the source data packet (for example, a UDP message) from the upper layer. Then, the IP layer determines whether to segment the data packet according to the MTU of the data link layer. After being segmented, the data is called a segment. Data packets or segments are encapsulated into IP packets in a fixed format. Later, IP packets are transmitted to the data link layer and then transmitted over Internet. At the receive end, the IP layer receives data from the data link layer, reassembles the payload to obtain the original data packet, and then transmits the data to the upper layer for further processing.

Figure 4-12 shows the IP packet format. An IP packet is conceptually divided into the header and the payload. The header contains addressing and control fields, whereas the payload carries the actual data to be sent over the Internet.

Figure 4-12 IP packet format



The fields of the IP packet are described as follows:

- Version: contains the IP version number. For example, 4 indicates IPv4.
- IHL (Internet Header Length): specifies the size of an IP packet header, which is obtained using 32 bits as the divisor. The minimum size of the IP packet header is 20 bytes (excluding the size of the Options field), and the maximum size is 60 bytes (including the size of the Options field).
- TOS: specifies the type of service. As indicated in RFC 1122, IP precedence uses three most significant bits to define eight service classes. IETF proposes a generic scheme to replace IP precedence. In this scheme, six bits of the TOS field define the DSCP, which indicates the service priority and has a value range from 0 to 63.
- Total Length: specifies the total size of an IP packet.

- Identification: identifies the datagram sent by a host. The value of this field is common to each of the fragments belonging to a particular packet. This field is used by the receive end to reassemble packets without accidentally mixing fragments from different packets.
- Flags: is used for segmentation and reassembly. It specifies whether the current data is the last fragment of an IP packet and whether fragmentation is allowed.
- Fragment Offset: specifies the position of a fragment in an IP packet. This field is used to fragment and reassemble datagrams.
- Time to live (TTL): specifies how long the IP packet is allowed to "live" on the network, in terms of router hops. Each router reduces the value of the TTL field by one before transmitting it. When the value of TTL becomes zero, the IP packet is discarded and an ICMP packet is sent to the transmit end.
- Protocol: specifies the type of the upper-layer protocol (including TCP, UDP, ICMP, and IGMP) carried in the IP packet.
- Header Checksum: contains the checksum of the header.
- Source IP Address: specifies the source IP address.
- Destination IP Address: specifies the destination IP address.
- Options: specifies whether measures such as error correction, measurement, and security are taken. This field has a variable size, from 1 byte to 40 bytes, depending on the selected items. Padding may be added to ensure that the size of this field is a multiple of four bytes.
- Data: contains the data to be transmitted in the datagram, either an entire higher-layer message or a fragment of one message.

4.4.1.2 IP Address

The IP protocol defines the IP address, which uniquely identifies a host or a port of a router on the Internet. An IP address consists of two parts: the network part, which uniquely identifies a network or subnet, and the host part, which uniquely identifies the host on that network or subnet.

The IP protocol specifies that standard IP addresses have three classes: A, B, and C. The IP protocol also defines some special IP addresses. For example, an IP address with all 0s in the host part indicates a network segment address. A network segment address identifies a subnet. All hosts on the same subnet have the same subnet mask and network segment address.

In the old model, IP transmission supports the following types of IP addresses:

- Ethernet port IP address and Ethernet trunk IP address
- PPP link/MLPPP group IP address
- Loopback interface IP address (logical IP address)
- Sub-port IP address
- Ethernet CI port IP address

NOTE

- For details on the Ethernet trunk, see **6.4 Ethernet Link Aggregation**.
- A 31-bit mask can be configured for the IP address of the Ethernet port. In this case, note that during transport network planning, network segments other than the 31-bit network segment cannot be all 0s or all 1s.

Table 4-5 lists the IP address types supported by base stations and base station controllers.

Table 4-5 IP address types supported by base stations and base station controllers

Category	Description	Related MO (BSC6900/ BSC6910)	Related MO (BSC6960)	Related MO (Base Station)
Ethernet port IP address	Each Ethernet port must be configured with at least one IP address.	**ETHIP**	**IPADDR4**	**DEVIP**
Ethernet trunk IP address	Several Ethernet ports can be bound to form an Ethernet trunk port. An Ethernet trunk port must be configured with at least one IP address.	**ETHTRKIP**	**IPADDR4**	**DEVIP**
PPP link IP address	Each PPP link and each MLPPP group must be configured with at least one IP address, which can be the device IP address.	**PPPLNK**	Not supported	**PPPLNK**
MLPPP link group IP address	Each PPP link and each MLPPP group must be configured with at least one IP address, which can be the device IP address.	**MPGRP**	Not supported	**MPGRP**
Loopback interface IP address	A loopback interface is a virtual interface that supports loopback and can be configured with an IP address.	**DEVIP**	**IPADDR4**	**DEVIP**
Sub-port IP address	A physical port can be configured with multiple sub-ports. Sub-ports are supported only by the base station.	-	Not supported	**DEVIP**
Ethernet CI port IP address	Ethernet CI port IP addresses are applicable to only Cloud BB base stations.	-	-	**DEVIP**

When the new transmission configuration model is used, IP transmission supports the following types of IP addresses:

- Interface IP address: IP address of an Ethernet port, Ethernet CI port, or Ethernet trunk
- Loopback interface IP address (logical IP address)

Table 4-6 IP address types supported by a base station

Category	Description	Related MO (Base Station Controller)	Related MO (Base Station)
Interface IP address	Each interface must be configured with at least one IP address.	-	**IPADDR4**
Loopback interface IP address	A loopback interface is a virtual interface that supports loopback and can be configured with an IP address.	-	**IPADDR4**

NOTE

IP addresses of the base station or base station controller cannot be on the same network segment.

4.4.1.3 Interface

An **INTERFACE** MO is used to configure IP-layer interfaces, including normal interfaces and VLAN subinterfaces. In this MO, the IP addresses and packet filtering of interfaces are specified.

4.4.1.4 IP Route

Two IP routing modes are supported:

- Destination-based IP routing: With this mechanism, IP addresses of egress ports and gateways for IP packets are determined based on the destination IP addresses. For static routes with the same destination address and subnet mask but different next-hop IP addresses, the route configuration takes effect according to the following principles:
  - If the static routes have the same priority, they are equal-cost routes. Packets are forwarded on equal-cost routes based on flows or on a per packet basis. Huawei wireless devices support only flow-based packet forwarding.
  - If the static routes have different priorities, they are active and standby routes. In such cases, packets are forwarded on the route with the highest priority.

  A direct route is a type of destination route. A base station automatically generates the direct route whose destination address is located on the network segment of the interface IP address. A direct route does not require manual maintenance and has the highest priority. It can be queried by running the **DSP IPRT** (in the old model)/**DSP IPROUTE4** (in the new model) command.
- Source-based IP routing: With this mechanism, IP addresses of egress ports and gateways for IP packets are determined based on the source IP addresses. Both the base station and base station controller support source-based routing and source-based routing backup. For static routes with the same

destination address and subnet mask but different next-hop IP addresses, the route configuration takes effect according to the same principles as destination-based routing.

- If the static routes have the same priority, they are equal-cost routes. Packets are forwarded on equal-cost routes based on flows or on a per packet basis. Huawei wireless devices support only flow-based packet forwarding.
- If the static routes have different priorities, they are active and standby routes. In such cases, packets are forwarded on the route with the highest priority.

NOTE

The equal-cost route is not recommended. This is because one-way audio or silence may occur if a route matching a traffic flow is abnormal, for example, incorrectly configured, or if the route path is abnormal, for example, link disconnection or packet loss occurs.

If both the destination-based and source-based routings are configured, only the source-based routing takes effect.

Table 4-7 IP routing modes when the old transmission configuration model is used

Routing Mode	Supported By	Related Command
Destination-based IP routing	Base station and base station controller	ADD IPRT
Source-based IP routing	Base station and base station controller	ADD SRCIPRT

Table 4-8 IP routing modes when the new transmission configuration model is used

Routing Mode	Supported By	Related Command
Destination-based IP routing	Base station	ADD IPROUTE4
	Base station controller (BSC6900/BSC6910)	ADD IPRT
Source-based IP routing	Base station and base station controller (BSC6960)	ADD SRCIPROUTE4

Routing Mode	Supported By	Related Command
	Base station controller (BSC6900/ BSC6910)	ADD SRCIPRT

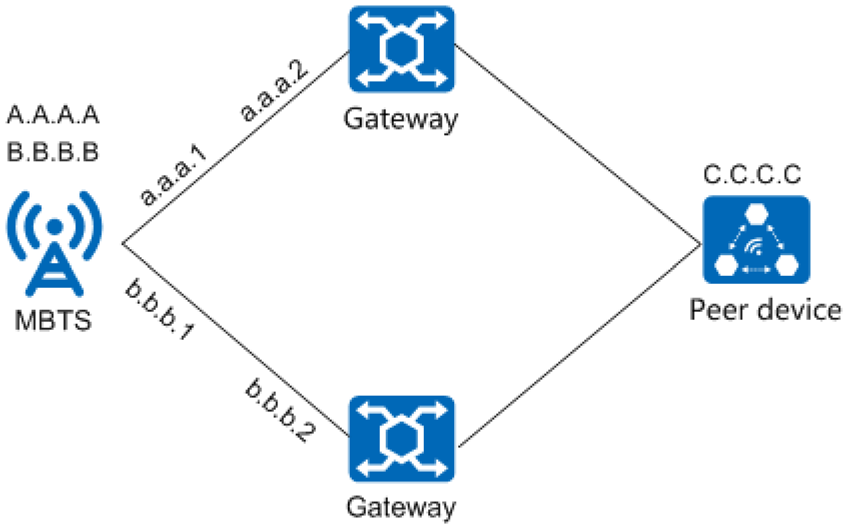
For information about source-based routing supported by the base station controller, see *Transmission Resource Pool in RNC Feature Parameter Description* and *Transmission Resource Pool in BSC Feature Parameter Description*.

Source-based routing applies to the following scenarios:

- Necessity of source-based routing on the base station

  If a base station has different service IP addresses and next-hop IP addresses, but the same destination IP address, the source-based routing must be used on the base station, as shown in **Figure 4-13**.

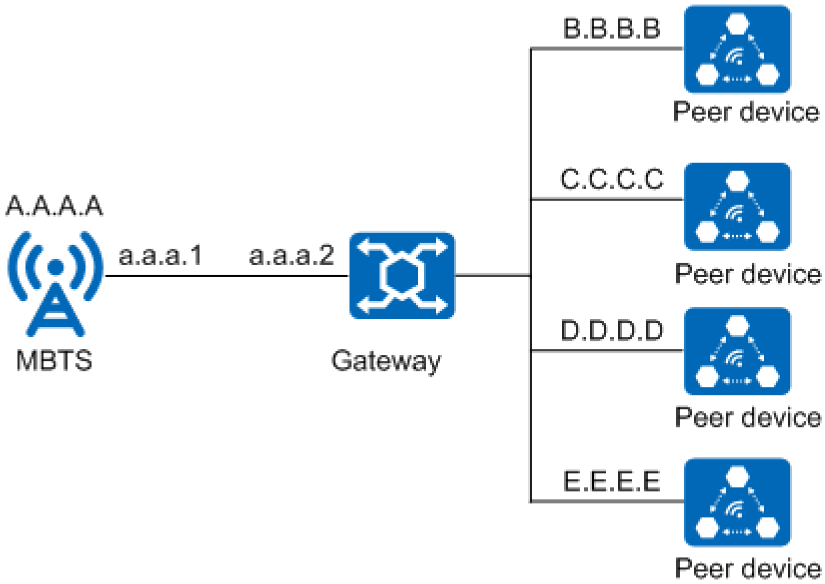
  Figure 4-13 Necessity of source-based routing on the base station



- Recommendation of source-based routing on the base station

  Routes to be configured on a base station have the same service IP address and next-hop IP address but different destination IP addresses. In this case, source-based routing is recommended for simplifying route configurations on the base station side. **Figure 4-14** is an exemplary diagram of scenario in which source-based routing on the base station is recommended.

Figure 4-14 Recommendation of source-based routing on the base station



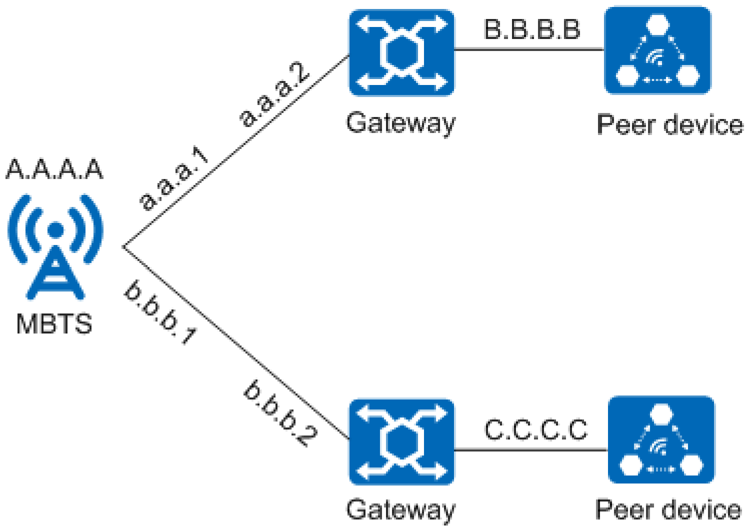
For example,

– In base station controller transmission pool scenarios, a UMTS-only base station is used or a co-MPT base station works in UMTS mode.
– In LTE MME pool scenarios, an LTE-only base station is used or a co-MPT base station works in LTE mode.
– In base station controller expansion or swapping scenarios, a GSM-only or UMTS-only base station is used or a co-MPT base station works in GSM or UMTS mode.

- Lack of support for source-based routing on the base station

  Routes to be configured on a base station have the same service IP address but different next-hop IP addresses and destination IP addresses. **Figure 4-15** is an exemplary diagram of scenario in which source-based routing on the base station is not supported.

Figure 4-15 Lack of support for source-based routing on the base station



For example, if the LTE X2 interface uses layer 2 networking and does not support source-based routing on the base station, and the S1 interface uses Layer 3 networking and requires source-based routing on the base station, two IP addresses must be configured for the S1 and X2 interfaces, respectively, on the base station.

4.4.2 IP Layer Fragmentation

The IP layer supports fragmentation of an IP packet if the packet exceeds the lower layer MTU size. It also supports reassembly of fragmented packets received from the lower layer. The MTU varies depending on the communication interface. An MTU includes the IP packet header bytes, but excludes the header bytes of layers below the IP layer.

NOTE

The base station/BSC6960 supports fragmentation and reassembly of IP packets with a maximum of five fragments. In addition, fragmentation and reassembly degrade network performance and transmission efficiency. Therefore, properly plan the MTU to avoid fragmentation.

4.4.2.1 MTU over an Ethernet Port

The MTU over an Ethernet port has two types: port MTU and route MTU.

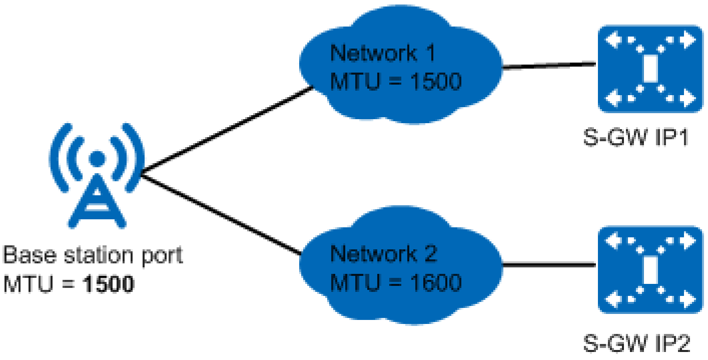
Port MTU

When the old transmission configuration model is used, a port MTU defines the size of the maximum transmission unit that can be sent over a port. It is specified by the **MTU** parameter in the **ETHPORT**, **ETHTRK**, or **ETHCIPORT** MO. Both the base station and base station controller support port MTU.

When the new transmission configuration model is used, a port MTU of a base station defines the size of the maximum transmission unit that can be sent over an interface. It is specified by the **INTERFACE**.***MTU4 (LTE eNodeB, 5G gNodeB)*** parameter. A port MTU of the BSC6910/BSC6900 defines the size of the maximum transmission unit that can be sent over a port. It is specified by the **MTU** parameter in the **ETHPORT** MO. A port MTU of the BSC6960 defines the size of the maximum transmission unit that can be sent over an interface. It is specified by the **INTERFACE**.***MTU4*** parameter.

If a base station or a base station controller connects to multiple backhaul networks that have different MTUs, it is recommended that the port MTU configured on the base station or base station controller side be the minimum MTU of these backhaul networks. This helps reduce packet fragmentation of packets on the backhaul networks. For the backhaul networks whose MTUs are greater than the port MTU, fragmentation increases on the base station side, which degrades the transmission performance of networks.

Figure 4-16 Configuring a port MTU



In the example shown in **Figure 4-16**, the port MTU of the base station is set to 1500 bytes. If the size of an IP packet (for example, 1550 bytes) to be sent to networks 1 and 2 exceeds 1500 bytes, the base station fragments the packet before sending it.

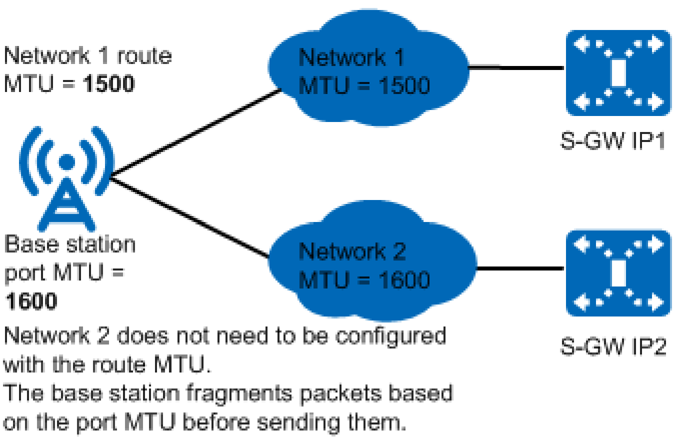
Route MTU

A route MTU defines the size of the maximum transmission unit that can be sent over a route. It is configured using the **IPRT**.***MTUSWITCH (5G gNodeB, LTE eNodeB)*** and **IPRT**.***MTU (5G gNodeB, LTE eNodeB)*** parameters in the old model, or the **IPROUTE4**.***MTUSWITCH (5G gNodeB, LTE eNodeB)*** and **IPROUTE4**.***MTU (5G gNodeB, LTE eNodeB)*** parameters in the new model. Base stations support the route MTU for destination routes.

If a base station connects to multiple backhaul networks that have different MTUs, it is recommended that the following MTU configurations be adopted to reduce packet fragmentation on the backhaul network and base station and improve transmission performance:

- The port MTU of the base station is set to the maximum MTU of the backhaul networks.
- Different MTUs are configured for the destination routes to the backhaul networks.
- The route MTU is less than or equal to the port MTU.

Figure 4-17 Configuring a route MTU



In the example shown in **Figure 4-17**, the port MTU of the base station is set to 1600 bytes. The route MTU for routes that link the base station to backhaul network 1 is set to 1500 bytes. If the base station sends a packet of 1550 bytes to backhaul networks 1 and 2 separately, then:

- For network 1

  The base station uses the route MTU (the minimum one of the port MTU and route MTU) for fragmentation decisions.

  The packet size exceeds the route MTU size of the route to backhaul network 1. Therefore, the base station fragments the packet before sending it.

- For network 2

  The base station uses the port MTU for fragmentation decisions.

  The packet size is less than the route MTU size of the route to backhaul network 2. Therefore, the base station directly sends the packet without fragmenting it.

In a secure network, the base station can only use the route MTU for routes based on outer destination IP address of an IP packet and the port MTU for fragmentation decisions.

4.4.2.2 MTU over an E1/T1 Port

The value of MTU over an E1/T1 port is the smaller MTU value of the two ends, namely the negotiated maximum receive unit (MRU).

The MRU indicates the capability of an interface to receive packets. The initial negotiated MRU is the configured MTU or the default port MTU. For details about MRUs in IP over E1 (PPP) scenarios, see *RFC1661 - The Point-to-Point Protocol (PPP)*.

On the base station side, an MRU can be configured by using the **MRU** parameter in the **PPPLNK** MO or in the **MPLNK** MO. On the base station controller side, the MRU cannot be configured.

4.5 Transport Layer

4.5.1 UDP

The User Datagram Protocol (UDP) is used to transmit user data stream. UDP runs over the IP layer and is a connectionless transport layer protocol.

UDP is mainly used to convert data between network data streams and UDP datagrams. At the transmit end, UDP encapsulates network data streams into datagrams and then transmits the datagrams. At the receive end, UDP decapsulates datagrams to obtain the original data.

The user planes of the following interfaces are carried on the UDP: A and Gb interfaces of GSM; Iub, Iur, Iu, and uX2 interfaces of UMTS; S1, X2, and eX2 interfaces of LTE; S1, X2, NG, Xn, and eXn interfaces of NR.

4.5.2 SCTP

The Stream Control Transmission Protocol (SCTP) establishes an SCTP association between two SCTP endpoints to implement connection-oriented reliable transmission. For details, see RFC 4960. Only one SCTP association is established between two endpoints.

SCTP, based on IP, is a signaling bearer protocol for the following interfaces: A and Abis (eGBTS) of GSM, Iub/Iur/Iu-CS/Iu-PS/uX2 of UMTS, S1/X2/eX2 of LTE, and NG/X2/Xn of NR. An SCTP path is identified by the local IP address, local SCTP port number, peer IP address, and peer SCTP port number.

The SCTP layer fragments the packet whose length exceeds the maximum SCTP data unit (excluding the IP header byte) and reassembles the received fragments. This function is implemented before IP layer fragmentation and after IP layer reassembly.

For base station controllers, the maximum SCTP data unit is specified by the **SCTPLNK**.***MTU*** parameter in non-IP-transmission-resource-pool scenarios, and is specified by the **SCTPPROF**.***MTU*** parameter in IP transmission resource pool scenarios. For base stations, the maximum SCTP data unit is specified by the **SCTPLNK**.***MAXSCTPPDUSIZE (5G gNodeB, LTE eNodeB)*** parameter in link mode, and is specified by the **SCTPTEMPLATE**.***MAXSCTPPDUSIZE (5G gNodeB, LTE eNodeB)*** parameter in endpoint mode. You are advised to set this parameter to the IP layer MTU minus the IP header byte. This ensures that packets are not fragmented again at the IP layer after being fragmented at the SCTP layer.

The **TRANSFUNCTIONSW**.***SCTPSERVERSW (5G gNodeB, LTE eNodeB)*** parameter can be used to specify whether the server function is supported for SCTP of any of the following service types: RSL, NCP, CCP, S1-AP, R6-AP, R8-AP, ALCAP-AP, M2-AP, Se-AP, gNBNG-AP, DsaS1-AP, and M3-AP.

4.5.3 TCP

TCP is a reliable connection-oriented transport layer protocol. TCP is used for carrying OM channels for GSM (eGBTS and BSC), UMTS, NR and LTE.

4.5.4 GTP-U (UMTS/LTE/NR)

GTP-U is used to transmit user data on the following interfaces:

- Iu-PS for UMTS
- S1/X2 for LTE
- S1/X2/NG/Xn for NR

GTP-U complies with 3GPP TS 29.281.

A GTP-U path contains multiple GTP-U tunnels between two IP addresses.

A GTP-U tunnel carries a radio service. Each tunnel is identified by an IP address and a tunnel endpoint identifier (TEID). Two TEIDs can be used to identify the uplink and downlink of a GTP-U tunnel.

4.5.5 RTP (GSM/UMTS)

The Real-Time Transport Protocol (RTP) is specially designed for transmitting Internet multimedia data. RTP is usually used in combination with UDP to transmit real-time data. UDP is responsible for transmitting datagrams, but it does not ensure the time sequence of datagram transmission. RTP adds fields such as Timestamp and Sequence Number to the datagrams to provide time information and implement stream synchronization. RTP, however, does not provide reliable transmission mechanisms, flow control, or congestion control. RTP is used for the A interface of GSM and user plane of the Iu-CS interface of UMTS.

4.5.6 RTCP (GSM/UMTS)

The Real-Time Transport Control Protocol (RTCP) works together with RTP, is stipulated in 3GPP TS 29.414 v7.2.0 and TR 29.814 V7.1.0, and can be used for configuring the multiplexing function.

RTCP is used for the A interface of GSM and user plane of the Iu-CS interface of UMTS.

4.5.7 M3UA (GSM/UMTS)

The MTP3-User Adaptation Layer (M3UA) protocol is the user adaptation protocol for MTP3 or MTP-3b. It provides primitive-based communication services for the MTP3 users on the IP network and (on the signaling gateway) at the network edge. It enables the interworking between TDM-based SS7 and IP and provides connectionless services. M3UA is used for the A interface of GSM and the Iur, Iu-CS, and Iu-PS interfaces of UMTS.

The data configurations of M3UA are similar to those of MTP-3b. The local and destination entities of M3UA can be considered as the originating signaling point (OSP) and destination signaling point (DSP) of SS7.

The parameters of the base station controller are as follows:

- **Local entity No.** (***LENO***) uniquely identifies a local entity within the OSP.
- **Destination entity No.** (***DENO***) uniquely identifies a destination entity within the DSP.
- **Local entity type** (***ENTITYT***) specifies the types of local entities. The types can be **M3UA_ASP** or **M3UA_IPSP**. If the local and destination entities are connected by a signaling transfer point, **M3UA_ASP** is recommended. If the local and destination entities are directly connected, **M3UA_IPSP** is recommended. **Destination entity type** (***ENTITYT***) specifies the types of destination entities. The types can be **M3UA_SGP**, **M3UA_IPSP**, **M3UA_SS7SP**, or **M3UA_SP**. For details about ASP, SGP, IPSP, SS7SP, and SP, see RFC4666.
- **Routing Context** (***RTCONTEXT***) specifies the routing context of the M3UA local entity or the M3UA destination entity. The routing context connects an ASP and an AS. An ASP can serve multiple ASs by using the same SCTP connection. When the M3UA local entity sends the ASPActive or DATA message to a peer end, the corresponding routing context must also be sent. When the M3UA local entity is M3UA_ASP, the routing context at local and peer ends must be consistent after negotiation. If they are inconsistent, the route cannot be pinged. When all the bits of the routing context are 1, no routing context needs to be configured. The routing context of the M3UA local entity must be unique. For details, see RFC4666.
- **Signaling link set index** (***SIGLKSX***) uniquely identifies a signaling link set within the base station controller.

- **Signaling link mask** (***LNKSLSMASK***) specifies the signaling link mask corresponding to an M3UA link set. It is used for the M3UA link load sharing.

  The number of 1s (n) determines the maximum number (2^n) of links for load sharing. For example:

  – B0000 has no 1s, indicating that a maximum of one (2^0) link is used for load sharing.

  – B0001 and B1000 both have one 1s, indicating that a maximum of two (2^1) links are used for load sharing.

  The AND operation between this value and the value of the signaling route mask (***SLSMASK***) in the **ADD N7DPC** command must be equal to 0.

- **Traffic mode** (***TRAMODE***) specifies the traffic mode of an M3UA link set.

  The value of this parameter can be **M3UA_OVERRIDE_MOD** or **M3UA_LOADSHARE_MOD**. For details about these two modes, see RFC4666.

- **Work mode** (***WKMODE***) specifies the working mode of an M3UA link set.

  The value of this parameter can be **M3UA_ASP** or **M3UA_IPSP**. For details about ASP and IPSP, see RFC4666.

- **PENDING timer** (***PDTMRVALUE***) specifies the length of the PENDING timer for an M3UA link set.
- **Signaling link ID** (***SIGLNKID***) uniquely identifies a signaling link within a signaling link set.
- **SCTP link No.** (***SCTPLNKN***) specifies the SCTP link that carries an M3UA link.
- **Signaling link priority** (***PRIORITY***) specifies the priority of a signaling link. The value **0** indicates the highest priority.
- **Initial bearing traffic active tag** (***LNKREDFLAG***) specifies the initial bearer flag of an M3UA link.

  The value of this parameter can be **M3UA_MASTER_MOD** or **M3UA_SLAVE_MOD**. For details about the bearer flag, see RFC4666.

- **Route priority** (***PRIORITY***) specifies the priority of a route. The value **0** indicates the highest priority.

4.5.8 SCCP (GSM/UMTS)

The Signaling Connection Control Part (SCCP) provides additional functions for MTP-3b and M3UA. It enables the SS7 signaling points to transmit circuit-related messages and non-circuit-related messages. SCCP is used for the A interface of the GSM and the Iur, Iu-CS, and Iu-PS interfaces of the UMTS. SCCP supports connection-oriented and connectionless services and provides address resolution. In the IP transmission architecture, SCCP is used on the Iu and Iur interfaces to transmit RANAP and RNSAP messages respectively. Messages related to a single user are transmitted in connection-oriented mode, and other messages are transmitted in connectionless mode.

SCCP performs addressing according to the destination point code (DPC) and the subsystem number (SSN). Both communication parties use the SSN to identify the upper-layer application, such as RANAP or RNSAP.

The parameters of the base station controller are as follows:

- **OSP index** (***SPX***): uniquely identifies an OSP in the SS7 signaling network.

- **DSP index** (*DPX*): uniquely identifies a DSP in the SS7 signaling network.

4.6 Protocol Stacks for Transport Interfaces

4.6.1 Protocol Stacks for GSM Transport Interfaces

Abis Interface

Abis over IP supports two transmission modes: IP over Ethernet and IP over E1/T1/Ch-STM-1. The IP protocol version adopted is IPv4.

Figure 4-18 Protocol stack for Abis over IP (when the GBTS is used)

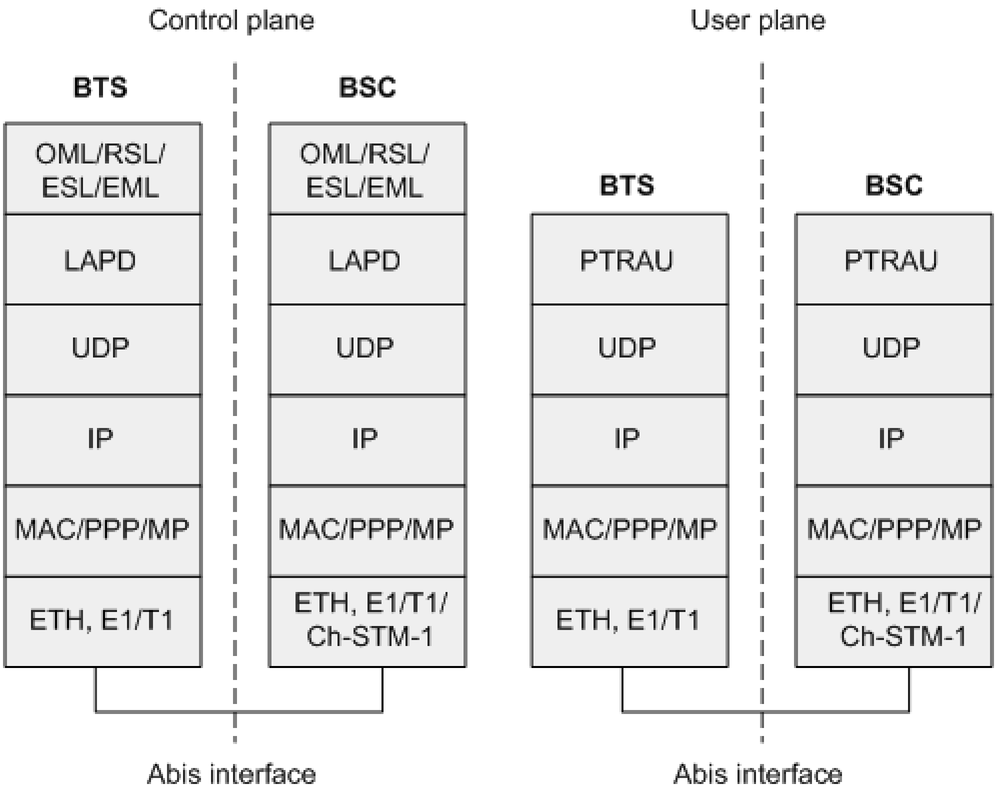


Figure 4-19 RSL carrying SCTP for Abis over IP (when the eGBTS is equipped with the UMPT)

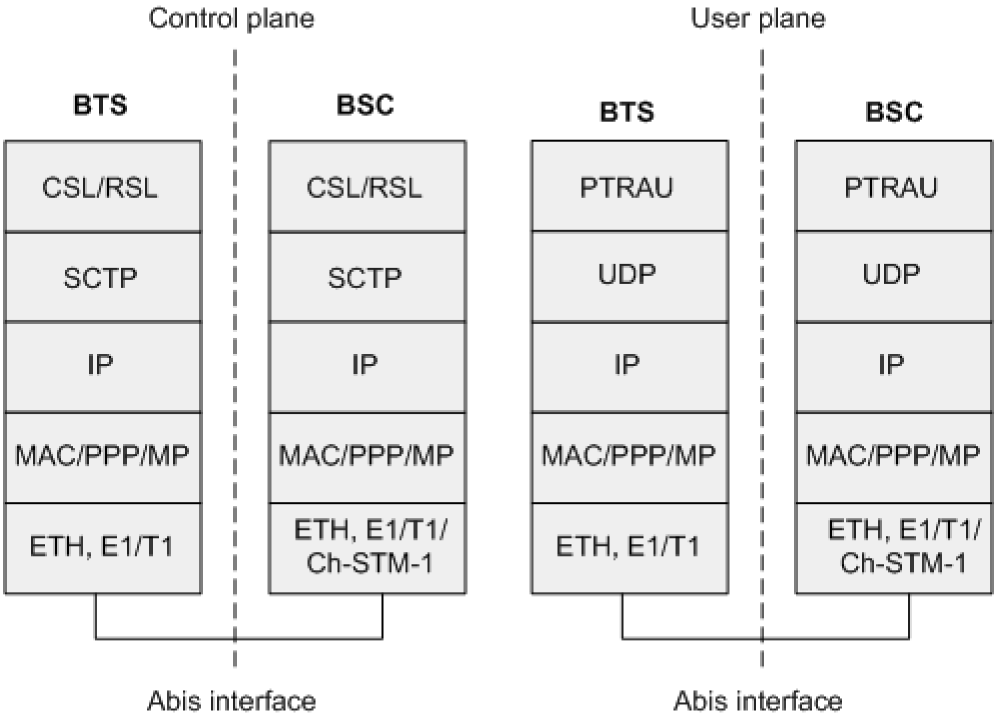


Figure 4-20 RSL carrying LAPD for Abis over IP (when the eGBTS is equipped with the UMPT)

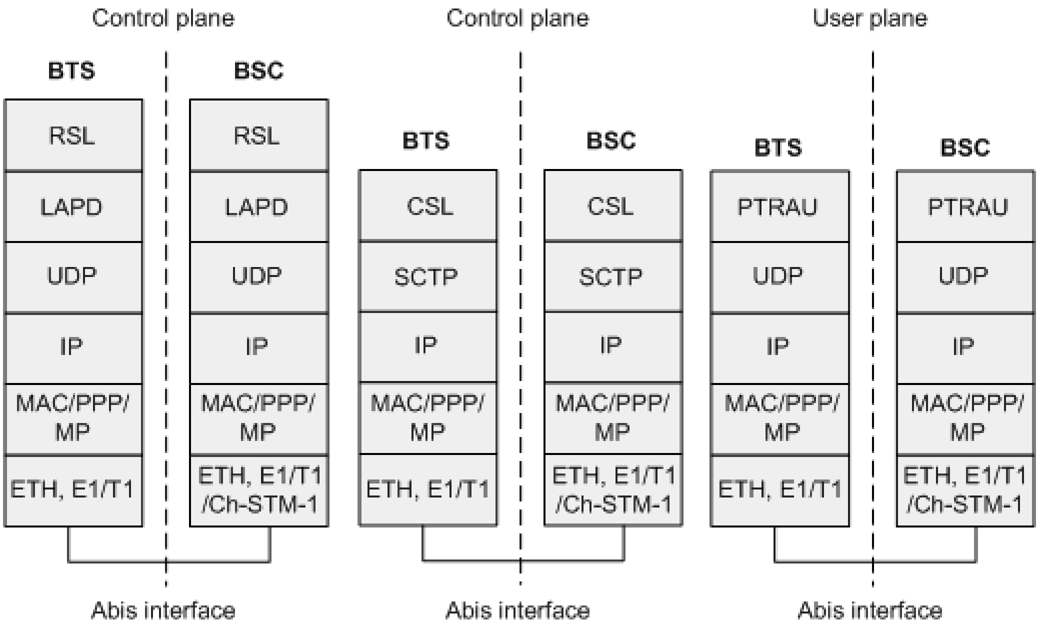
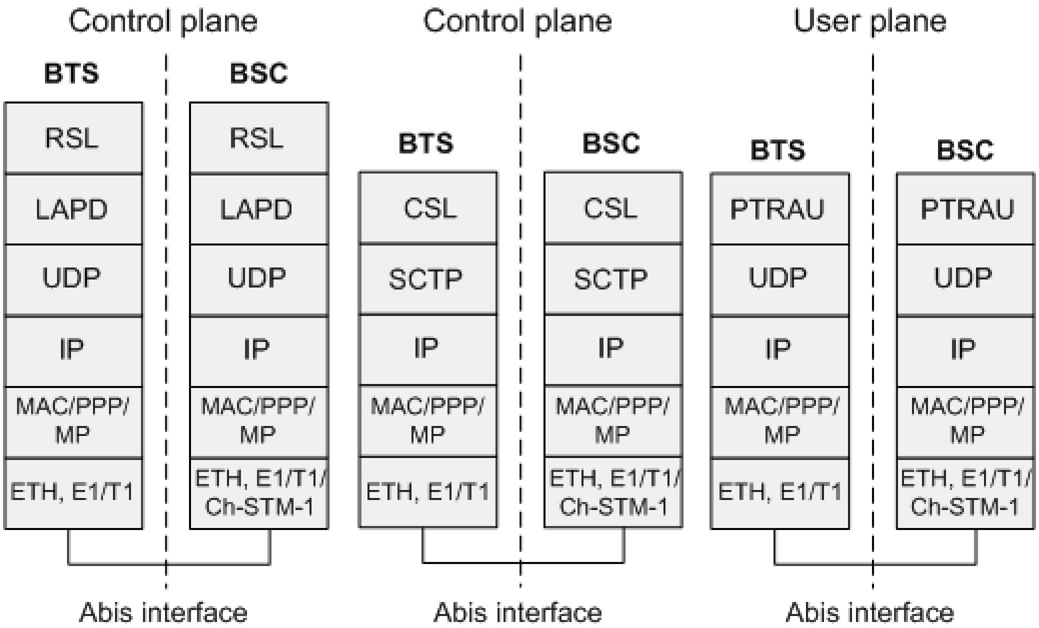


Figure 4-21 Protocol stack for Abis over IP (when the eGBTS is equipped with the UMPT)

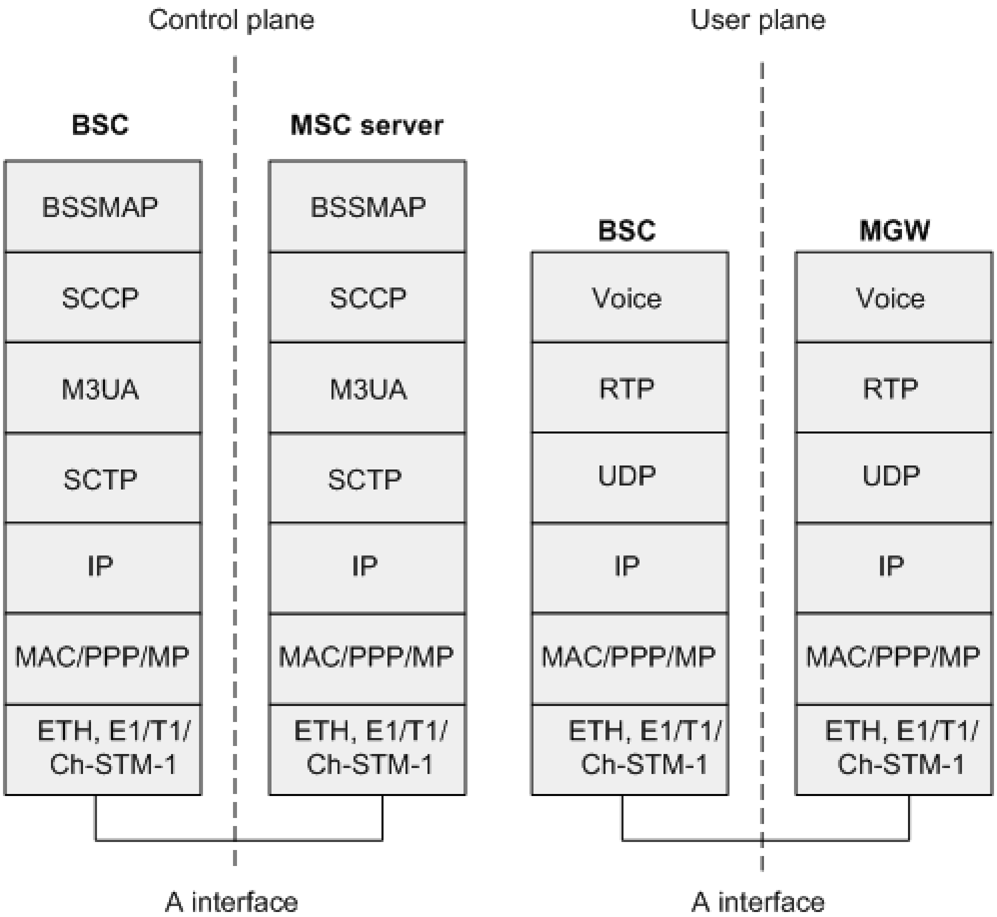


A Interface

A over IP indicates that the IP network is used for transmitting signaling and voice signals over the A interface. A over IP supports two transmission modes: IP over Ethernet and IP over E1/T1/Ch-STM-1. The IP protocol version adopted is IPv4.

On the A interface, the signaling plane uses the SCCP/M3UA/SCTP/IP protocol stack to carry signaling, and the user plane uses the RTP/UDP/IP protocol stack to carry voice signals, as shown in **Figure 4-22**.

Figure 4-22 Protocol stack for A over IP

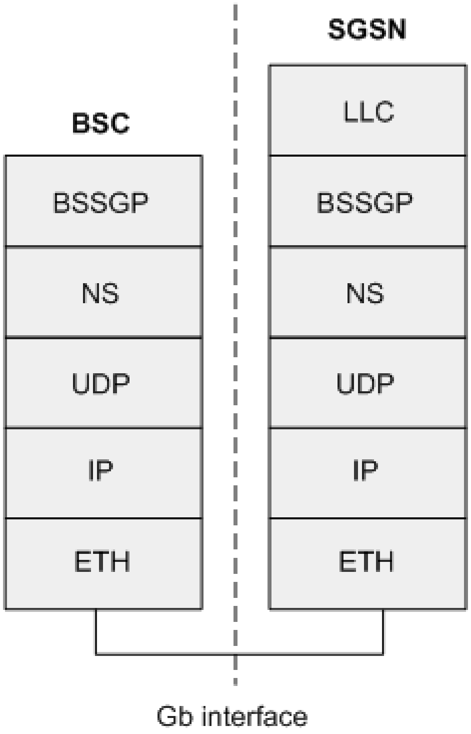


Gb Interface

Gb over IP supports only IP over Ethernet and does not support IP over E1/T1/Ch-STM-1. The IP protocol version adopted is IPv4.

In Gb over IP, the physical layer uses the Ethernet transmission, the layer 2 is based on the NS protocol, the sub NS layer of the NS protocol complies with the IP protocol, and the application layer of the NS protocol complies with the BSSGP protocol, as shown in **Figure 4-23**.

Figure 4-23 Protocol stack for Gb over IP



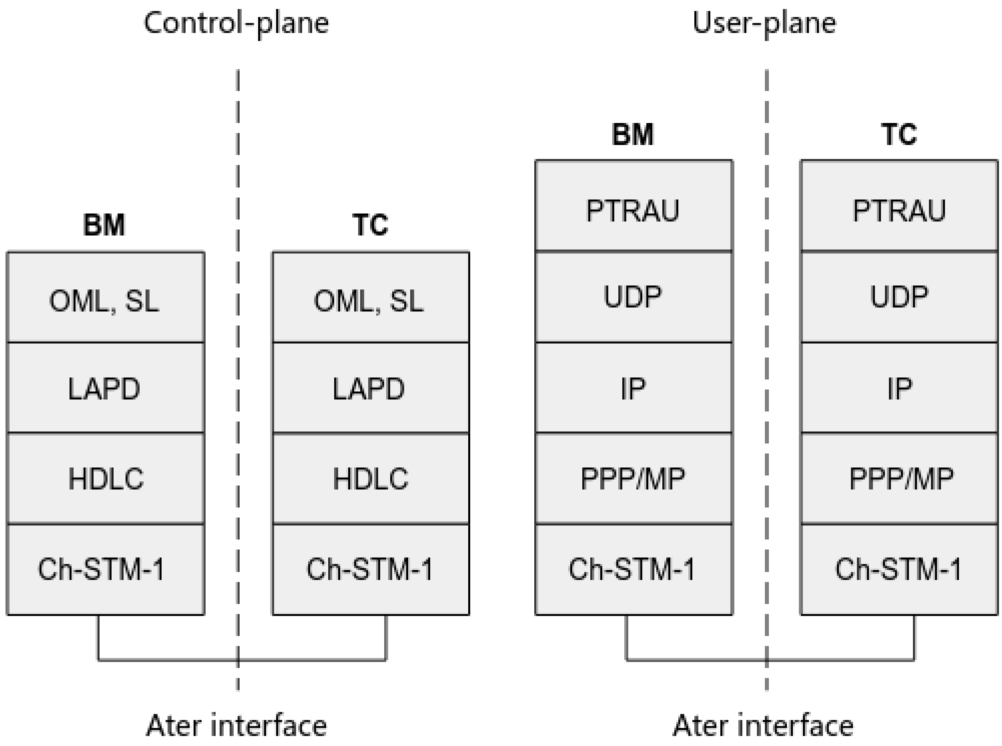
Ater Interface

Ater over IP supports only IP over PPP over E1/T1/Ch-STM-1. The IP protocol version adopted is IPv4. The BSC6910 does not provide an Ater interface.

Currently, the Huawei Ater interface supports only IP over Ch-STM-1 and does not support IP over E1/T1.

In Ater over IP mode, the physical layer uses Ch-STM-1 optical transmission, the user plane uses the PTRAU/UDP/IP protocol stack, and the signaling plane uses the LAPD/HDLC/TDM protocol stack. **Figure 4-24** shows the protocol stack of the Ater interface.

Figure 4-24 Protocol stack for the Ater interface

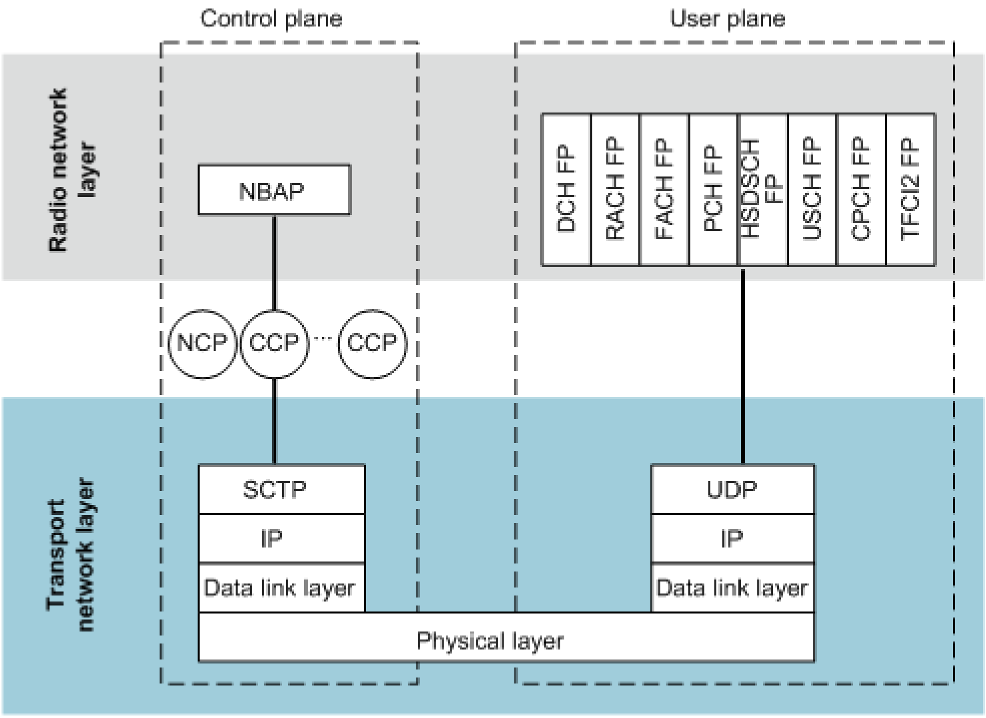


4.6.2 Protocol Stacks for UMTS Transport Interfaces

Iub Interface

Figure 4-25 shows the protocol stack for Iub over IP.

Figure 4-25 Protocol stack for Iub over IP



- Control plane

  The application protocol for the control plane of the Iub interface is the NodeB Application Part (NBAP). NBAP is responsible for the transport of control plane messages between the NodeB and the CRNC at the radio network layer.
- User plane

  The application protocols for the user plane of the Iub interface are a series of frame protocols: DCH FP, RACH FP, FACH FP, PCH FP, HS-DSCH FP, and E-DCH FP. These protocols are responsible for the transport of data and control frames between the NodeB and the CRNC. These frames contain Uu interface user data and user-related control data.

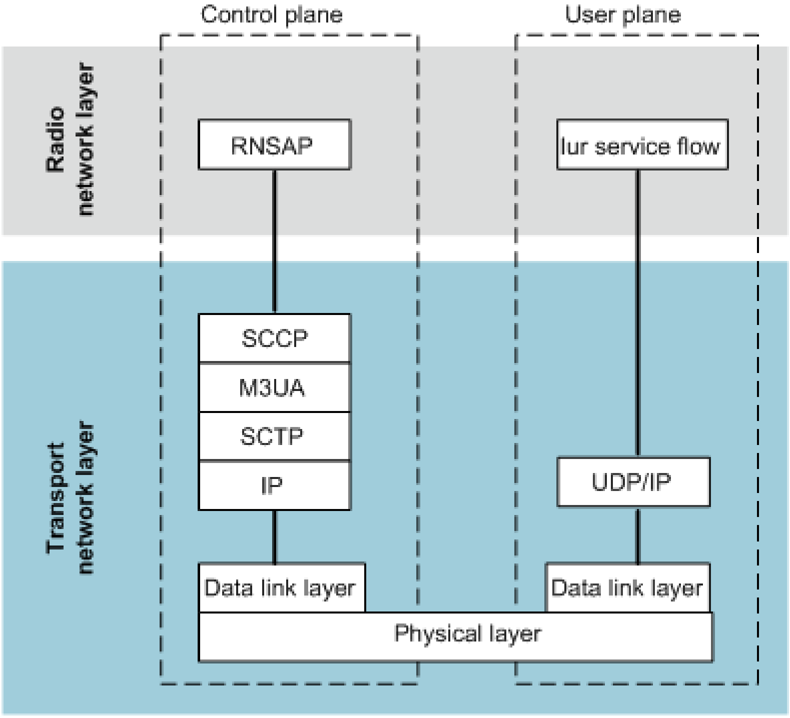
NOTE

For details about the protocol stack for ATM&IP dual-stack transmission, see *ATM&IP Dual Stack Feature Parameter Description*.

Iur Interface

Figure 4-26 shows the protocol stack for Iur over IP.

Figure 4-26 Protocol stack for Iur over IP



- Control plane

  The application protocol for the control plane of the Iur interface is the Radio Network Subsystem Application Part (RNSAP). The UE communicates with the DRNC through RNSAP signaling messages.

  When Iur over IP is used, RNSAP is carried on the SCTP link.

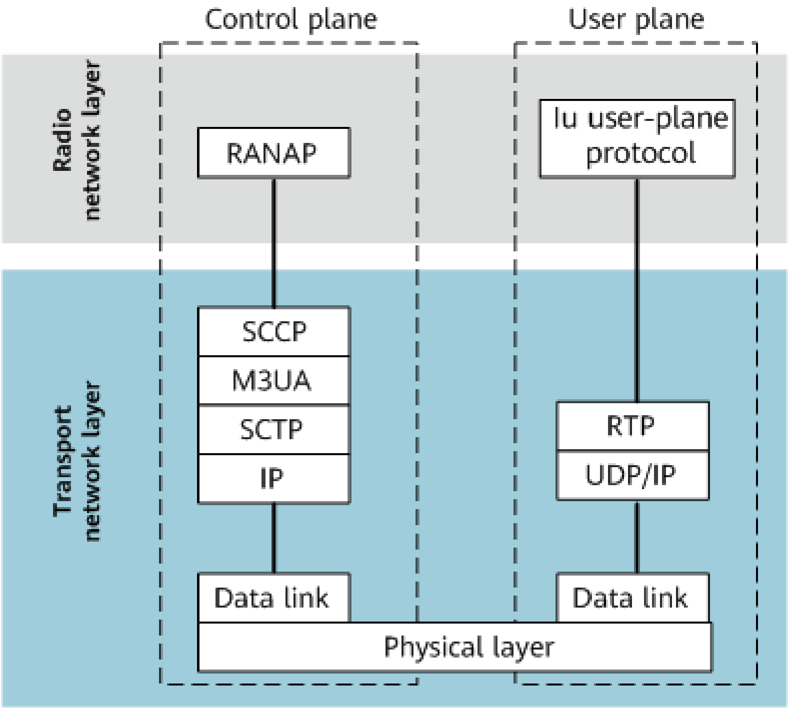
- User plane

  The Iur data stream carries the data forwarded by the DRNC between the SRNC and the NodeB.

Iu-CS Interface

Figure 4-27 shows the protocol stack for Iu-CS over IP.

Figure 4-27 Protocol stack for Iu-CS over IP



- Control plane

  The application protocol for the control plane of the Iu-CS interface is the Radio Access Network Application Part (RANAP). The UE communicates with a circuit switched (CS) node on the core network (CN) through RANAP signaling messages.

  When Iu-CS over IP is used, RANAP is carried on the SCTP link.

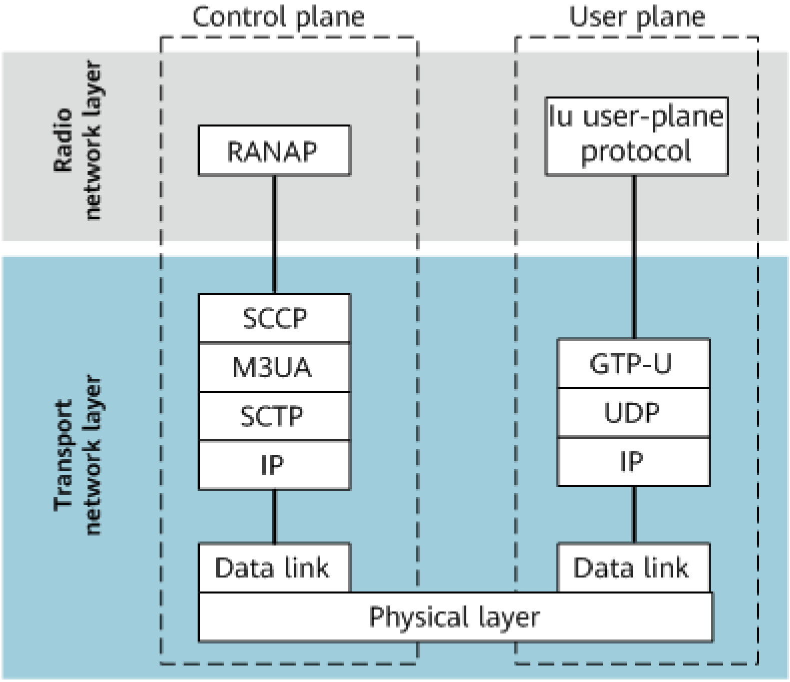
- User plane

  The application protocol for the user plane of the Iu-CS interface is the Iu-UP. Iu-UP is responsible for the transport of the user data carried on the radio access bearer (RAB).

Iu-PS Interface

Figure 4-28 shows the protocol stack for Iu-PS over IP.

Figure 4-28 Protocol stack for Iu-PS over IP



- Control plane

  The application protocol for the control plane of the Iu-PS interface is the Radio Access Network Application Part (RANAP). UE communicates with a packet switched (PS) node on the CN through RANAP signaling messages.

  When Iu-PS over IP is used, RANAP is carried on the SCTP link.
- User plane

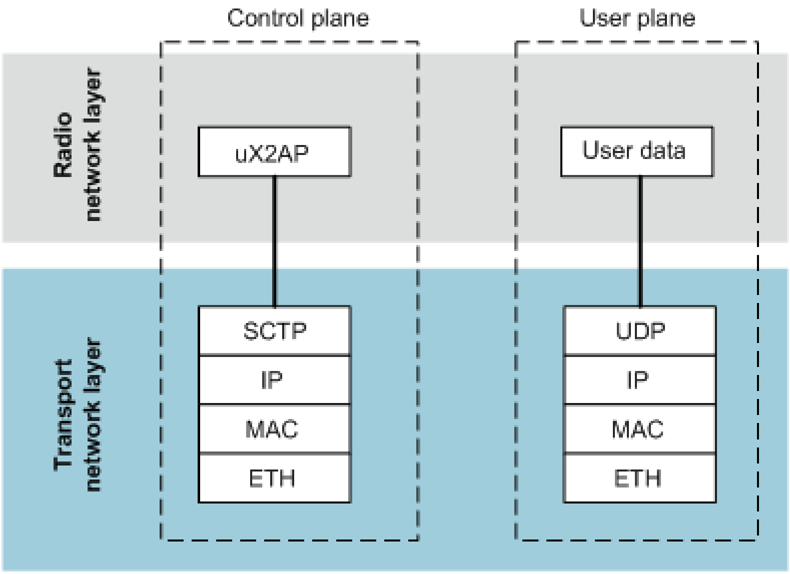
  The application protocol for the user plane of the Iu-PS interface is the Iu-UP. Iu-UP is responsible for the transport of the user data carried on the RAB.

uX2 Interface

Figure 4-29 shows the protocol stack on the uX2 interface. The control plane and user plane use IP, and the data link layer and physical layer use Ethernet.

- The control plane of the uX2 interface uses SCTP at the transport layer.
- The user plane of the uX2 interface uses UDP/IP at the transport layer.

Figure 4-29 Protocol stack for the uX2 interface



4.6.3 Protocol Stacks for LTE Transport Interfaces

Control- and user-plane protocols for the S1 and X2/eX2 interfaces are essential to the communications between the eNodeB and its peer devices.

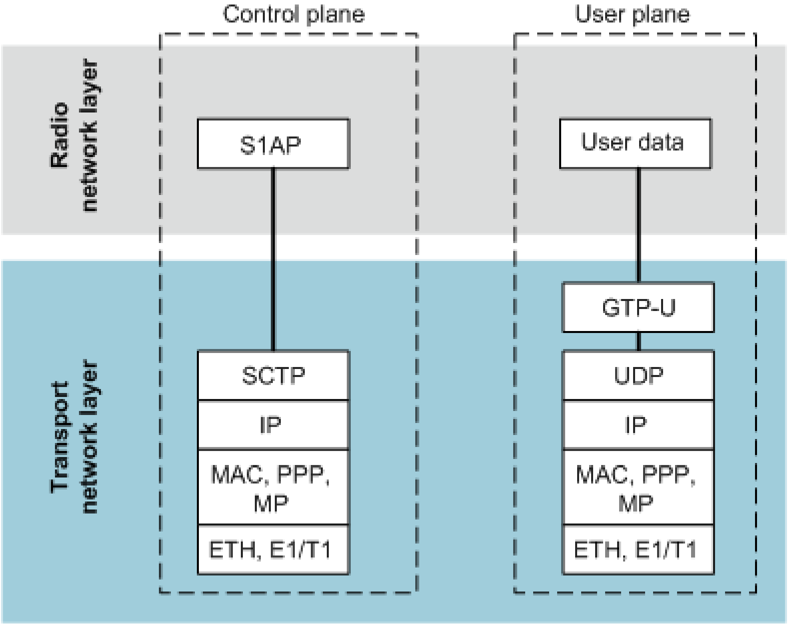
NB-IoT supports only the S1 and X2-C interfaces. It does not support the X2-U or eX2 interface and transmission over these interfaces.

S1 Interface

As shown in **Figure 4-30**, the control plane and user plane use IP. The data link layer and physical layer use Ethernet or PPP over E1/T1.

- The control plane of the S1 interface uses SCTP at the transport layer.
- The user plane of the S1 interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-30 Protocol stack for the S1 interface

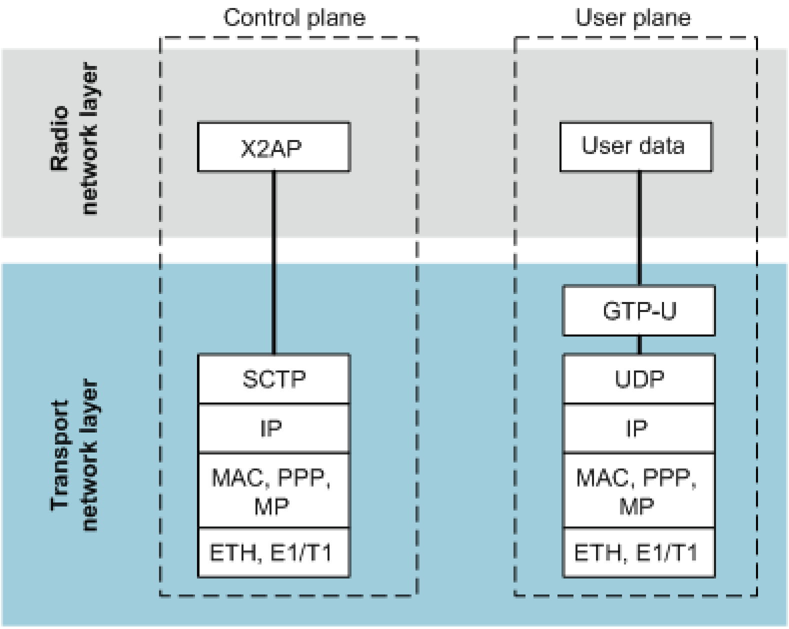


X2 Interface

As shown in **Figure 4-31**, the control plane and user plane use IP. The data link layer and physical layer use Ethernet or PPP over E1/T1.

- The control plane of the X2 interface uses SCTP at the transport layer.
- The user plane of the X2 interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-31 Protocol stack for the X2 interface

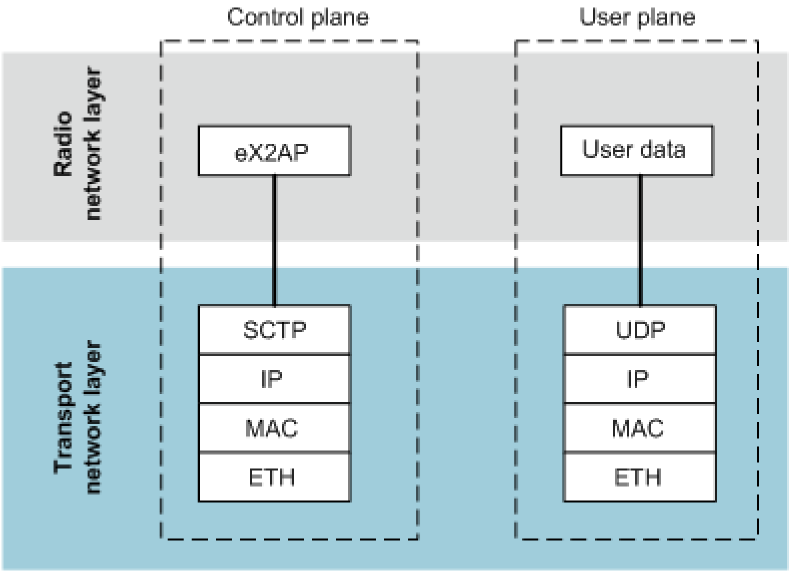


eX2 Interface

As shown in **Figure 4-32**, the control plane and user plane of an eX2 interface use IP. The data link layer and physical layer use Ethernet.

- The control plane of the eX2 interface uses SCTP at the transport layer.
- The user plane of the eX2 interface uses UDP/IP at the transport layer.

Figure 4-32 Protocol stack for the eX2 interface



4.6.4 Protocol Stacks for NR Transport Interfaces

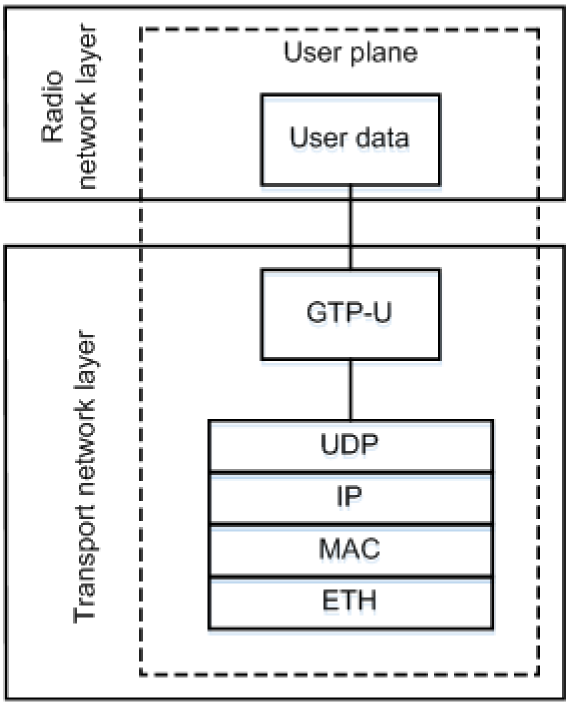
Transmission interfaces on the NSA NR RAN include the S1-U, X2, and eXn-U interfaces. Transmission interfaces on the SA NR RAN include the NG, Xn, and eXn-U interfaces. Control- and user-plane protocols for these interfaces are essential to the communications between a gNodeB and its peer devices.

S1 Interface

As shown in **Figure 4-33**, the user plane of the S1 interface uses IP. The data link layer and physical layer use Ethernet.

The user plane of the S1 interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-33 Protocol stack for the S1 interface user plane

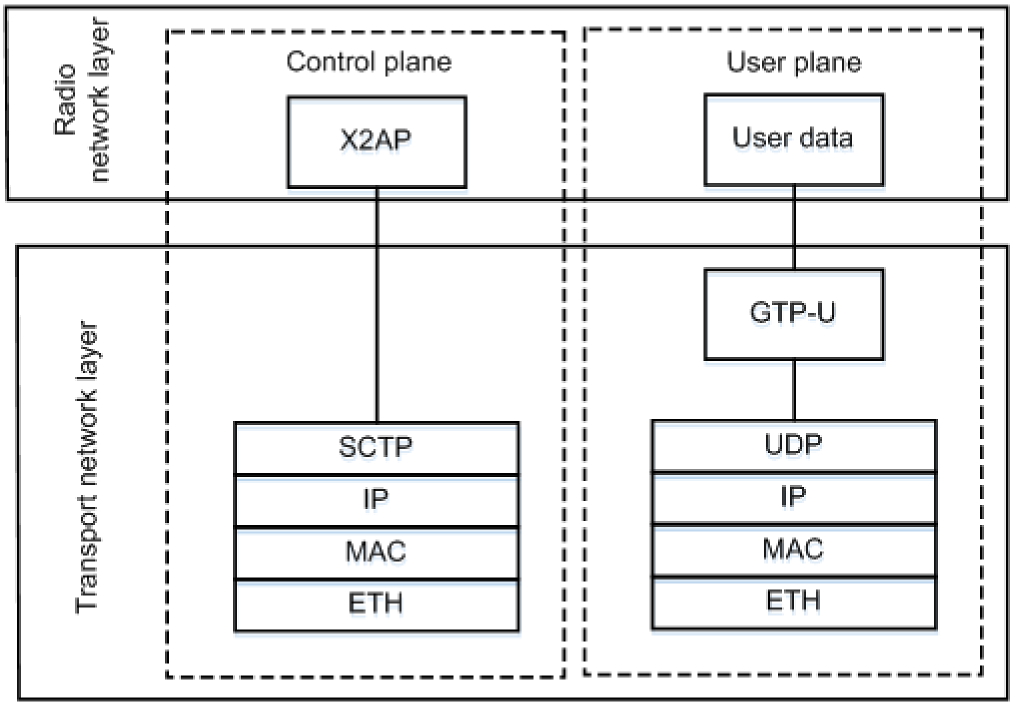


X2 Interface

As shown in **Figure 4-34**, the control plane and user plane of the X2 interface use IP. The data link layer and physical layer use Ethernet.

- The control plane of the X2 interface uses SCTP at the transport layer.
- The user plane of the X2 interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-34 Protocol stack for the X2 interface

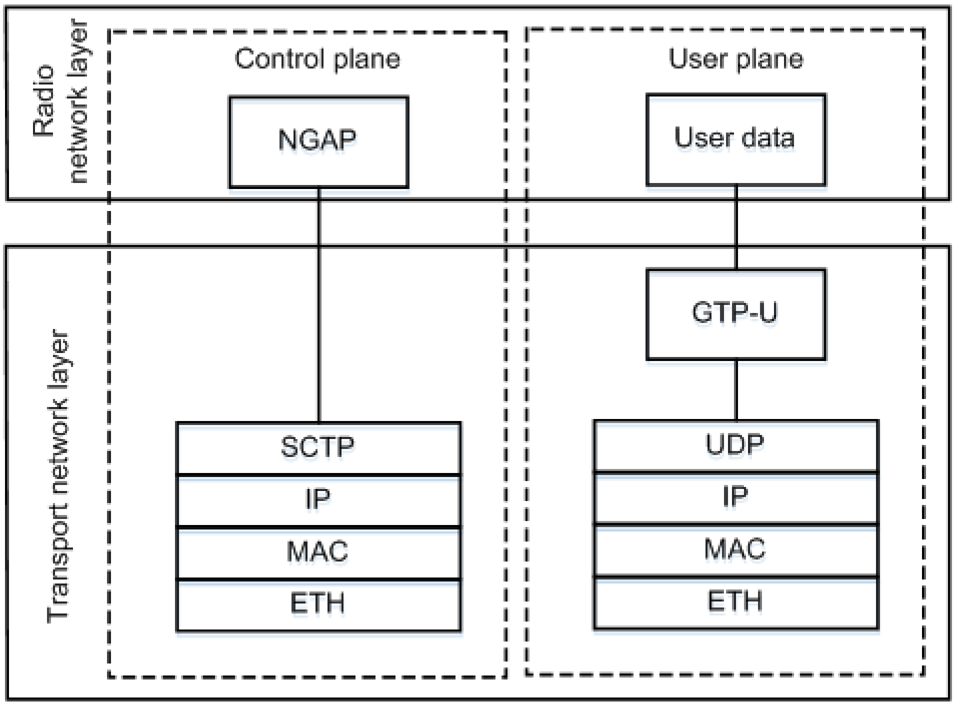


NG Interface

As shown in **Figure 4-35**, the control plane and user plane of the NG interface use IP. The data link layer and physical layer use Ethernet.

- The control plane of the NG interface uses SCTP at the transport layer.
- The user plane of the NG interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-35 Protocol stack for the NG interface

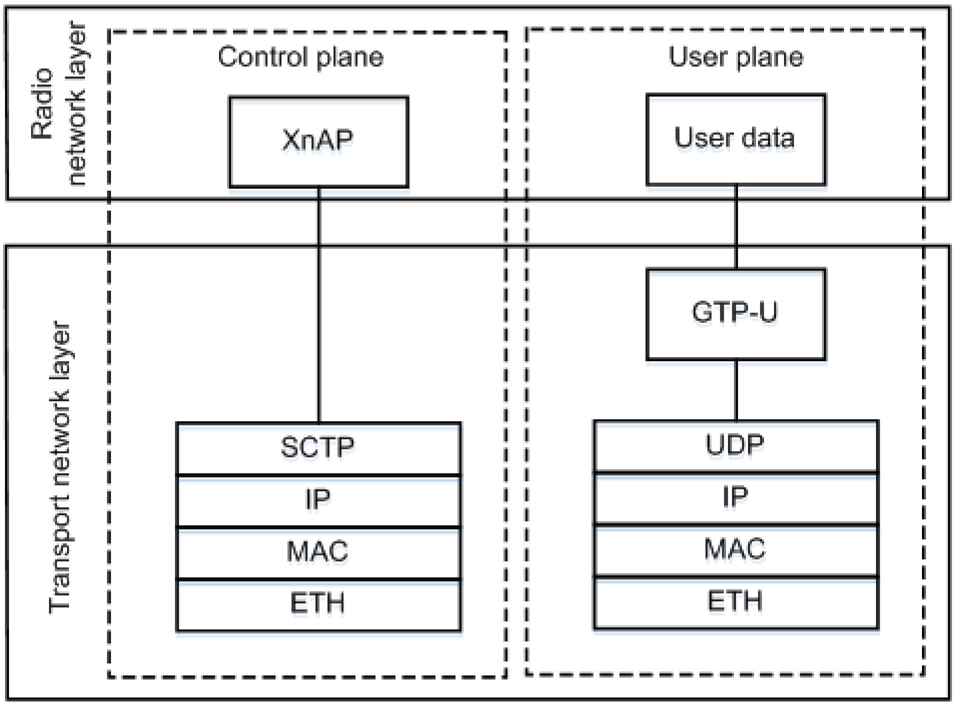


Xn Interface

As shown in **Figure 4-36**, the control plane and user plane of the Xn interface use IP. The data link layer and physical layer use Ethernet.

- The control plane of the Xn interface uses SCTP at the transport layer.
- The user plane of the Xn interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-36 Protocol stack for the Xn interface

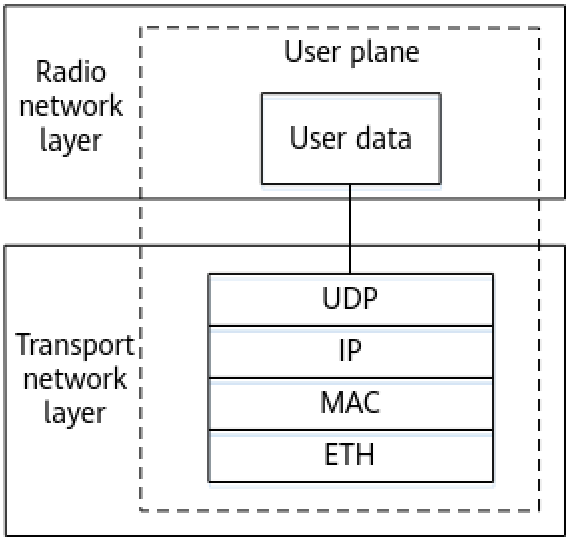


eXn Interface

As shown in **Figure 4-37**, the user plane of the eXn interface uses IP. The data link layer and physical layer use Ethernet.

The user plane of the eXn interface uses UDP/IP at the transport layer.

Figure 4-37 Protocol stack for the eXn interface



4.6.5 Protocol Stacks for Other Transport Interfaces

This section describes protocol stacks for the management-plane interface, IP clock interface, and co-transmission interface of GSM, UMTS, LTE, and NR.

OM Interfaces

The OM interfaces include:

- Interfaces between the OSS and the eGBTS, NodeB, eNodeB, gNodeB, and multimode base station
- Interfaces between the OSS and the RNC, BSC, and multimode base station controller
- Management interfaces between the GBTS and BSC

The OM interfaces for wireless NEs (except the GBTS) mainly use TCP over IP. Some protocols such as the NTP can also use UDP over IP. The data link layer of the OM interface for the eGBTS, NodeB, eNodeB, gNodeB, and multimode base station support Ethernet (FE/GE ports) and PPP or MLPPP (E1/T1 ports). If the eGBTS, NodeB, eNodeB, or MBTS uses the E1/T1 ports, the router or base station controller must convert IP over E1/T1 to IP over Ethernet because the MAE supports only Ethernet transmission.

The OMCH can be configured using the **OMCH** MO. The local IP address of the OMCH (specified by **OMCH**.***IP (LTE eNodeB, 5G gNodeB)***) can be shared with other interfaces or exclusively used by the OMCH.

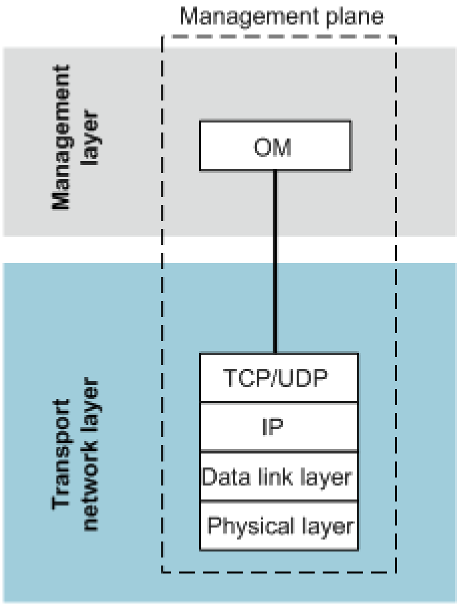
The local IP address of the OMCH can be configured using either of the following methods:

1. Set the **OMCH**.***IP (LTE eNodeB, 5G gNodeB)*** parameter. A **DEVIP** (in the old model)/**IPADDR4** (in the new model) MO does not need to be configured.
2. Configure a **DEVIP** (in the old model)/**IPADDR4** (in the new model) MO. Then set the **OMCH**.***IP (LTE eNodeB, 5G gNodeB)*** parameter to the same value as **DEVIP**.***IP (LTE eNodeB, 5G gNodeB)*** (in the old model)/**IPADDR4**.***IP (LTE eNodeB, 5G gNodeB)*** (in the new model). This method can be used when the IP address of the OMCH must be an interface IP address.

In both configuration methods, ensure that the route between the local IP address of the OMCH and the MAE is reachable.

When the primary and secondary subracks are configured, the local IP address of the OMCH cannot be the same as the IP address defined in the **DEVIP** (in the old model)/**IPADDR4** (in the new model) in the secondary subrack.

Figure 4-38 Protocol stack for the OM interface

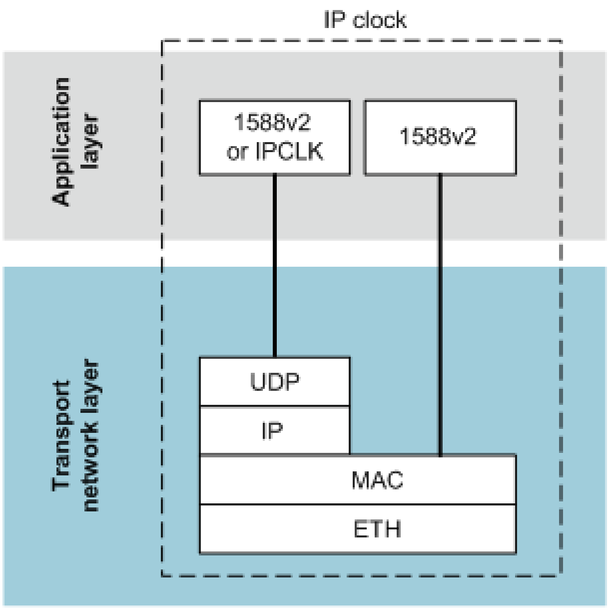


IP Clock Interface

The IP clock interface is used to provide IP-based clock synchronization for base stations. The IP clock is based on IEEE 1588v2, synchronous Ethernet, or Huawei proprietary IPCLK protocol. For more information, see *Synchronization*.

Figure 4-39 shows the protocol stack for the IP clock interface using 1588v2 Layer 3 unicast, 1588v2 Layer 2 multicast, or IPCLK.

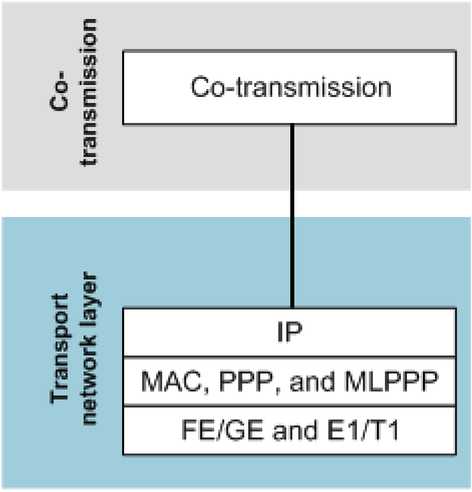
Figure 4-39 Protocol stack for the IP clock interface



Co-Transmission Interface

Co-transmission is implemented by cascading the hub node and leaf nodes. The hub node needs to provide leaf nodes with functions including IP routes, DHCP relay, and transmission bandwidth management. Co-transmission uses basic protocols under the IP layer, as shown in **Figure 4-40**. For more information about co-transmission, see *Common Transmission Feature Parameter Description*.

Figure 4-40 Protocol stack for the co-transmission interface



5 Interface Networking

5.1 Introduction

This chapter describes IP-based networking for the interfaces of GSM, UMTS, LTE, and NR.

5.2 GSM Interface Networking

5.2.1 Abis Interface Networking

The Abis interface supports IP over Ethernet transmission and IP over E1/T1 transmission.

The IP over Ethernet transmission for the Abis interface uses layer 2 or layer 3 networking, depending on the data exchange technologies used by the transport network.

The IP over E1/T1 transmission for the Abis interface uses end-to-end networking or non-end-to-end networking. The BTS supports IP over E1/T1 cascading networking.

NOTE

The Abis interface supports IP over E1/T1/Ch-STM-1. Currently, the BTS does not support IP over Ch-STM-1; therefore, only IP over E1/T1 is described in this document.

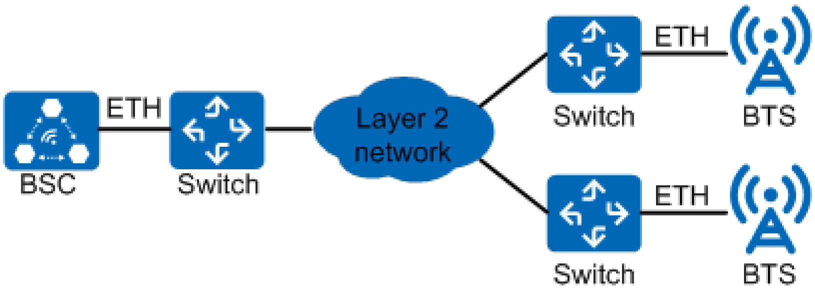
5.2.1.1 IP over Ethernet Networking

Layer 2 Networking

All the network devices in layer 2 networking are located on the same network segment. In this case, data is switched using layer 2 devices. Layer 2 networking can either be data-based (using Ethernet switches) or MSTP-based (using the multi-service transmission platform).

Figure 5-1 shows the data-based Layer 2 networking.

Figure 5-1 Layer 2 data networking



- The BTS and BSC are connected through the Layer 2 network, and they are connected to the bearer network through Ethernet ports.
- The clock reference source of the base station can be the IP clock or GNSS clock.
- VLAN configurations are recommended.

Figure 5-2 shows the MSTP-based networking.

Figure 5-2 MSTP-based networking

In the MSTP-based networking, it is recommended that the Ethernet ports work in active/standby mode to ensure reliability.

- The Layer 2 network is an MSTP network and uses the ring topology to improve transmission reliability.
- The BSC uses the Ethernet interface boards. The networking reliability of interface boards must be considered.
- The BTS provides a transmission bandwidth of up to 100 Mbit/s over the FE port. This facilitates BTS upgrade and capacity expansion and enables quick deployment of data services for the BTS.
- The MSTP device uses the Ethernet interface boards with port-level protection.

The MSTP-based networks are classified into two types: Ethernet virtual private line (EVPL) network and Ethernet virtual private LAN service (EVPLAN) network.

- EVPL advantages:
  - A VLAN is configured for each base station. VLANs are used to distinguish base stations and paths, facilitating fault locating.
  - The VLAN isolation function ensures network security and prevents broadcast storms.
- EVPL disadvantages:
  - A large number of VLANs are required. If OM links require independent VLANs, excessive VLAN resources are consumed.
  - Each base station has a unique VLAN and the VLAN needs to be reused across areas. Therefore, the planning and configuration are difficult.

  - The wireless side and transmission side are closely coupled based on VLANs, leading to larger error possibilities, poor flexibility, and heavy workload in capacity expansion.
- EVPLAN advantages:
  - The number of required VLANs is reduced because the network is based on MAC address switching, making it convenient for base station deployment and migration.
  - There is no need to establish a unique end-to-end VLAN channel for each base station, which reduces the configuration difficulty on the wireless side and transmission side.
- EVPLAN disadvantages:
  - On an MSTP network, it is difficult to locate and resolve a problem related to MAC address-based forwarding.
  - Loopback must be disabled on all the ports (including Ethernet ports and optical ports), or the MSTP network must be upgraded so that Ethernet OAM can automatically disable the port loopback to prevent the loopback from affecting services.

The EVPLAN network is recommended for Layer 2 data networking and MSTP-based networking because the network requires fewer VLANs and reduces the configuration complexity.

The following suggestions are provided for network construction if Huawei devices are used on the RAN and transmission sides:

- The MSTP uses two-level convergence. Both level 1 and level 2 use EPLAN. The optical data board maps MAC addresses of the data on the board to virtual channels (VCs).
- The converged optical switch node (OSN) and router use board protection switching (BPS) and Virtual Router Redundancy Protocol (VRRP). The Ethernet port on the transmission device at the access layer uses the hybrid mode in commissioning before service provisioning.
- It is recommended that one VLAN be planned for the base stations connected to the two pairs of optical network data boards at the level-1 convergence node (one pair of optical network data boards at the level-1 convergence node supports a maximum of 64 base stations) and that the IP addresses be on the same small network segment so that the router can identify the VLAN based on the IP addresses.
- The BSC at the level 2 convergence node is used with one pair of optical network boards. In deployment, each pair of optical network boards is configured with one pair of Ethernet ports. In expansion, each pair of optical network boards is configured with two pairs of Ethernet ports.

VLAN Configuration for Layer 2 Networking

When the old transmission configuration model is used, the single VLAN mode is recommended for the BTS side so that data with the same next-hop IP address is tagged with the same VLAN. When the new transmission configuration model is used, the interface VLAN mode is recommended for the BTS side so that data on the same interface is tagged with the same VLAN.

VLAN tagging based on the next-hop IP address is recommended for the BSC side.

Different VLAN priorities can be assigned to different service types to implement differentiated services. If the network is congested, high-priority packets are scheduled, and low-priority packets are buffered or discarded.

For details about VLAN configurations, see **4.3.5 VLAN**.

IP Address Configuration for Layer 2 Networking

To simplify data configuration, the port IP addresses configured on the GBTS and BSC are used as their communicating IP addresses.

For the eGBTS, the port IP address configured on the BSC is used as the control- and user-plane IP address for the Abis interface. The eGBTS can use this or another IP address as the OM IP address.

IP Route Configuration for Layer 2 Networking

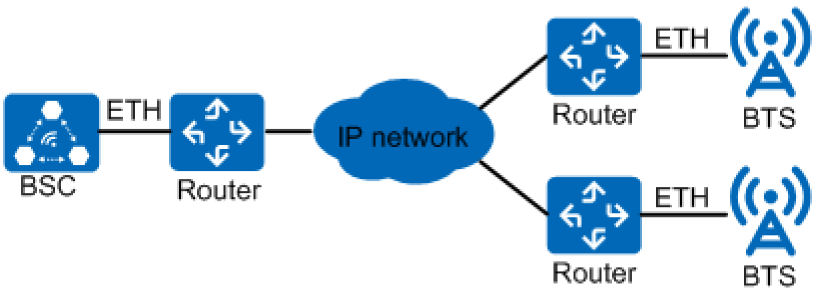
If the preceding IP addresses are used, you do not need to configure routes. When the device IP address is used for transmission on the base station controller side, the base station must be configured with a route to the device IP address of the base station controller.

If the eGBTS is not on the same network segment as the MAE, a route to the MAE must be configured on the eGBTS and a route to the eGBTS OM IP address must be configured on the MAE.

Layer 3 Networking

In Layer 3 networking, the network devices are located on different network segments. In this case, data is switched using Layer 3 devices (for example, routers), as shown in **Figure 5-3**. The Layer 3 network provides differentiated services according to the DSCP value in the IP header. If the network is congested, high-priority packets are scheduled, and low-priority packets are buffered or discarded.

Figure 5-3 Layer 3 data networking



- The BTS and BSC are connected to each other through the Layer 3 network, and they are connected to the bearer network through Ethernet ports.
- VLAN tags can be attached by the exchange device on the access side or by the BTS on the outgoing side.
- Routes must be configured on the router for the BTS and BSC. The router on the BTS side must be configured with DHCP relay.

VLAN Configuration for Layer 3 Networking

When the old transmission configuration model is used, the single VLAN mode is recommended on the base station side. All data to the same next hop is tagged with the same VLAN. VLAN tagging based on the next-hop IP address is recommended for the BSC side.

When the new transmission configuration model is used, the interface VLAN mode is recommended for the base station side so that data on the same interface is tagged with the same VLAN.

Different VLAN priorities can be assigned to different service types to implement differentiated services. If the network is congested, high-priority packets are scheduled, and low-priority packets are buffered or discarded.

For details about VLAN configurations, see **4.3.5 VLAN**.

IP Address Configuration for Layer 3 Networking

It is recommended that the device IP address and port (or logical) IP address be used as the communicating IP addresses for the BSC and GBTS, respectively.

For the eGBTS, the device IP address configured on the BSC is used as the control- and user-plane IP address for the Abis interface. The eGBTS can use this or another IP address as the OM IP address.

NOTE

A Layer-2-enabled router supports the configuration of VLANs. One VLAN has several Ethernet ports and can be configured with a Layer 3 IP address (VLAN interface IP address) for the interworking between VLANs.

IP Route Configuration for Layer 3 Networking

A route to the base station must be configured on the base station controller.

The GBTS must be configured with a route to the BSC. The eGBTS must be configured with a route to the MAE, and the MAE must be configured with a route to the eGBTS OM IP address.

Routes on the bearer network need to be configured based on the actual networking.

5.2.1.2 IP over E1/T1 Networking

End-to-End IP over E1/T1 Networking

Figure 5-4 shows the end-to-end networking for IP over E1/T1 between a BTS and a BSC that are directly connected. (The BTS and BSC are regarded as directly connected even if they are connected through an SDH/PDH network.) The BSC provides E1/T1, Ch-STM-1, and Ethernet ports, and the BTS provides E1/T1 ports.

In end-to-end IP over E1/T1 networking:

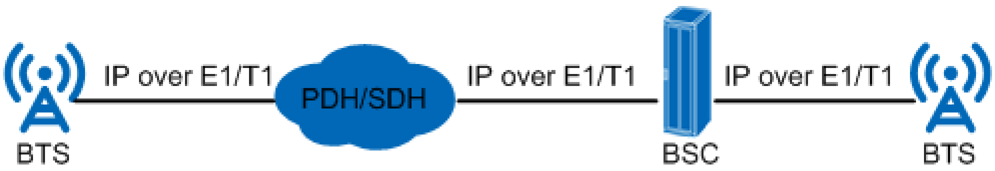
- The BSC is directly connected to the BTS using E1 links based on the PPP/MLPPP.

- The BSC and BTS require at least one pair of E1 cables. Use more E1 cables based on site requirements to increase bandwidth.
- A PPP/MLPPP link can be carried on one E1 link or multiple timeslots of an E1 link.
- When only one E1 link is used between the BSC and the BTS, the E1 link can be set as a PPP link or an MLPPP link group. The MLPPP link group is recommended to facilitate capacity expansion or cascaded BTS addition.
- When multiple E1 links are used between the BSC and the BTS, the E1 links must be set as an MLPPP link group to carry all the services.

NOTE

There are cases where the BSC uses the Abis interface board as the optical interface board. An O/E conversion device is used between the BTS and BSC to convert E1 optical signals to E1 electrical signals and then transmit the signals to the BTS.

Figure 5-4 Networking for IP over E1/T1 (direct connection between GBTS and BSC)



When you configure the IP addresses:

- It is recommended that the PPP/MLPPP IP address be used as the communicating IP address for the BSC and GBTS.
- For the eGBTS, it is recommended that the PPP/MLPPP IP address be configured as the control- and user-plane IP address for the Abis interface on the BSC. On the eGBTS side, the PPP/MLPPP IP address can be configured as the control- and user-plane and OM IP address, or only as the control- and user-plane IP address with another IP address being configured as the OM IP address.

With the preceding IP address configurations, no routes need to be configured on the GBTS, but the following routes must be configured: route from the eGBTS to the MAE, routes from the BSC to the MAE and eGBTS (because the PPP/MLPPP link terminates at the BSC), and route from the MAE to the eGBTS OM IP address.

Routes on the bearer network need to be configured based on the actual networking.

Non-End-to-End IP over E1/T1 Networking

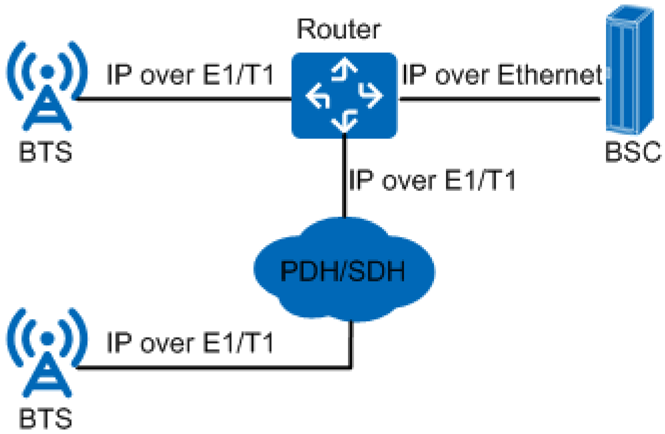
Figure 5-5 shows the networking for IP over E1/T1 between a BTS and a BSC that are connected through a router.

The BTS uses IP over E1/T1 transmission, while the BSC uses IP over Ethernet transmission. The router converts IP over E1/T1 to IP over Ethernet for data transmission. The BSC provides Ethernet ports, and the BTS provides E1/T1 ports.

In non-end-to-end IP over E1/T1 networking:

- The router must be directly connected to the BTS using the E1 cable. Therefore, the router must support an E1 port and the creation of an MLPPP link group.
- The router can be connected to the BSC through an Ethernet port.
- Routes and DHCP relay need to be configured on the router for the BTS and BSC.

Figure 5-5 Networking for IP over E1/T1 (connection between GBTS and BSC through a router)



In Abis over IP mode, it is recommended that the BTS connect to the BSC through Ethernet ports. If the BTS does not provide Ethernet ports, it is recommended that the BTS be connected to the router through E1/T1 ports and then the router be connected to the BSC through Ethernet ports.

You can configure IP addresses for the GBTS and eGBTS as follows:

- It is recommended that the device IP address and PPP/MLPPP IP address be used as the communication IP addresses for the BSC and GBTS, respectively.
- For the eGBTS, the device IP address configured on the BSC is used as the control- and user-plane IP address for the Abis interface. The eGBTS can use this or another IP address as the management-plane IP address.

The following routes must be configured:

- Routes from the BSC to the GBTS and eGBTS
- Route from the GBTS to the BSC
- Route from the eGBTS to the MAE IP address
- Route from the MAE to the eGBTS OM IP address

Routes on the bearer network need to be configured based on the actual networking.

IP over E1/T1 Networking for Cascaded BTSs

Figure 5-6 shows the networking for cascaded BTSs.

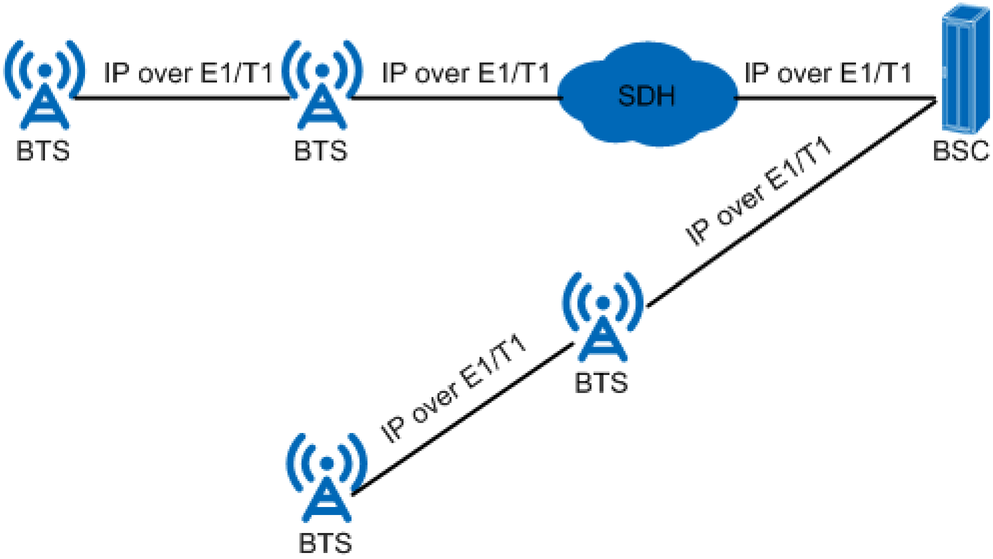
- If cascaded BTSs are connected using one E1 link, the E1 link can be set as a PPP link or an MLPPP link group.

- If cascaded BTSs are connected using multiple E1 links, these E1 links must be set as an MLPPP link group.
- The number of lower-layer BTSs connecting to an upper-layer BTS is determined by the PPP link quantity and MLPPP link group quantity on the upper-layer BTS. The GBTS supports a maximum of four PPP links (including MLPPP links) or two MLPPP link groups.

NOTE

The BTS does not support the IP over E1/T1+TDM hybrid networking mode. IP over E1/T1 does not support the ring topology.

Figure 5-6 Networking for IP over E1/T1



PPP links terminate at each BTS, and a BTS performs route forwarding for its cascaded BTS.

Currently, PPP and TDM hybrid networking for cascaded BTSs is not supported.

IP address configurations in this networking are the same as those in IP over E1/T1 end-to-end or non-end-to-end networking. The PPP/MLPPP IP addresses the BTS uses for communicating with the BSC and its cascaded BTS must be on different network segments.

The following routes must be configured:

- Routes from the BSC to the cascaded BTSs
- Route from the GBTS to the BSC
- Route from the eGBTS to the MAE
- Routes from the hub base station to all its leaf base stations

Routes on the bearer network need to be configured based on the actual networking.

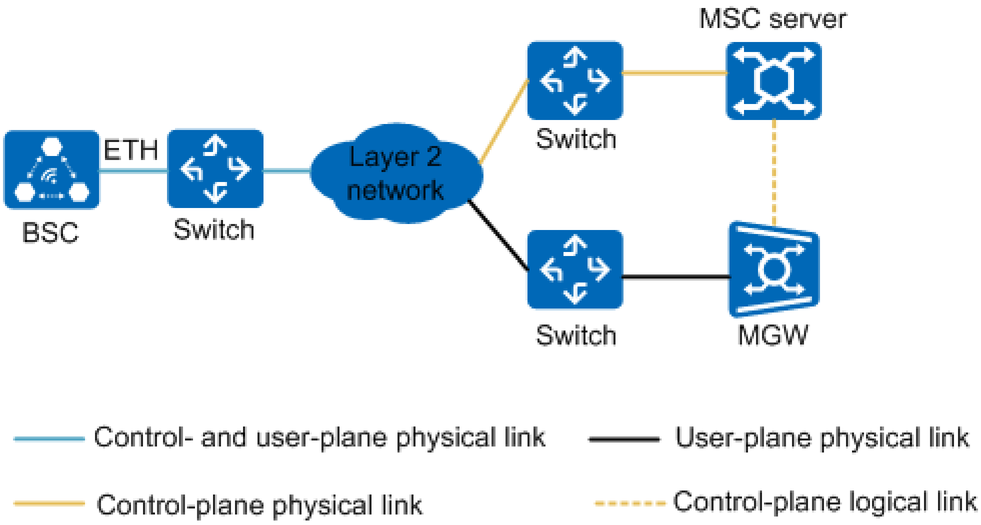
5.2.2 A Interface Networking

A over IP supports layer 2 networking, layer 3 networking, and direct networking. The BSC6960 does not support layer 2 networking.

Layer 2 Networking

Layer 2 networking is applicable only to Ethernet transmission. **Figure 5-7** shows the layer 2 networking through switches.

Figure 5-7 Layer 2 networking at the IP layer of the A interface



In layer 2 networking, it is recommended that the Ethernet ports work in active/standby mode to ensure network reliability. The BSC6910 does not support Ethernet ports working in active/standby mode.

To configure the Ethernet ports to work in active/standby mode, run the **ADD ETHREDPORT** command to specify ***SRN***, ***CIUSLOTNO***, and ***E1PORTNOINBTSCONBSC*** for both the active and standby boards.

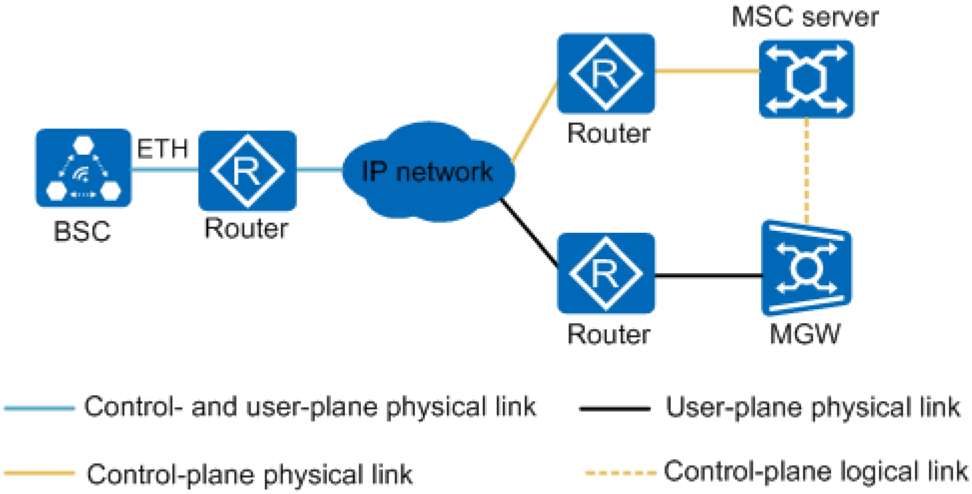
Layer 2 networking has the following advantages:

- Provides a large-capacity bandwidth and a reliable transmission bearer.
- Meets the requirements for the bearer network in future GSM evolution.
- Reduces the impact of data burst on the network.
- Supports flexible networking on a large scale.

Layer 3 Networking

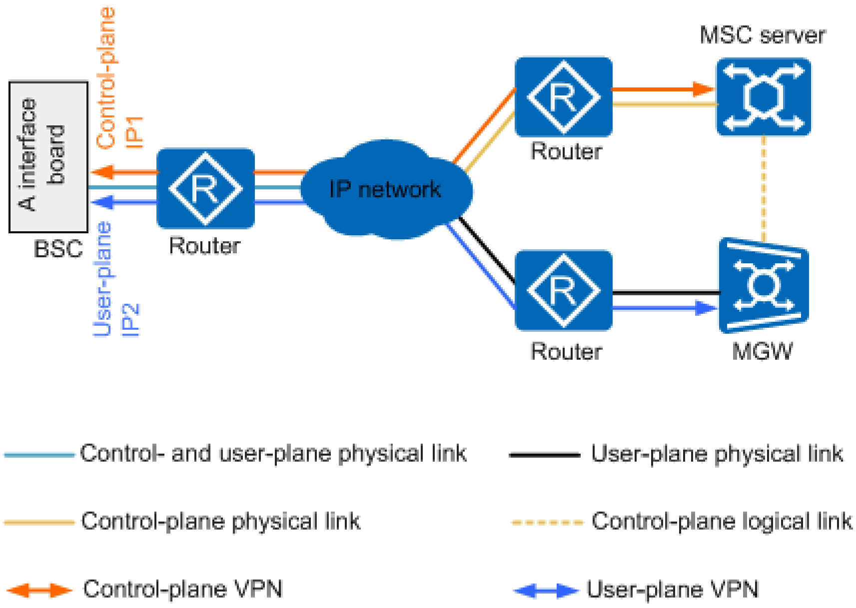
Layer 3 networking is applicable only to Ethernet transmission. **Figure 5-8** shows layer 3 networking through routers.

Figure 5-8 Layer 3 networking at the IP layer of the A interface



For information about reliability in layer 3 networking, see *BSC6900 GSM Technical Description*, *BSC6960 GSM Technical Description*, and *BSC6910 GSM Technical Description*.

Figure 5-9 Layer 3 (IP layer) networking with multiple IP addresses on one port of the A interface board



The layer 3 networking has the following advantages:

- Provides a large-capacity bandwidth and a reliable transmission bearer.
- Meets the requirements for the bearer network in future GSM evolution.
- Reduces the impact of data burst on the network.
- Supports flexible networking on a large scale.

Direct Networking

There are two direct networking scenarios: IP over Ethernet and IP over E1/T1/Ch-STM-1. In IP over E1/T1/Ch-STM-1 mode, only the direct networking is considered. The IP over E1/T1/Ch-STM-1 networking through routers is not recommended.

The direct networking has the following advantages:

- High reliability, low network construction cost, and easy maintenance
- High QoS and easy-to-control call admission

For details about direct networking, see *BSC6900 GSM Technical Description* and *BSC6910 GSM Technical Description*.

5.2.3 Gb Interface Networking

After Gb over IP is applied, the signals between the BSC and the SGSN are transmitted over the IP network. Routers are used to provide layer 3 routing services for the BSC and SGSN. As a result of IP application, the bandwidth on the Gb interface increases and the OM cost decreases.

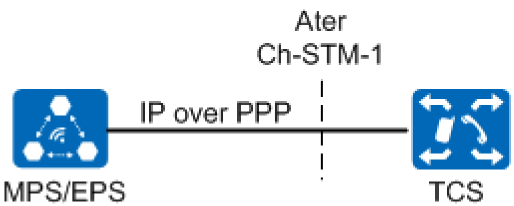
Figure 5-10 shows the networking for Gb over IP.

Figure 5-10 Networking for Gb over IP

5.2.4 Ater Interface Networking

Figure 5-11 shows the networking for Ater over IP.

Figure 5-11 Networking for Ater over IP



The voice services between the MPS/EPS and the terminal customer service system (TCS) are transmitted through IP over PPP. The transport network is still the SDH/PDH network in TDM mode. The control-plane and OM links between the MPS/EPS and TCS still use the original TDM transmission mode. The BSC6910 does not provide an Ater interface.

5.3 UMTS Interface Networking

5.3.1 Iub Interface Networking

The Iub interface supports star and chain topologies. From the perspective of transport bearers, the Iub interface supports ATM transport networking, IP transmission networking (including hybrid IP transmission), and hybrid ATM/IP transmission networking.

The following mainly describes Iub over IP networking. For information about Iub over ATM&IP networking and hybrid IP transmission, see *ATM&IP Dual Stack Feature Parameter Description* and *Hybrid Iub IP Transmission Feature Parameter Description*, respectively.

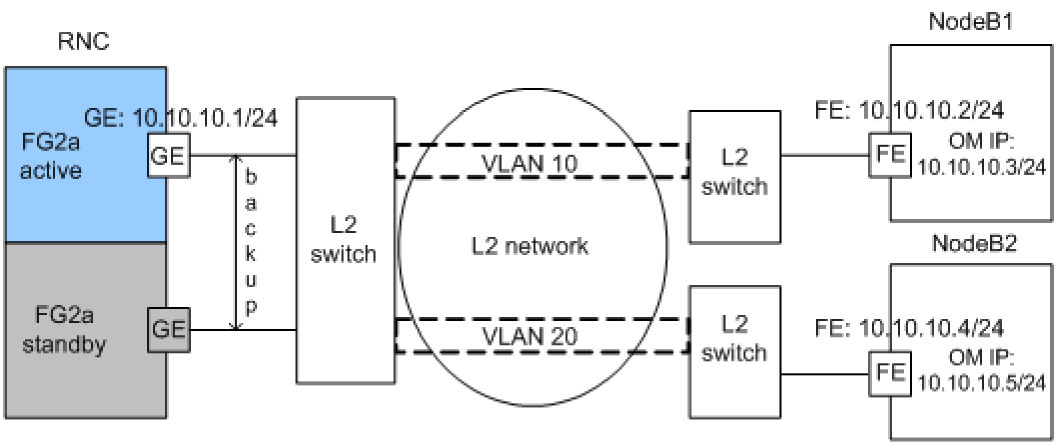
Based on the data switching technology that is used, IP networking can be classified into Layer 2 networking and Layer 3 networking. In addition, the Iub interface can use hybrid IP transmission networking to improve transport reliability for the Iub interface.

Layer 2 Networking

Layer 2 networking is relatively simple because all the network devices are located on the same network segment. In this case, data is switched by using Layer 2 devices (for example, LAN switches), and IP routes do not need to be configured.

Figure 5-12 shows an example of Layer 2 networking in the case of Iub over IP.

Figure 5-12 Example of Layer 2 networking in the case of Iub over IP



As shown in **Figure 5-12**, the NodeBs and the RNC are connected through a Layer 2 network. To improve transport reliability for the Iub interface, the RNC uses port backup or trunk groups working in active/standby or load sharing mode.

NOTE

The BSC6910 does not support port backup. It supports only trunk groups working in active/standby or load sharing mode.

Layer 2 networking is not recommended for the BSC6960.

The Layer 2 network distinguishes data from different NodeBs according to the virtual local area network (VLAN) ID. In addition, the Layer 2 network provides differentiated services according to the PRI field in the VLAN tag. If the network is congested, high-priority packets are scheduled, and low-priority packets are buffered or discarded. The RNC supports VLAN tagging based on the next hop IP address or data streams.

NOTE

The BSC6910/BSC6960 does not support VLAN tagging based on data streams.

IP Address Configuration for Layer 2 Networking (in Non-Transmission Resource Pool Mode)

To facilitate data configuration:

- On the RNC side, a port IP address is configured as the common IP address of the control plane and user plane.
- On the NodeB side, a port IP address is configured as the IP address of both the control and user planes, and another OM IP address is configured.

  3900 and 5900 series base stations support a single IP address for a NodeB. In other words, the control plane, user plane, and OMCH of a NodeB can share one IP address.

IP Address Configuration for Layer 2 Networking (in Transmission Resource Pool Mode)

The IP addresses of different ports on the RNC side are on different network segments, and one IP address pool contains only one port IP address. The NodeB whose IP address is on the same network segment as an RNC port IP address must be connected to this RNC by using the IP address pool including the RNC port IP address.

On the RNC side (BSC6900/BSC6910):

- The **ADD IPPOOL** command is used to add an IP address pool. The **ADD IPPOOLIP** command is used to add a port IP address to an IP address pool. The port IP address can be an Ethernet port IP address or a trunk group IP address. The former is added using the **ADD ETHIP** command. The latter is added using the **ADD ETHTRKIP** command.
- The **ADD SCTPLNK** command is used to add a control plane IP address. This IP address can be an IP address in an IP address pool.

The IP address configuration principles on the NodeB side are the same between the transmission resource pool mode and non-transmission-resource-pool mode. For details, see *Transmission Resource Pool in RNC Feature Parameter Description*.

IP Route Configuration for Layer 2 Networking

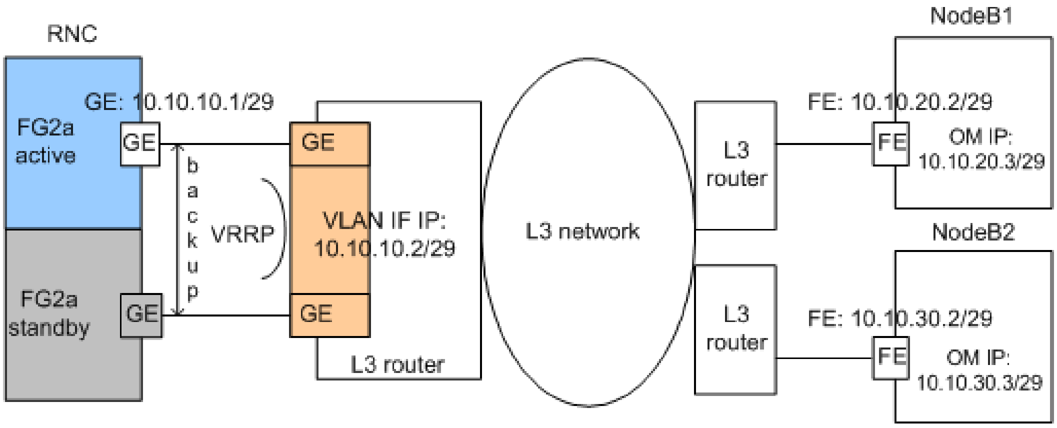
Routes are not required for Layer 2 networking.

Layer 3 Networking

In Layer 3 networking, the network devices are located on different network segments. In this case, data is switched by using Layer 3 devices (for example, routers), and IP routes need to be configured.

Figure 5-13 shows an example of Layer 3 networking in the case of Iub over IP.

Figure 5-13 Example of Layer 3 networking in the case of Iub over IP



As shown in **Figure 5-13**, the RNC and the NodeBs are connected through a Layer 3 network. To improve transport reliability for the Iub interface, the RNC uses active and standby ports to connect to the peer ports on the Layer 3 router. In addition, the Layer 3 network provides differentiated services according to the DSCP value in the IP header. If the network is congested, high-priority packets are scheduled, and low-priority packets are buffered or discarded.

NOTE

The RNC can also use the trunk groups working in active/standby or load sharing mode to improve Iub transport reliability. The BSC6910 does not support port backup. It supports only trunk groups working in active/standby or load sharing mode.

The BSC6960 supports only the manual active/standby trunk networking.

IP Address Configuration for Layer 3 Networking

The two ports on the router are configured in the same VLAN and share one VLAN port IP address (VLAN IF IP shown in **Figure 5-13**).

The active and standby ports on the RNC side share one IP address, which is on the same network segment as the VLAN IF IP address of the router. The port IP address on the RNC side functions as the common IP address of the control plane, user plane, and OMCH for the Iub interface.

On the NodeB side, a port IP address is configured as the IP address of both the control and user planes, and another IP address is configured for the OMCH. The port IP address and the OM IP address can be located either on the same network segment or on different network segments. If they are located on the same network segment, the ARP proxy function of the FE port (in the old model) or of the interface (in the new model) must be enabled. That is, the ***ARPPROXY*** parameter on the NodeB side must be set to **ENABLE**. The BTS3900/BTS5900/BTS3900A/BTS5900A/DBS3900/DBS5900 supports a single IP address for a NodeB. In other words, the control plane, user plane, and OMCH of a NodeB can share one IP address.

NOTE

A Layer-2-enabled router supports the configuration of VLANs. One VLAN has several Ethernet ports and can be configured with a Layer 3 IP address (VLAN interface IP address) for the interworking between VLANs.

IP Route Configuration for Layer 3 Networking

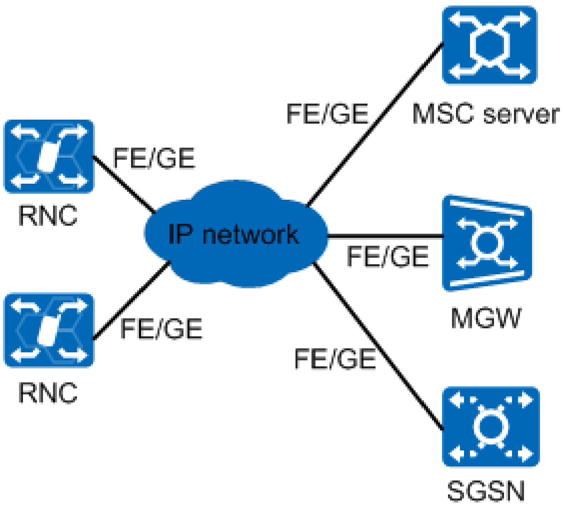
On the RNC side, routes to the NodeBs need to be configured. The routes can be configured to access either each single NodeB or the network segment where several NodeBs are located. For example, as shown in **Figure 5-13**, only the route from the RNC to the network segment 10.10.0.0/16 is configured. Through this route, the RNC communicates with the two NodeBs.

Routes to the RNC should be configured for each NodeB. Routes on the bearer network need to be configured based on the actual networking.

5.3.2 Iu/Iur Interface Networking

Figure 5-14 shows an example of Iu/Iur over IP networking.

Figure 5-14 Example of Iu/Iur over IP networking



The IP network shown in **Figure 5-14** can be either of the following networks:

- Layer 2 network, for example, metro Ethernet, VPLS network, and MSTP network
- Layer 3 network, for example, IP network and multiprotocol label switching virtual private network (MPLS VPN)

The Iu or Iur interface can also use the transmission resource pool networking. For details, see *Transmission Resource Pool in RNC Feature Parameter Description*.

5.3.3 uX2 Interface Networking

On a live network, the NodeB generally uses a transport network to connect to adjacent NodeBs. Layer 2 and Layer 3 networks are commonly used in UMTS IP networking.

Layer 2 Networking

A Layer 2 network, also called an Ethernet switching network, mainly consists of Ethernet switches and is used for uX2 transmission. The NodeB accesses this network through the FE/GE/10GE/25GE/50GE port.

Figure 5-15 Layer 2 network

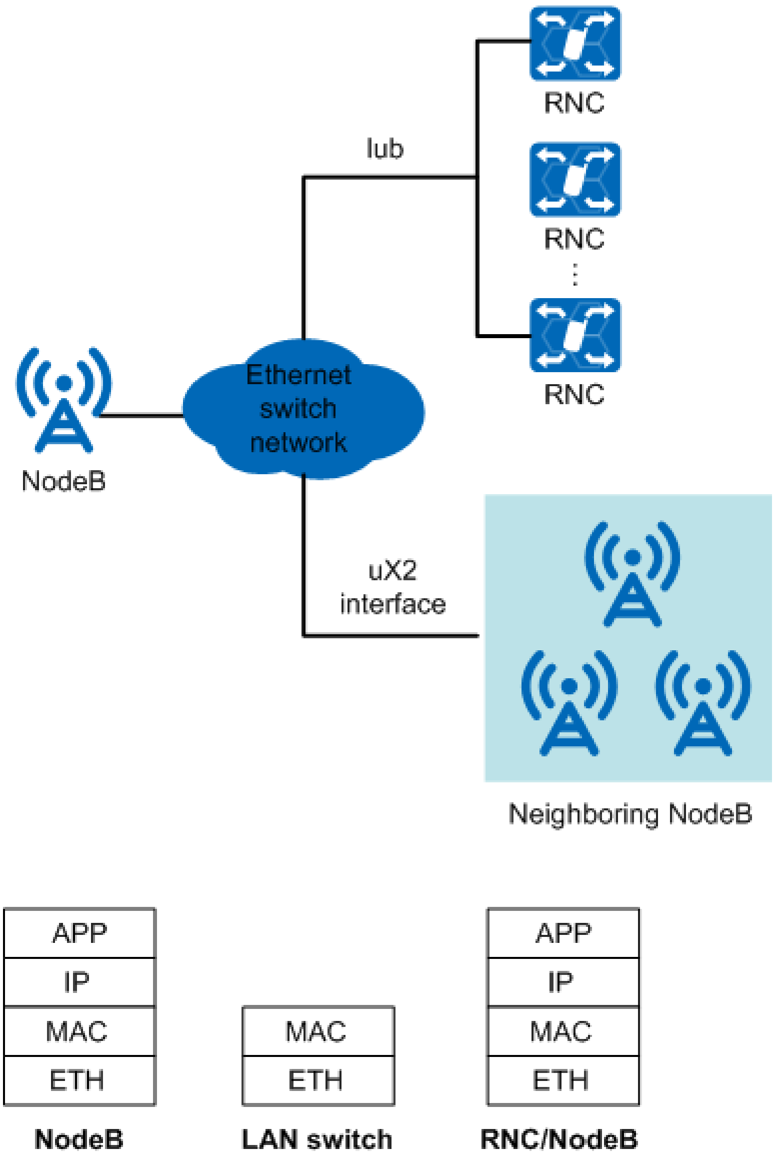


Figure 5-15 shows a Layer 2 network that provides the bearer function at the MAC layer. A NodeB does not provide the Layer 2 Ethernet switching function, and supports only packet forwarding based on IP addresses. After the NodeB encapsulates packets into Ethernet frames, the Layer 2 switch forwards packets based on the source and target MAC addresses.

Users must configure the Ethernet port, MAC layer, and IP layer, as described in **Table 5-1**.

Table 5-1 Layer 2 network configuration items

Configuration Item	Configuration Description	Interface
Ethernet port	Negotiation parameters, such as the duplex mode and rate, must be set. Automatic negotiation is used by default.	FE/GE/10GE/25GE/50GE
MAC layer	The functions defined in the following protocols must be configured: ● IEEE 802.1p/q ● IEEE 802.3ad (related to Ethernet port trunks) ● IEEE 802.3ah (related to Ethernet OAM) ● IEEE 802.1ag (related to Ethernet OAM) ● Y.1731	N/A
IP layer	IP addresses must be configured.	IP address

When the old transmission configuration model is used, a NodeB can use the following VLAN tagging methods at the MAC layer:

- VLAN group: VLANs and VLAN priorities are configured based on next-hop IP addresses, subnet masks, and packet types.
- Single VLAN: VLANs are configured based on next-hop IP addresses and subnet masks, and VLAN priorities are configured based on the mapping defined by the **DSCPMAP** MO.

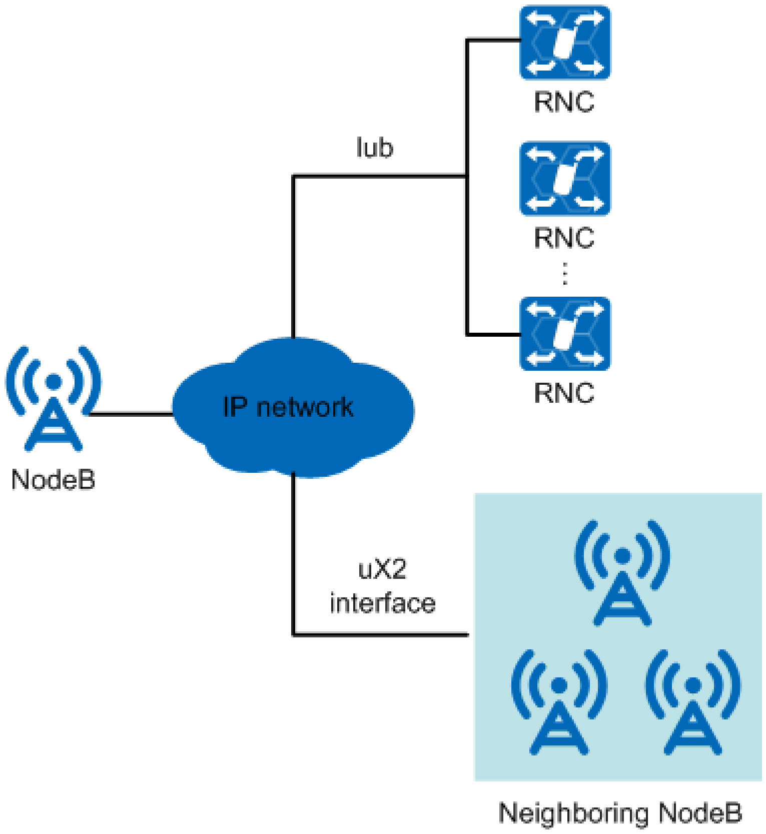
When the new transmission configuration model is used, a NodeB can use the following VLAN tagging methods at the MAC layer:

- VLAN group: VLANs and VLAN priorities are configured based on next-hop IP addresses, subnet masks, and packet types.
- Single VLAN: VLANs are configured based on next-hop IP addresses and subnet masks, and VLAN priorities are configured based on the mapping defined by the **DSCP2PCPMAP** MO.
- Interface VLAN: VLANs are configured based on local interfaces, and VLAN priorities are configured based on the mapping defined by the **DSCP2PCPMAP** MO.

Layer 3 Networking

A Layer 3 network is an IP routing and switching network and mainly consists of routers. The NodeB accesses this network through the FE/GE/10GE/25GE/50GE port.

Figure 5-16 Layer 3 network



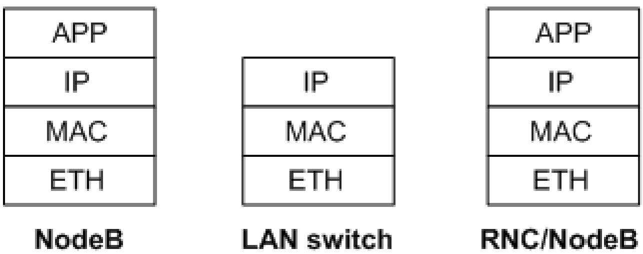


Figure 5-16 shows a Layer 3 network that provides the bearer function at the IP layer of the uX2 interface. Users must configure the physical layer, data link layer, and IP layer. **Table 5-2** lists the configuration of a Layer 3 network over the FE/GE/10GE/25GE/50GE port.

Table 5-2 Configuration of a Layer 3 network over the FE/GE/10GE/25GE/50GE port

Configuration Item	Configuration Description	Interface
Physical layer	N/A	FE/GE/10GE/25GE/50GE
MAC layer	The configuration is the same as that on a Layer 2 network. For details, see **Table 5-1**.	N/A

Configuration Item	Configuration Description	Interface
IP layer	IP addresses, IP routing tables, and DiffServ values must be configured.	IP address

5.4 LTE Interface Networking

On a live network, the eNodeB generally uses a transport (backhaul) network to connect to adjacent NEs, such as the S-GW, MME, eNodeB, IP clock server, and MAE. Layer 2 and Layer 3 networks are commonly used in LTE IP networking.

eNodeB cascading is supported as follows:

- For non-cascaded eNodeBs (without passing-by data flows):

  Set **Forward Mode** to **HOST** and **Same Port Forward Switch** to **DISABLE** in the **GTRANSPARA** MO. Disabling inter-port or intra-port data routing and forwarding prevents loopback-triggered network storms when default routes are configured.
- For cascaded eNodeBs (with passing-by data flows):

  Set **Forward Mode** to **ROUTE** in the **GTRANSPARA** MO.

  When an eNodeB is cascaded to a lower-level eNodeB through a downstream port, intra-port data routing and forwarding is not required. In this case, set **Same Port Forward Switch** to **DISABLE** in the **GTRANSPARA** MO.

  When an eNodeB is cascaded to multiple lower-level eNodeBs through the transport network, and the downstream port of the eNodeB is required to forward data to lower-level eNodeBs, set **Same Port Forward Switch** to **ENABLE** in the **GTRANSPARA** MO.

Layer 2 Networking

A Layer 2 network, also called an Ethernet switch network, mainly consists of Ethernet switches and is commonly used for LTE transmission. The eNodeB accesses this network through the FE/GE/10GE/25GE/50GE port.

Figure 5-17 Layer 2 network

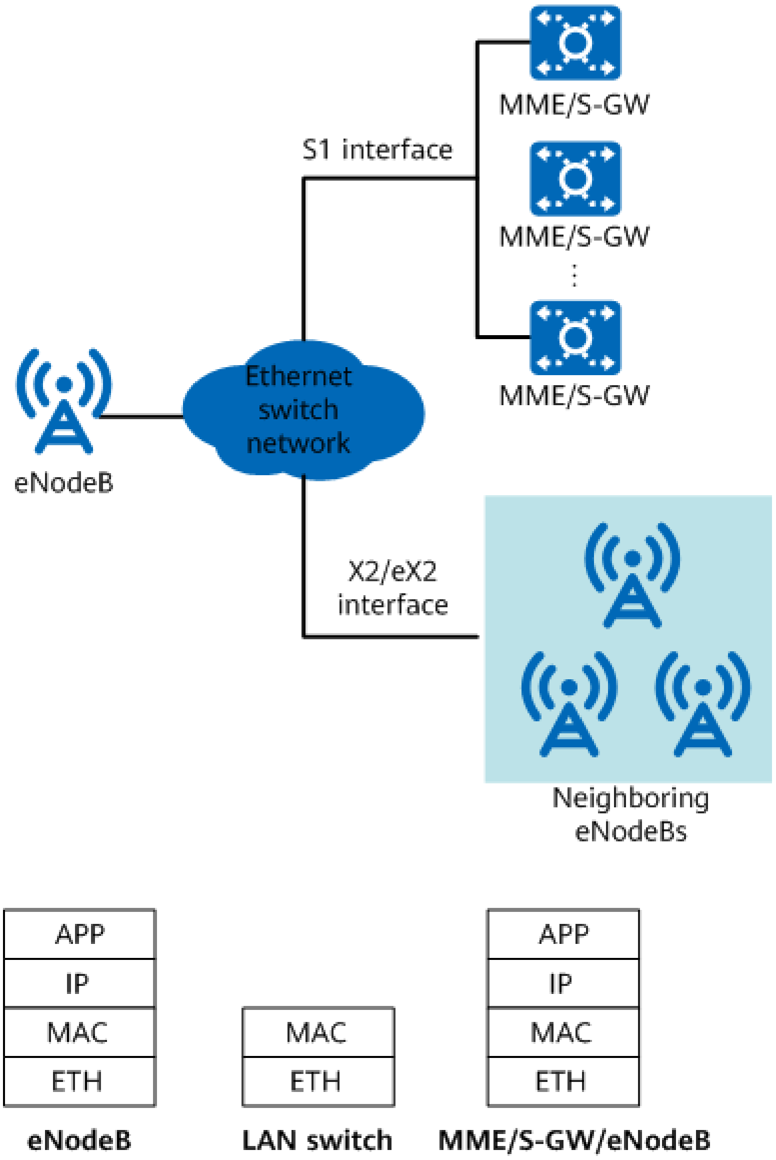


Figure 5-17 shows a Layer 2 network that provides the bearer function at the MAC layer. An eNodeB does not provide the Layer 2 Ethernet switching function, and supports only packet forwarding based on IP addresses. After the eNodeB encapsulates packets into Ethernet frames, the Layer 2 switch forwards packets based on the source and target MAC addresses.

Users must configure the Ethernet port, MAC layer, and IP layer, as described in **Table 5-3**.

Table 5-3 Layer 2 network configuration items

Configuration Item	Configuration Description	Interface
Ethernet port	Ethernet negotiation parameters, such as the duplex mode and rate, must be set. Automatic negotiation is used by default.	FE/GE/10GE/25GE/50GE
MAC layer	The configured functions include IEEE 802.1p/q, IEEE 802.3ad (Ethernet port trunk), IEEE 802.3ah (Ethernet OAM), IEEE 802.1ag (Ethernet OAM), and Y.1731.	N/A
IP layer	IP addresses must be configured.	IP address

When the old transmission configuration model is used, an eNodeB can use the following VLAN tagging methods at the MAC layer:

- VLAN group: VLANs and VLAN priorities are configured based on next-hop IP addresses, subnet masks, and packet types.
- Single VLAN: VLANs are configured based on next-hop IP addresses and subnet masks, and VLAN priorities are configured based on the mapping defined by the **DSCPMAP** MO.

When the new transmission configuration model is used, an eNodeB can use the following VLAN tagging methods at the MAC layer:

- VLAN group: VLANs and VLAN priorities are configured based on next-hop IP addresses, subnet masks, and packet types.
- Single VLAN: VLANs are configured based on next-hop IP addresses and subnet masks, and VLAN priorities are configured based on the mapping defined by the **DSCP2PCPMAP** MO.
- Interface VLAN: VLANs are configured based on local interfaces, and VLAN priorities are configured based on the mapping defined by the **DSCP2PCPMAP** MO.

Layer 3 Networking

For LTE, a Layer 3 network is an IP routing and switching network and mainly consists of routers. The eNodeB accesses this network through the FE/GE/10GE/25GE/50GE port.

Figure 5-18 Layer 3 network

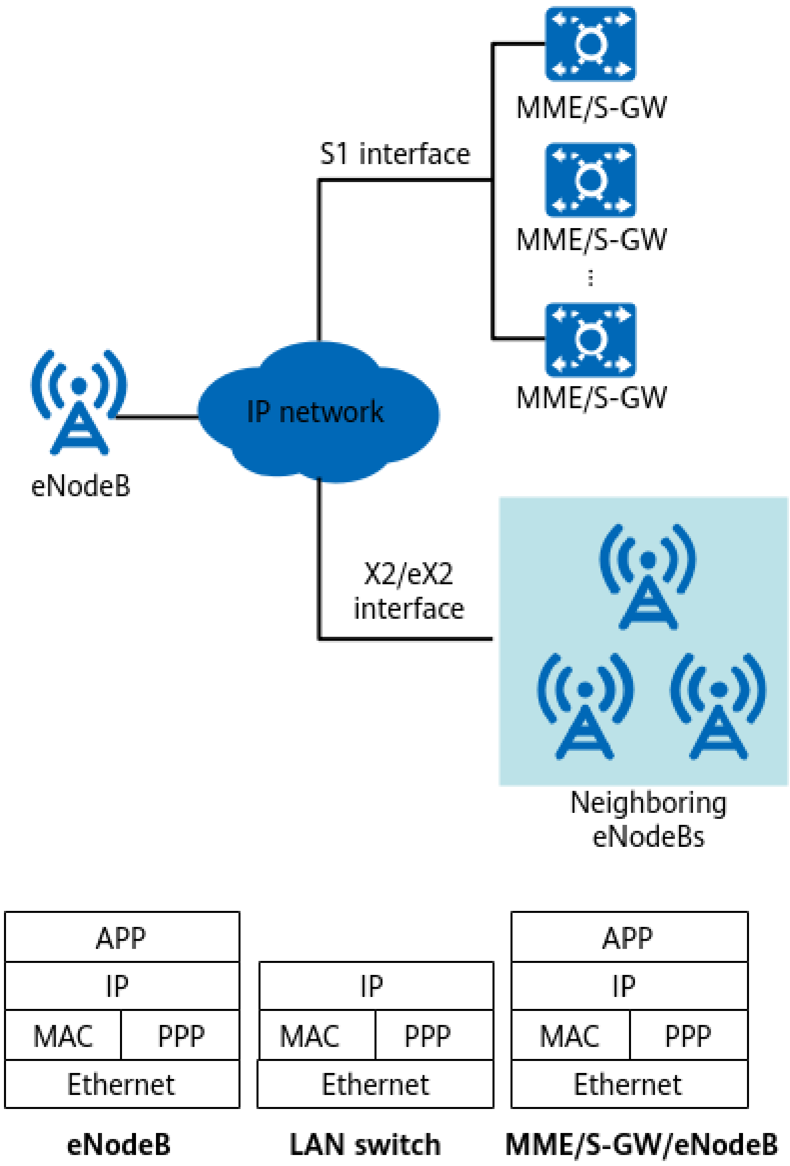


Figure 5-18 shows a Layer 3 network that provides the bearer function at the IP layer, which is commonly applied to LTE. Users must configure the physical layer, data link layer, and IP layer. **Table 5-4** lists the configuration of a Layer 3 network over the FE/GE/10GE/25GE/50GE port.

Table 5-4 Configuration of a Layer 3 network over the FE/GE/10GE/25GE/50GE port

Configuration Item	Configuration Description	Interface
Physical layer	N/A	FE/GE/10GE/25GE/50GE
MAC layer	The configuration is the same as that on a Layer 2 network. For details, see **Table 5-3**.	N/A

Configuration Item	Configuration Description	Interface
IP layer	IP addresses, IP routing tables, and DiffServ values must be configured.	IP address

5.5 NR Interface Networking

On a live network, the gNodeB generally uses a transport network to connect to adjacent NEs, such as the S-GW, eNodeB, IP clock server, and MAE. In IP-based networking, a transport network is classified into Layer 2 and Layer 3 networks.

Base station cascading is supported. The configuration details are as follows:

- For non-cascaded base stations (without passing-by data flows):

  Set **GTRANSPARA**.***FORWARDMODE*** to **HOST** and **GTRANSPARA**.***SAMEPORTFORWARDSW*** to **DISABLE**. Inter-port (on the same panel) and intra-port data routing and forwarding are disallowed to prevent loopback-triggered network storms when default routes are configured.
- For cascaded base stations (with passing-by data flows):

  Set **GTRANSPARA**.***FORWARDMODE*** to **ROUTE**. When a base station is cascaded to one lower-level base station through a downstream port, intra-port data routing and forwarding is not required. In this case, set **GTRANSPARA**.***SAMEPORTFORWARDSW*** to **DISABLE**.

  When a base station is cascaded to multiple lower-level base stations through the transport network, the downstream port of the base station may be required to forward data to lower-level base stations. In this case, set **GTRANSPARA**.***SAMEPORTFORWARDSW*** to **ENABLE**.

Layer 2 Networking

A Layer 2 network is an Ethernet switching network and mainly consists of Ethernet switches. The gNodeB accesses this network through the FE/GE/10GE/25GE/50GE port.

Figure 5-19 Layer 2 network

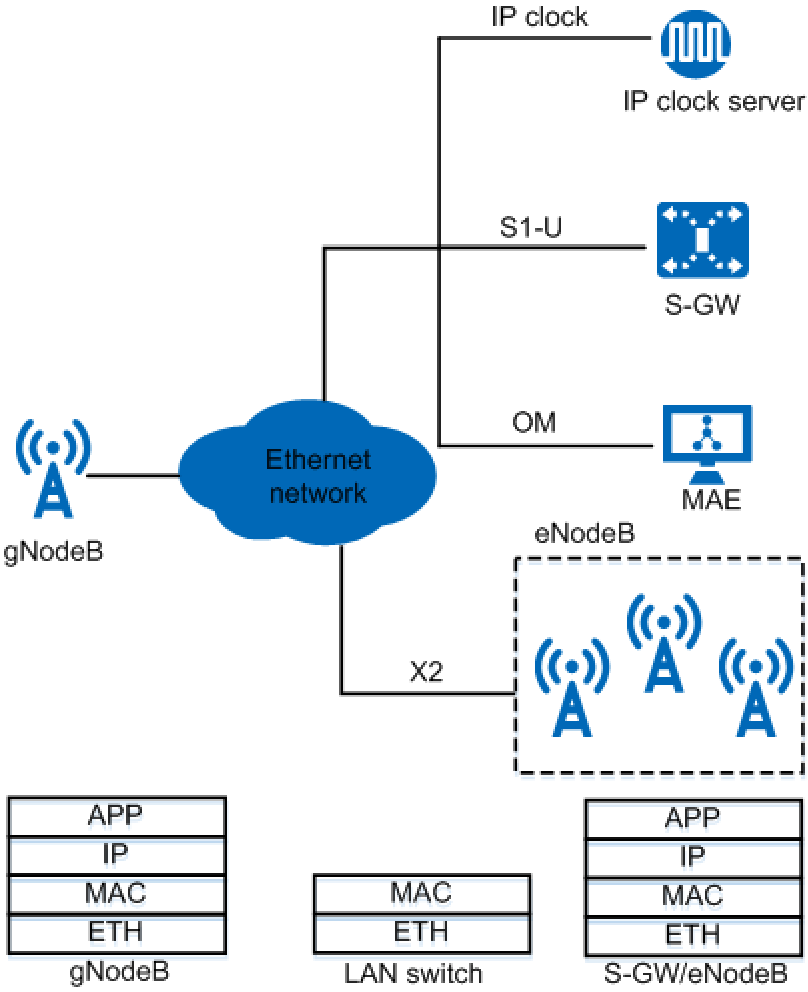


Figure 5-19 shows a Layer 2 network that provides the bearer function at the MAC layer. A gNodeB does not provide the Layer 2 Ethernet switching function, and supports only packet forwarding based on IP addresses. After the gNodeB encapsulates packets into Ethernet frames, the Layer 2 switch forwards Ethernet frames based on the source and destination MAC addresses.

Users must configure the Ethernet port, data link layer, and IP layer, as described in **Table 5-5**.

Table 5-5 Layer 2 network configuration items

Configuration Item	Configuration Description
Ethernet port	Negotiation parameters, such as the duplex mode and rate, must be set. Automatic negotiation is used by default.

Configuration Item	Configuration Description
Data link layer	The functions defined in the following protocols must be configured: ● IEEE 802.1p/q ● IEEE 802.3ad (related to Ethernet port trunks) ● IEEE 802.3ah (related to Ethernet OAM) ● IEEE 802.1ag (related to Ethernet OAM) ● Y.1731
IP layer	IP addresses must be configured.

Layer 3 Networking

A Layer 3 network is an IP routing and switching network and mainly consists of routers. The gNodeB accesses this network through the FE/GE/10GE/25GE/50GE port.

Figure 5-20 Layer 3 network

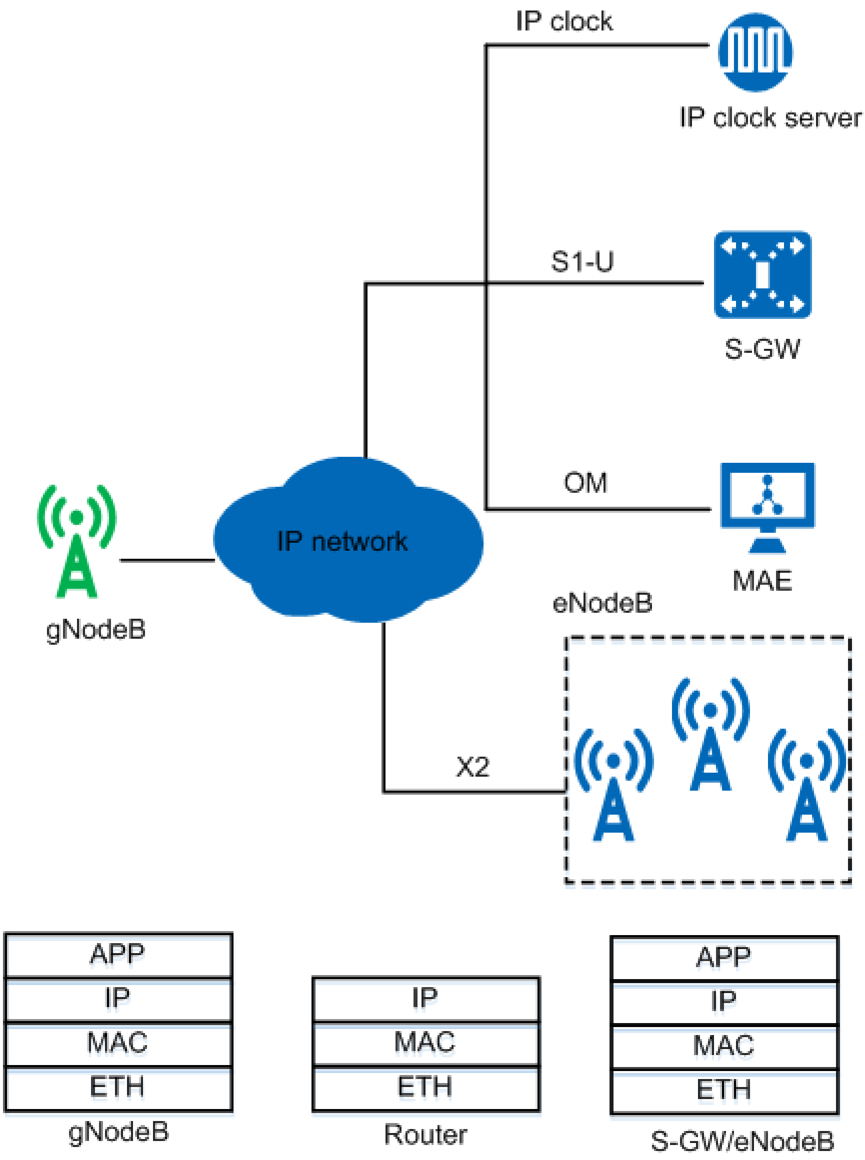


Figure 5-20 shows a Layer 3 network that provides the bearer function at the IP layer, which is commonly applied to NR. Users must configure the physical layer, data link layer, and IP layer. **Table 5-6** lists the configuration of a Layer 3 network over the FE/GE/10GE/25GE/50GE port.

Table 5-6 Layer 3 network configuration items

Configuration Item	Configuration Description
Physical layer	The configuration is the same as that on a Layer 2 network. For details, see **Table 5-5**.
Data link layer	The configuration is the same as that on a Layer 2 network. For details, see **Table 5-5**.

Configuration Item	Configuration Description
IP layer	IP addresses, IP routing tables, and DiffServ values must be configured.

5.6 Configuration Modes for Transmission Interfaces

On the Base Station Controller Side

IP transmission resource pools are configured on the base station controller side. For details about IP transmission resource pools, see *Transmission Resource Pool in RNC* in *RAN Feature Documentation*.

The MBSC supports the following two configuration modes:

- Link configuration mode, including IPPATH mode and SCTPLNK mode
- Endpoint configuration mode, including transmission resource pool mode and SCTPSRV mode. The SCTPSRV mode applies to the Abis interface (connecting to the eGBTS) or to the Iub interface.

On the Base Station Side

The base station supports the following two configuration modes:

- Link configuration mode: In this mode, configurations are based on a link from the source end to the destination end. When configuring a control- or user-plane port, users must specify the parameters related to the source and destination ends of the link in one MO. The parameters include source and destination IP addresses and source and destination port numbers.
- Endpoint configuration mode: This mode is introduced to simplify the configuration of a large number of links after a base station supports the pool networking. No link specified by the parameters of the source and destination ends needs to be configured. Only MOs of the local end are specified. The parameters for the peer end can be automatically obtained through signaling or manually configured.

NR supports only the endpoint configuration mode. For details about the basic principles of the endpoint mode, see *NG and Xn Self-Management* in *5G RAN Feature Documentation*.

It is recommended that the endpoint configuration mode be used for LTE. For details about the basic principles of the endpoint mode, see *S1 and X2 Self-Management* in *eRAN Feature Documentation*.

For the Iub interface of the NodeB, the link configuration mode is recommended for the control plane, and also for the user plane when no transmission resource pools are used. The endpoint configuration mode is recommended for the user plane of the Iub interface of the NodeB when transmission resource pools are used. For the uX2 interface, the endpoint configuration mode is used at both the control plane and the user plane. For details, see *uX2 Interface Management* in *RAN Feature Documentation*.

For GSM, the link configuration mode is used for the Abis interface. The endpoint configuration mode is not recommended.

NOTE

In endpoint configuration mode, each RAT must be configured with its own endpoint groups. If different RATs reference the same endpoint group (for example, the S1 and Iub interfaces reference the same endpoint group), ALM-26247 Configuration Failure will be reported with the cause value being **EPGROUP Conflict**.

6 Transmission Reliability

6.1 Overview

Transmission reliability can be ensured by redundancy mechanisms which prevent a board, port, or link failure from affecting services.

The transmission port and link redundancy mechanisms include Ethernet port backup, Ethernet route backup, Ethernet link aggregation, E1/T1/STM-1 board backup, and MLPPP ports. The service link redundancy mechanisms include control-plane SCTP multihoming, OM channel backup, and user-plane IP transmission resource pool.

The base station controller requires board backup. The following descriptions assume that board backup is applied on the base station controller.

NOTE

Principles and parameter configurations of transmission reliability for the BSC and RNC are the same. The following only describes transmission reliability for the BSC.

Transmission reliability can be ensured by functions listed in **Table 6-1** to prevent a board, port, or link failure from affecting services.

Table 6-1 Redundancy mechanisms for transmission reliability

Port/Link Backup	Protocol Layer	Base Station Controller	Base Station
Ethernet port/link backup	Physical layer	Port backup	N/A
	Data link layer	Link aggregation for the active and standby boards Link aggregation within one board	Ethernet link aggregation
	IP layer	Route backup	IP route backup

Port/Link Backup	Protocol Layer	Base Station Controller	Base Station
E1/T1/STM-1 port/ link backup	Physical layer	Port backup and multiplex section protection (MSP)	N/A
	Data link layer	(Intra-board) MLPPP port	(Intra-board) MLPPP port
Service link backup	Application layer	Control-plane SCTP multihoming	Control-plane SCTP multihoming
		User-plane IP transmission resource pool	N/A
		N/A	OM channel backup

NOTE

For details on user-plane IP transmission resource pools, see *Transmission Resource Pool in BSC Feature Parameter Description* in *GBSS Feature Documentation* and *Transmission Resource Pool in RNC Feature Parameter Description* in *RAN Feature Documentation*.

Active/standby routes rather than equal-cost routes are recommended for route backup at the IP layer. Besides, it is a good practice to enable BFD together with route backup to detect link failures. Otherwise, a route switchover cannot be triggered when the link becomes faulty.

6.2 E1/T1/STM-1 Board Backup for the Base Station Controller (GSM/UMTS)

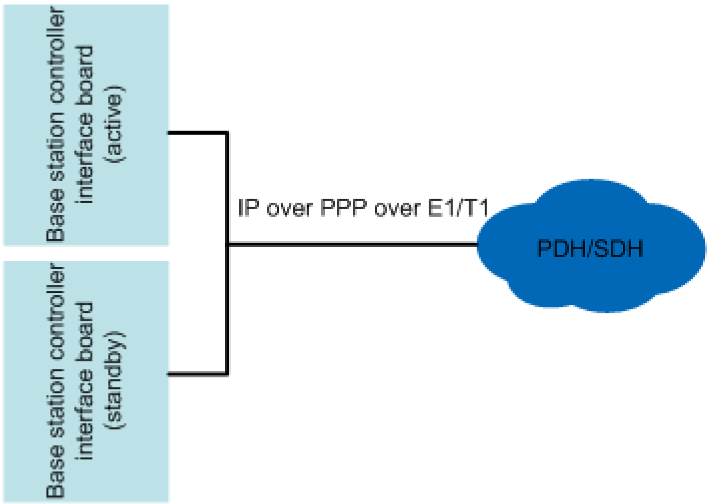
With board backup, if the active board is faulty or reset but the standby board is functional, an active/standby board switchover is triggered.

Board backup applies to all IP interface boards, including FE/GE/10GE Ethernet interface boards and IP over E1/T1/Ch-STM-1 (using PPP ports) interface boards. IP over E1/T1/Ch-STM-1 (using PPP ports) interface boards are recommended.

To implement board backup for the base station controller, set ***RED*** to **YES** when adding a board.

The base station controller must be configured with active and standby interface boards in IP over PPP over E1/T1 networking. The E1/T1 ports on the active and standby boards are connected using Y-shaped cables. Only the E1/T1 port on the active board transmits and receives IP packets. The E1/T1 port on the standby board is used only for link detection. The standby board does not process services, but saves the PPP link status, negotiation session information, and some stable user data of the active board. In this way, PPP renegotiation is not required during an active/standby board switchover. This prevents service interruption.

Figure 6-1 Networking for board backup in IP over PPP over E1/T1 mode



NOTE

When the base station controller uses the IP over PPP ports, PPP links terminate only at the base station, not at the intermediate transmission device.

When multiple E1/T1 links are configured between the base station controller and the base station, these E1/T1 links are generally bound together to form an MLPPP link to improve reliability and efficiency. In this scenario, the failure of a single E1/T1 link only reduces transmission bandwidth without interrupting ongoing services.

In IP over PPP over Ch-STM-1 mode, the base station controller interface boards can also work in active/standby mode. The STM-1 ports on the active and standby boards connect to the optical transmission equipment at the peer end through optical fibers. If the working channel becomes faulty, the base station controller negotiates with the optical transmission device about the protection channel according to the multiplex section protection (MSP) protocol. The base station controller then performs MSP by switching the services to the protection channel.

NOTE

For information about STM-1 optical port backup (with MSP), see *BSC6900 GU Technical Description* and *BSC6910 GU Technical Description*.

6.3 Ethernet Port Backup for the Base Station Controller (GSM/UMTS)

Based on the board backup, the Ethernet ports on the active and standby boards of the base station controller are configured to work in active/standby mode.

In Ethernet port backup mode, the active port transmits and receives data and processes services. The standby port does not process services. The standby port can be used for link detection as required.

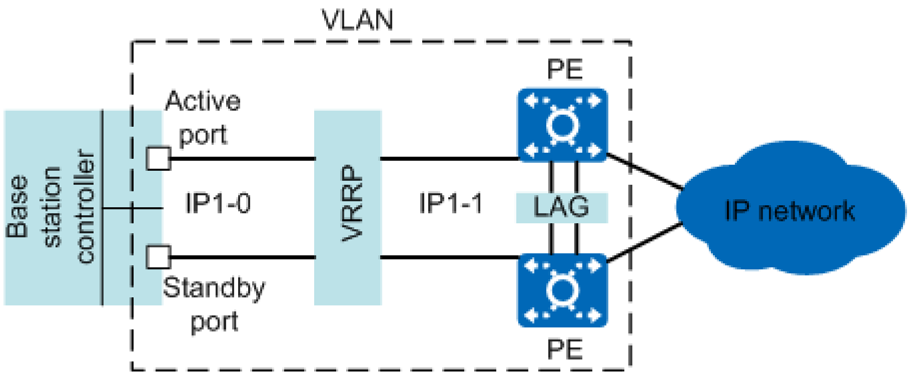
If the active and standby Ethernet ports do not depend on the active and standby boards, the working Ethernet ports can be located on either the active or standby board.

NOTE

The BSC6910 does not support Ethernet ports working in active/standby mode. Instead, it supports manual active/standby link aggregation that involves Ethernet ports.

The active and standby FE/GE ports on the active/standby board of the base station controller connect to two routers enabled with VRRP. The routers provide a virtual gateway IP address (IP1-1), and the active and standby FE/GE ports of the base station controller share one IP address (IP1-0), as shown in **Figure 6-2**.

Figure 6-2 Ethernet active and standby ports on the active/standby board of the base station controller



BFD is enabled between the active port of the base station controller and the two real IP addresses of the VRRP-enabled routers. The active link is considered faulty when both BFD sessions fail. Active/standby port switchover is performed when any of the following conditions is met:

- The active link is faulty, but the standby link is functional.
- The active port is faulty, but the standby port is functional.
- The active board is faulty, but the standby board is functional.
- The board where the active port is located is reset, but the board where the standby port is located is functional.

After the port switchover, the previous standby port automatically uses the configuration data of the previous active port. After the active port fault is rectified, services will not be automatically switched back.

NOTE

Do not associate the BFD status with route advertisement on the router side. This is because the BFD status changes from faulty to restored during an active/standby port switchover. If a router associates the BFD status with route advertisement, routes are withdrawn quickly when a BFD fault occurs. When the BFD fault is rectified, route re-advertisement and learning are slow. As a result, the end-to-end service interruption time during an active/standby port switchover is affected by the route re-advertisement and learning time.

You can use ARP to check the status of the standby port and standby link.

6.4 Ethernet Link Aggregation

Ethernet link aggregation, defined in IEEE 802.3ad, groups two or more Ethernet links into one logical path. If a link is faulty, service on this link will be switched over to functional links, preventing service interruption and improving reliability.

6.4.1 Principles

Links in a link aggregation group must be configured with the same attributes, including optical/electrical mode, data rate, and duplex mode. The number of links on the local and peer ends of a link aggregation group must be the same. Each port in a link aggregation group must be configured with a priority. Port priorities are used for determining the actually used ports by using Link Aggregation Control Protocol (LACP).

The MAC address of a link aggregation group is the MAC address of the first port added to the group. A link aggregation group can work in active/standby mode to improve reliability or in load sharing mode to improve bandwidth efficiency. The link aggregation group can be static or manual.

- For a static link aggregation group, users need to add member ports to the group. LACP must be enabled between two ends. The actually used ports are negotiated with the peer end by using LACP.
- For a manual link aggregation group, users add member ports to the group. LACP must be disabled between two ends. All added ports will be used without exchanging the aggregation status with the peer end.

Link aggregation groups at the local and peer ends must be of the same type.

The base station controller supports inter-board and intra-board link aggregation. The base station supports only intra-board link aggregation in load sharing mode.

According to the IEEE 802.3ad standard, the interval for sending LACP PDUs has long timeout and short timeout. Both the base station and base station controller support short timeout. Their peer ends must also support short timeout to enable quick link fault detection.

NOTE

The GBTS does not support link aggregation.

6.4.2 Application on the Base Station Controller (GSM/UMTS)

Link aggregation on the base station controller applies to the following scenarios:

- The active and standby boards are connected to the Layer 3 routers.
- The active and standby boards are connected to the Layer 2 transmission devices.

NOTE

Intra-board link aggregation applies only when IP transmission resource pools are used.

Table 6-2 and **Table 6-3** list the networking modes supported by the BSC6900 Ethernet link aggregation groups (LAGs). **Table 6-4** and **Table 6-5** list the networking modes supported by the BSC6910 Ethernet LAGs. **Table 6-6** lists the networking mode supported by the BSC6960 Ethernet LAGs.

Table 6-2 Networking modes supported by the BSC6900 Ethernet LAGs (manual)

Networking Mode	Manual Active/ Standby LAG	Manual Inter-Board Load-Sharing LAG	Manual Intra-Board Load-Sharing LAG
IP path	Supported	Supported	Supported
IP pool	Supported	Not supported	Supported

Table 6-3 Networking modes supported by the BSC6900 Ethernet LAGs (static)

Networking Mode	Static Active/Standby LAG	Static Inter-Board Load-Sharing LAG	Static Intra-Board Load-Sharing LAG
IP path	Supported	Supported	Supported
IP pool	Supported	Not supported	Supported

Table 6-4 Networking modes supported by the BSC6910 Ethernet LAGs (manual)

Networking Mode	Manual Active/ Standby LAG	Manual Inter-Board Load-Sharing LAG	Manual Intra-Board Load-Sharing LAG
IP path	Supported	Supported	Supported
Layer 2 IP pool	Supported	Supported	Supported
Layer 3 IP pool	Supported	Not supported	Supported

Table 6-5 Networking modes supported by the BSC6910 Ethernet LAGs (static)

Networking Mode	Static Active/Standby LAG	Static Inter-Board Load-Sharing LAG	Static Intra-Board Load-Sharing LAG
IP path	Supported	Supported	Supported
Layer 2 IP pool	Supported	Supported	Supported
Layer 3 IP pool	Supported	Not supported	Supported

Table 6-6 Networking modes supported by the BSC6960 Ethernet LAGs (manual)

Networking Mode	Manual Active/ Standby LAG	Manual Inter-Board Load-Sharing LAG	Manual Intra-Board Load-Sharing LAG
IP path	Not supported	Not supported	Not supported
Layer 2 IP pool	Not supported	Not supported	Not supported
Layer 3 IP pool	Supported	Not supported	Not supported

NOTE

The BSC6960 does not support static Ethernet LAGs.

Connecting to Layer 3 Routers

Manual active/standby link aggregation is used for the two FE/GE/10GE ports on the active/standby board of the base station controller to connect to the two VRRP-enabled peer routers. Settings are as follows:

- ***LACPMODE*** is set to **MANUAL_AGGREGATION**
- ***WORKMODE*** is set to **ACTIVE_STANDBY**.
- ***RT*** is set to **NON-REVERTIVE**.

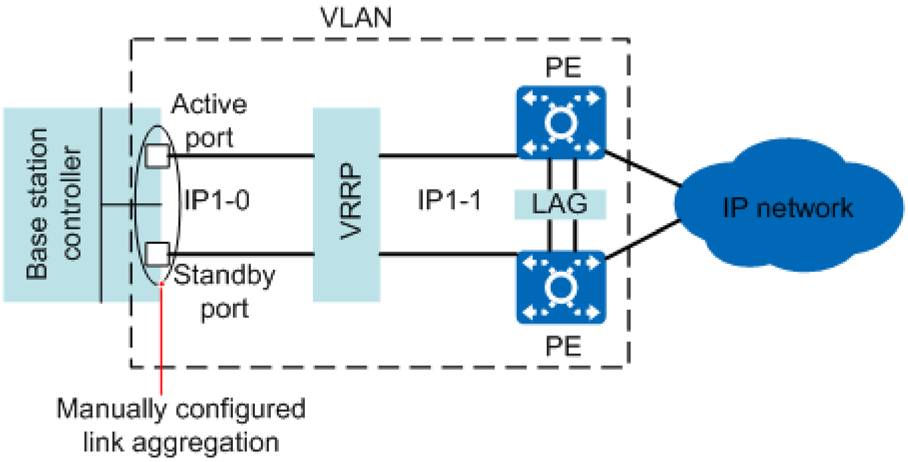
BFD is enabled between the active port of the link aggregation and the two real IP addresses of the VRRP-enabled routers. The base station controller considers the active link faulty when detecting that both BFD sessions fail. If this happens, link switchover is performed within the link aggregation group. The networking is similar to that for Ethernet active and standby ports on the active/standby board of the base station controller connected to the VRRP-enabled routers. The configurations on the router are unchanged. The only difference is that the active/ standby ports are changed to manual active/standby link aggregation.

NOTE

The BSC6900, BSC6960, and BSC6910 support manual active/standby link aggregation. Only the BSC6900 supports Ethernet port backup on the active/standby board.

Optionally, status detection of the standby port or standby link through ARP can be performed.

Figure 6-3 Active/standby board connecting to Layer 3 routers through link aggregation



Connecting to Layer 2 Transmission Devices (BSC6900/BSC6910)

Static active/standby link aggregation is used for the two FE/GE/10GE ports on the active/standby board of the base station controller to connect to the Layer 2 transmission devices (such as OSN and PTN devices). Settings are as follows:

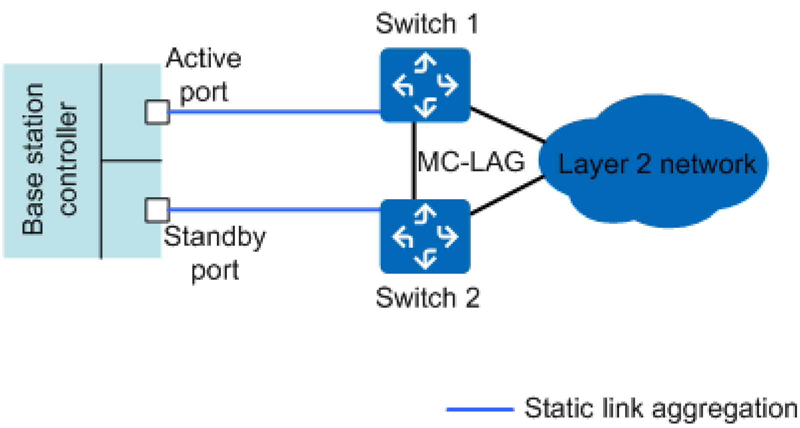
- ***LACPMODE*** is set to **STATIC_LACP**.
- ***WORKMODE*** is set to **ACTIVE_STANDBY(Active standby)**.
- ***RT*** is set to **NON-REVERTIVE(NON-REVERTIVE)**.

In static link aggregation mode, the active and standby port/link attributes between the local and peer ends keep consistent through the LACP exchange.

NOTE

Generally, the Layer 2 transmission devices support multi-chassis link aggregation group (MC-LAG) working in active/standby mode.

Figure 6-4 Active/standby board connecting to Layer 2 transmission devices through link aggregation



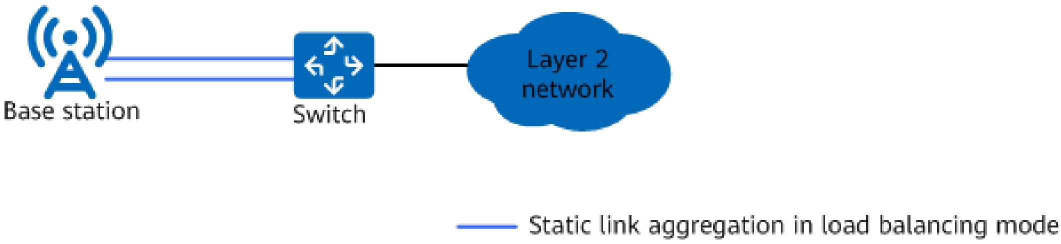
6.4.3 Application on the Base Station

The typical application scenario of link aggregation on the base station side is that the base station is connected to the link aggregation port of the Layer 2 or Layer 3 transmission device.

Multiple FE/GE/10GE/25GE/50GE ports on the main control board or transmission extension board of a base station form a link aggregation group and connect to a Layer 2 or Layer 3 transmission device. The link aggregation mode is static and the working mode is load balancing.

In static link aggregation mode, the LACP must be enabled.

Figure 6-5 Base station connecting to the Layer 2 or Layer 3 transmission device through link aggregation



The Ethernet link aggregation load balancing factor of Layer 2 packets on the base station side is source MAC address+destination MAC address.

Table 6-7 lists the load balancing factors for Layer 3 packets.

In heavy traffic scenarios, different load balancing effect can be obtained by selecting different factors. If Layer 4 or higher-layer packet information is used as the load balancing factor, traffic is evenly distributed to member ports.

Table 6-7 Ethernet link aggregation load balancing factor of Layer 3 packets on the base station side

Packet Type	Load Balancing Factor
GTP-U	If **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** is set to **LB_MODE_1(Load Balancing Mode 1)**, the load balancing factor is source IP address+destination IP address+TEID.
	If **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** is set to **LB_MODE_0(Load Balancing Mode 0)**, the load balancing factor is source IP address+destination IP address+DSCP.

Packet Type	Load Balancing Factor
UDP (including user-plane service packets and packets of specific features, such as the IKE, TWAMP, IP PM, BFD, UDP session, and clock packets)	If **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** is set to **LB_MODE_1(Load Balancing Mode 1)**, the load balancing factor is source IP address+destination IP address+source port number +destination port number.
	If **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** is set to **LB_MODE_0(Load Balancing Mode 0)**, the load balancing factor is source IP address+destination IP address+DSCP.
Other IP packets (such as TCP and SCTP packets)	The load balancing factor is source IP address +destination IP address+DSCP.

NOTE

- The base station's main control board and transmission extension board support only intra-board link aggregation, not inter-board link aggregation.
- The base station supports link aggregation only in load sharing mode.
- A port that is added to a link aggregation group on a base station cannot be configured with a non-AUTOPORT transmission resource group configured using **RSCGRP** in the old model or **IPRSCGRP** in the new model. If a port is manually or automatically configured with a non-AUTOPORT transmission resource group, delete the transmission resource group before adding the port to a link aggregation group.
- The rate of the trunk group is not greater than the board capability. The capability varies with the board type.
- The **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** parameter takes effect only on UMPT/UMDU boards in non-IPsec scenarios.

6.5 IP Route Backup

IP route backup requires two routes that have the same destination IP address but different next-hop IP addresses and priorities. The route priority determines whether a route is active or standby. The principles for activating route priorities are as follows:

- The route with a high priority is preferentially activated.
- If this high-priority route is faulty, the other route with a low priority is activated.
- After the high-priority route becomes functional, it is then automatically activated.
- Only the BSC6900 and BSC6910 support IP route backup. The BSC6960 does not support this.

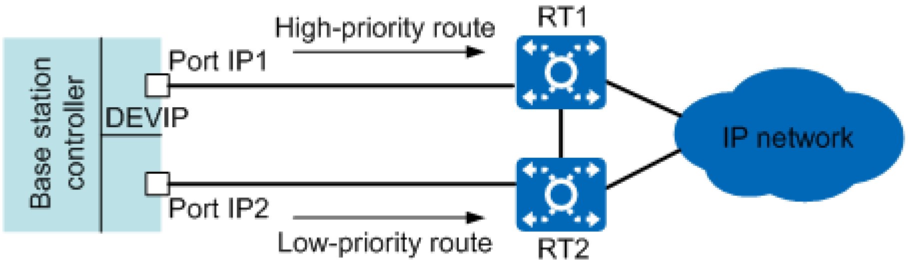
6.5.1 Application on the Base Station Controller (GSM/UMTS)

Ethernet route backup between the active and standby boards is typically applied to the base station controller.

As shown in **Figure 6-6**, routers RT1 and RT2 are configured with standby routes to the device IP address (defined by the **DEVIP** MO). RT1 is configured with standby routes to the port IP1 and RT2, and RT2 is configured with standby routes to the port IP2 and RT1. Both port IP1 and port IP2 can receive service packets.

BFD is enabled between each port on the base station controller and its peer port on the router. Both the base station controller and router associate the BFD status with the route status and they trigger active/standby route switchover on their own sides.

Figure 6-6 Inter-board route backup



NOTE

When inter-board route backup is used, the IP performance monitoring (PM) function is not recommended.

It is recommended that the route priority configured on the peer router be consistent with that on the base station controller. The consistent configuration ensures that the base station controller sends and receives packets over the same route.

If RT1 becomes faulty, although its BFD session with the base station controller can return normal soon after a restart, it needs some time to learn its routes to remote service NEs, such as base stations and core network NEs. Therefore, if **Route associated flag** is set to **YES**, it is recommended that **Route Switchover Delay** be specified so that a route switchback can be performed after RT1 has learned the routes to its remote service NEs.

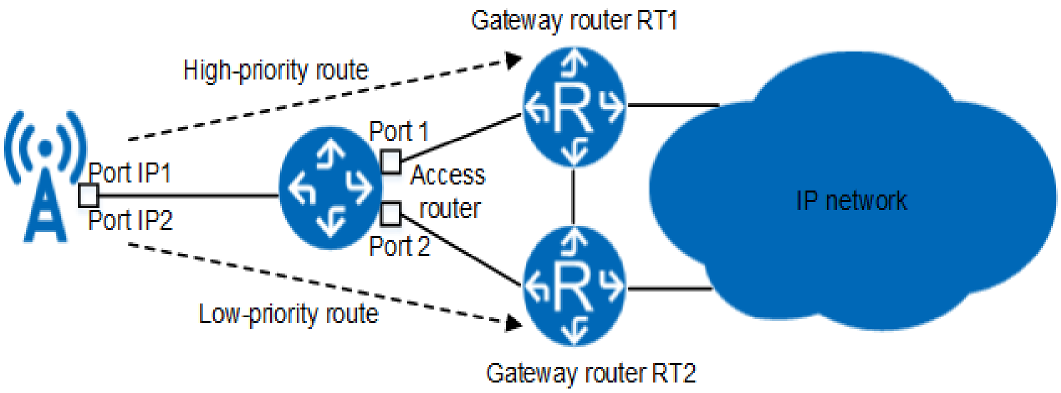
6.5.2 Application on the Base Station

IP route backup is applied when the base station is connected to dual gateway routers. As shown in **Figure 6-7**, gateway routers RT1 and RT2 that work in backup mode are configured with routes to the logical IP addresses of the base station.

- RT1 is configured with standby routes with the next hops as the port IP1 and RT2.
- RT2 is configured with standby routes with the next hops as the port IP2 and RT1.

BFD is enabled between the base station and gateway routers to detect the link status, and the BFD status is associated with the route status. In this way, route switchovers can be triggered when necessary.

Figure 6-7 Ethernet route backup on the base station



NOTE

If RT1 becomes faulty, although its BFD session with the base station can become normal soon after a restart, RT1 needs time to learn its routes to peer service NEs, such as the base station controller or core network NE. Therefore, route switchback delay is specified on the base station so that a route switchback can be performed after RT1 has learned the routes to its peer service NEs.

The base station does not support inter-board route backup.

The **OMCH** MO can be configured in route-unbinding mode or route-binding mode.

- In route-binding mode, a route for an OMCH must be specified by configuring the **IPRT** (in the old model)/**IPROUTE4** (in the new model) MO. In addition, set **OMCH.*BRT*** to **YES** and **OMCH.*RTIDX*** to the index of the route to be bound.
- In route-unbinding mode, a route for the OMCH is specified by adding an **IPRT** (in the old model)/**IPROUTE4** (in the new model) MO and the route does not need to be bound to the OMCH.

If there is only one OMCH, the route for the OMCH can be added in either route-unbinding mode or route-binding mode. In route-binding mode, ensure the route bound to the OMCH forwards only the traffic on the OMCH. This is because the route bound to an OMCH is invalid if the OMCH does not take effect. In this case, if the route is supposed to forward other traffic, the forwarding will fail.

If there are active and standby OMCHs, the route-binding mode must be used to bind separate routes to the two OMCHs. When the active OMCH takes effect, only the route bound to the active OMCH takes effect. When the standby OMCH takes effect, only the route bound to the standby OMCH takes effect.

The OMCH can share an IP address with another interface or use a separate IP address. The route from the base station to the MAE must be planned.

6.6 Control-Plane SCTP Multi-homing

For details about SCTP, see **4.5.2 SCTP**.

If an SCTP endpoint has multiple destination addresses, the SCTP endpoint is a multi-homing endpoint.

When both endpoints of an SCTP association are multi-homed, there are two types of SCTP transmission paths, as shown in **Figure 6-8** and **Figure 6-9**.

- Parallel paths: The path between the first IP address of the local end and the first IP address of the peer end serves as the default primary SCTP path. The path between the second IP address of the local end and the second IP address of the peer end serves as the secondary SCTP path.
- Cross paths: The path between the first IP address of the local end and the second IP address of the peer end, or the path between the second IP address of the local end and the first IP address of the peer end serves as the secondary SCTP path.

If one endpoint of an SCTP association is a multi-homing endpoint, the SCTP path to the second address of the peer end is used when the SCTP path to the first address of the peer end is faulty. See **Figure 6-10**.

Figure 6-8 Scenario where the two endpoints of an SCTP association are both multi-homed and only parallel paths are supported

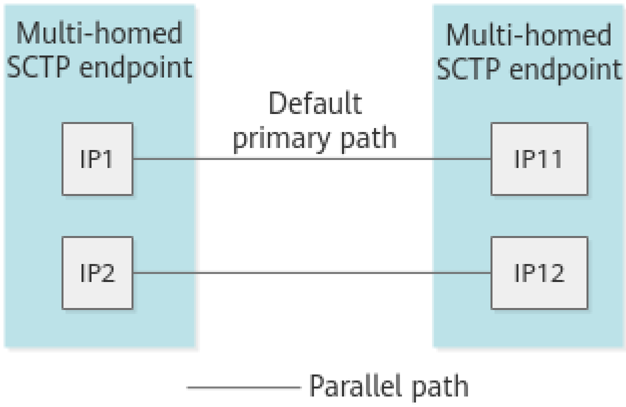


Figure 6-9 Scenario where the two endpoints of an SCTP association are both multi-homed and both parallel paths and cross paths are supported

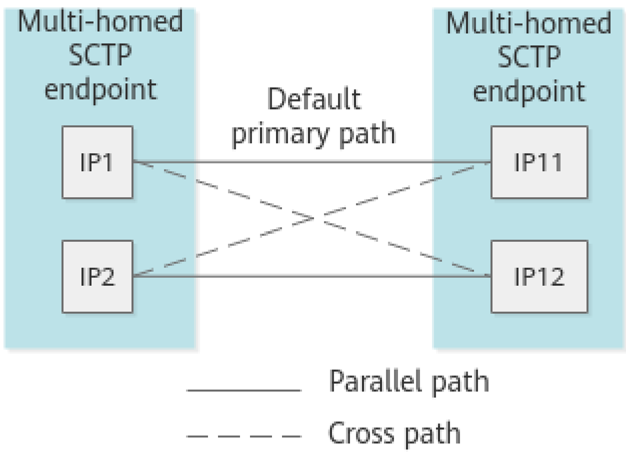
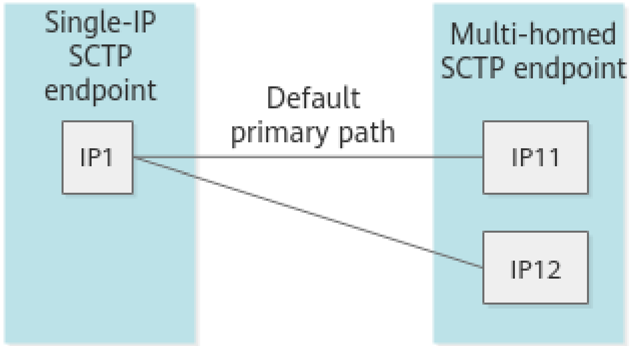


Figure 6-10 Scenario where one SCTP association endpoint is multi-homed



Configurations on the Base Station Controller (GSM/UMTS)

The base station controllers include the BSC6900, BSC6960, and BSC6910. The base station controllers support SCTP multi-homing.

The **SCTPLNK**.***CROSSIPFLAG*** parameter specifies whether SCTP multi-homing supports paths in cross mode. SCTP multi-homing between the base station controller and base station supports only parallel mode and that between the base station controller and CN supports both cross mode and parallel mode.

Configurations on the Base Station

The base stations include the NodeB, eNodeB, gNodeB, eGBTS, and co-MPT base station. In SCTP link configurations, the **SCTPLNK**.***AUTOSWITCH (LTE eNodeB, 5G gNodeB)*** parameter (in link configuration mode) or the **SCTPTEMPLATE**.***SWITCHBACKFLAG (LTE eNodeB, 5G gNodeB)*** parameter (in endpoint configuration mode) specifies whether services are switched back to the primary path after it becomes functional. The value **ENABLE(Enable)** is recommended.

The base station supports the following multi-homing scenarios:

- The two endpoints of an SCTP association are both multi-homed, and only parallel paths are supported. The transmission paths between the first local IP address and the first peer IP address and between the second local IP address and the second peer IP address must be set up. In this scenario, if a peer address is unreachable, ALM-25955 SCTP Link IP Address Unreachable is reported.
- An endpoint of an SCTP association is multi-homed. When the SCTP path between the endpoint and the first peer IP address is faulty, the SCTP path between the endpoint and the second peer IP address is used. This requires that the peer devices, such as the base station controller, core network, or base station, support this multi-homing mode.
  - The base station checks multiple destination IP addresses at the same time only when the peer SCTP endpoint is multi-homed. The base station reports ALM-25955 SCTP Link IP Address Unreachable only when some destination IP addresses are unreachable.
  - If the peer SCTP endpoint is a single IP endpoint, the local end checks the path from the first IP address to the peer IP address by default. If the path is unreachable, the local end checks the path from the second IP

address to the peer IP address. In this scenario, the base station does not report ALM-25955 SCTP Link IP Address Unreachable.

NOTE

If both endpoints of an SCTP association are multi-homed, configure SCTP links on the base station side as follows:

The first local IP address, second local IP address, first peer IP address, and second peer IP address are all set to valid IP addresses.

If one endpoint of an SCTP association is multi-homed, configure SCTP links on the base station side as follows:

- If the peer SCTP endpoint is multi-homed, the first local IP address is set to a valid IP address, the second local IP address is set to 0.0.0.0, and the first and second peer IP addresses are set to valid IP addresses.
- If the peer SCTP endpoint is a single IP endpoint, the first and second local IP addresses are set to valid IP addresses, the first peer IP address is set to a valid IP address, and the second peer IP address is set to 0.0.0.0.

SCTP multi-homing is not equivalent to multiple Transport Network Layer Associations (TNLAs). SCTP multi-homing is used only for reliability. Only one SCTP link is generated, and only one path to a peer IP address takes effect. In the case of multi-TNLA, multiple SCTP links are generated and all SCTP links take effect at the same time. For details about multi-TNLA, see *NG and Xn Self-Management Feature Parameter Description*.

6.7 User Plane Backup

User plane backup includes active/standby user plane and user-plane load sharing. The eNodeB, gNodeB, and LTE and NR co-MPT base station support user plane backup.

- When the transport network provides the active/standby transmission path, the base station can use the active/standby user plane function to support user plane E2E redundancy backup. The backup function does not depend on the active/standby IP route of the transport network.
- When the transport network provides multiple transmission paths for load sharing, the base station can use the user-plane load sharing function to support user plane E2E load sharing, improving transmission bandwidth and enhancing reliability.

NOTE

- Currently, the S1 and NG interfaces support active/standby user plane and user-plane load sharing. The X2, eX2, Xn, and eXn interfaces support only user-plane load sharing and do not support active/standby user plane.
- Both IPv4 transmission and IPv6 transmission support active/standby user plane and user-plane load sharing, and source-based routes need to be configured.
- Currently, only the UMPT board supports user-plane load sharing.

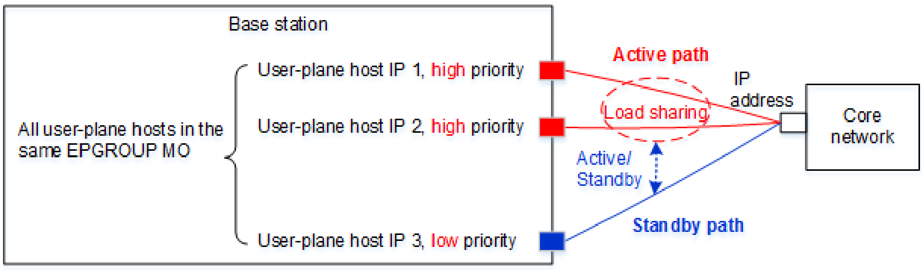
The S1 interface can be configured in link mode or endpoint mode. The NG interface supports only endpoint mode.

- Link mode: When multiple IP paths are configured over the S1-U interface, the base station selects available IP paths in polling mode based on the bearer granularity to set up S1-U bearers for the same S-GW. In this way, load sharing is achieved among multiple IP paths. For the same peer IP address, a maximum of six IP paths can be configured for load sharing. **Transmission**

Resource Type must be identical for the involved **IPPATH** MOs. Otherwise, load sharing cannot be achieved.

- Endpoint mode: As shown in **Figure 6-11**, for all user-plane hosts (**USERPLANEHOST** MOs) that are added to the same **EPGROUP** MO, load sharing is implemented among the user-plane hosts with the same priority, and active/standby backup is implemented between the user-plane hosts with different priorities.

Figure 6-11 User plane backup for a base station



The base station uses GTP-U to check the connectivity of load-sharing user-plane path to ensure that all S1-U links are available.

The base station uses the GTP-U to detect the connectivity of active and standby user-plane paths.

- When the base station detects that the active path is faulty but the standby path is normal, new services are established on the standby path.
- When the base station detects that the standby path is faulty, new services are established on the active path and services on the standby path are not switched over to the active path.

6.8 OM Channel Backup

On the base station side, OM channel backup is introduced between the base station and the MAE to provide redundancy backup for OM channels in scenarios such as hybrid transmission of high-QoS and low-QoS links and hybrid transmission of security and non-security domains. The eGBTS, NodeB, eNodeB, gNodeB, and co-MPT base station support OM channel backup.

NOTE

OM channel backup on an eGBTS uses Abis transmission backup enhancement (E1 backup). For details about Abis transmission backup enhancement, see *Abis Transmission Backup Feature Parameter Description*.

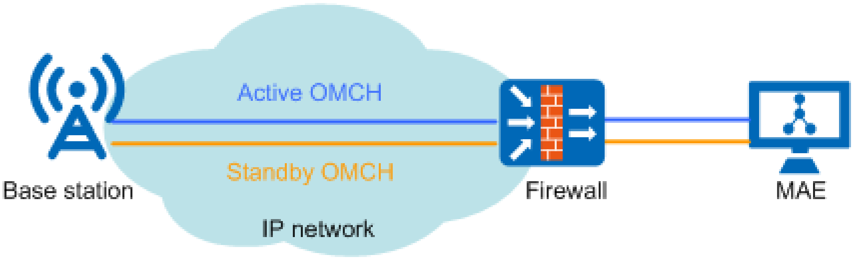
To configure OM channel backup on a NodeB, set the **SLAVENBIPOAMIP** and **SLAVENBIPOAMMASK** parameters on the base station controller side for the NodeB.

The active and standby OM channels do not share the same route with the service. This is because the OM channel switchover leads to the unavailability of the route of the active or standby OM channel.

The active and standby OM channels must correspond to the same MAE-Access.

The active OM channel takes priority over the standby OM channel, and therefore the MAE preferentially initiates connection over the active OM channel. The MAE detects the OM channel status by using a handshake mechanism at the application layer. If the active OM channel fails, the MAE instructs the base station through the standby OM channel to initiate an OM channel switchover.

Figure 6-12 OM channel backup for the base station



After the active OM channel recovers, the OM channel is automatically switched back to the active OM channel, which is called the OM channel switchback function.

- When **GTRANSPARA**.***OMCHSWITCHBACK (LTE eNodeB, 5G gNodeB)*** is set to **ENABLE** to enable the OM channel switchback function:

  The **OMCH**.***CHECKTYPE (LTE eNodeB, 5G gNodeB)*** parameter of the active OM channel must be set to **AUTO_UDPSESSION**. When the MAE is connected to the standby OM channel, the base station automatically creates a UDP session from the local IP address of the active OM channel to the MAE. The OM channel switchback function requires this UDP session detection. After detecting that the active OM channel recovers, the base station disconnects the TCP connection of the standby OM channel, which enables the MAE to initiate a connection on the active OM channel again.

  When the OM channel switchback function is enabled, the routes used by the active and standby OM channels are used at the same time. Since the same MAE uses the same IP address for the active and standby OM channels, the source-based routes must be separately configured for the local IP addresses of active and standby OM channels. It is recommended that neither OM channel be bound to destination-based routes. If the MAE uses the remote HA system, the source IP routes also need to be configured for the local IP addresses of active and standby OM channels. The source-based routes configured for the local IP addresses of the OM channels will always take effect, and will not be invalid after an OM channel switchover.

  When packet filtering (**PACKETFILTER** (old model)/**PACKETFILTERING** (new model) MO) is configured for the base station, you need to manually configure the ACL rule (**ACLRULE** MO) for the automatically established UDP session in the ACL (**ACL** MO) associated with the packet filtering. Otherwise, the OM channel switchback function does not take effect.

  On the base station, manually configure the **ACLRULE** MO corresponding to the UDP session to allow the MAE packets to reach the base station. The configuration requirements are as follows:

  – Set the source IP address (**ACLRULE**.***SIP (LTE eNodeB, 5G gNodeB)***) to the IP address of the MAE.

  – Set the destination IP address (**ACLRULE**.***DIP (LTE eNodeB, 5G gNodeB)***) to the local IP address of the active OM channel on the base station.

  - Set **ACLRULE.***PT (LTE eNodeB, 5G gNodeB)* to **UDP**, **ACLRULE**.*SPT1 (LTE eNodeB, 5G gNodeB)* to **31252**, and **ACLRULE**.*DPT1 (LTE eNodeB, 5G gNodeB)* to **65043**.
  - Set **ACLRULE**.*ACLID (LTE eNodeB, 5G gNodeB)* to the same value as **PACKETFILTER**.*ACLID (LTE eNodeB, 5G gNodeB)* (old model) or **PACKETFILTERING**.*ACLID (LTE eNodeB, 5G gNodeB)* (new model) of the corresponding packet filtering.
- When **GTRANSPARA**.*OMCHSWITCHBACK (LTE eNodeB, 5G gNodeB)* is set to **DISABLE** to disable the OM channel switchback function:

  After the fault in the active OM channel is rectified, the base station and MAE do not automatically switch back to the active OM channel. The MAE initiates connection on the active OM channel again only after the standby OM channel becomes faulty.

  When the base station uses the active and standby OM channels and destination-based routes need to be used, it is required to configure two destination-based routes and bind one to the active OM channel and the other to the standby OM channel. If the MAE uses the remote high reliability (HA) system and different routes are required by the active and standby MAE servers, bind routes to the active and standby MAE servers through the active and standby OM channels (using the **OMCH** MOs).

  After the base station starts, it activates the route of the active OM channel and deactivates the route of the standby OM channel. After the active OM channel is switched over to the standby OM channel, the base station automatically deactivates the route of the active OM channel and activates the route of the standby OM channel.

A security mechanism is introduced that allows the MAE and base station to perform authentication on OM channel switchover messages over the active or standby OM channel. This helps prevent a spoofed MAE from delivering fake switchover instructions and protect switchover messages against replay attacks or being tampered with.

NOTE

If the base station runs a software version of SRAN10.1 or later, the MAE uses this security mechanism to perform OM channel switchover. In addition, the MAE must run a matching version. Otherwise, the OM channel switchover will fail and the OM channel will be disconnected.

7 Transmission Maintenance and Detection

7.1 Overview

Maintenance and detection mechanisms for IP transmission include ARP probe, ICMP ping detection, BFD, LACP fault detection, IP path ping detection, trace route, IP link QoS detection, GTP-U echo, and LLDP, which are applied to different layers.

Transmission maintenance and detection improve network reliability.

Figure 7-1 describes the mapping between maintenance and detection mechanisms and their applicable layers.

Figure 7-1 Mapping between maintenance and detection mechanisms and their applicable layers

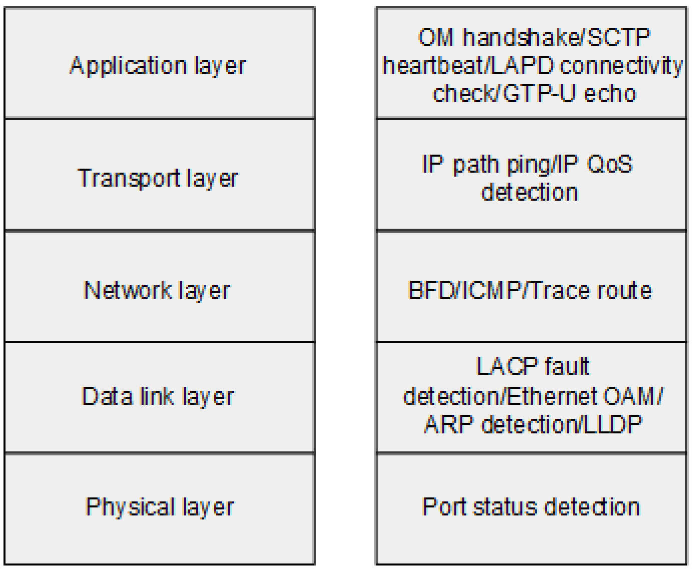


Table 7-1 describes the mapping between maintenance and detection mechanisms and their applicable layers.

Table 7-1 Transmission reliability and maintenance and detection methods supported by RAN equipment

Protocol Layer	Object	Maintenance and Detection Mechanism	Detection Interval
Transport layer/ Application layer	Base station OM channel	OM handshake protocol	The detection is performed once every minute. **NOTE** By default, if the MAE detects three consecutive handshake failures, ALM-25901 Remote Maintenance Link Failure is reported.
	Control-plane link	SCTP	By default, SCTP heartbeats are detected once every 5000 ms on the base station side. On the RNC, it is recommended that SCTP heartbeats be detected once every 5000 ms for the application types of NBAP, BBAP, and ABISCP, and once every 1000 ms for the application type of M3UA. The detection interval is configurable. **NOTE** Upon detecting a disconnected SCTP link, the base station reports ALM-25888 SCTP Link Fault, the BSC6960 reports ALM-21682 SCTP Link Fault, and the BSC6900/BSC6910 reports ALM-21541 SCTP Link Fault.
	Abis control-plane path	LAPD protocol	**NOTE** After detecting a disconnected LAPD link, the BSC reports ALM-21512 LAPD Link Fault.

Protocol Layer	Object	Maintenance and Detection Mechanism	Detection Interval
	S1/X2/eX2/NG/Xn/eXn/Iu-PS user-plane path	GTP-U echo	The base station checks the connectivity of the GTP-U tunnel once every minute. The detection timeout interval on the RNC side is configurable. **NOTE** By default, after detecting that a route is unreachable for three consecutive times, the base station reports ALM-25954 User Plane Fault. Upon detecting that all channels for PS user data transmission on the Iu-PS interface boards are faulty for two consecutive times, the BSC6900/BSC6910 reports ALM-22510 GTPU Faulty and the BSC6960 reports ALM-21697 GTPU Faulty.
	User plane on the base station controller side	IP path ping	The detection interval is configurable within the range of 1s to 60s. When adding an IP path, set ***PATHCHK*** to **ENABLED(Enabled)**.
Network layer	Route	BFD	By default, the base station detects the route reachability once every 100 ms. The detection interval is configurable within the range of 10 ms to 1000 ms. **NOTE** By default, after detecting that a route is unreachable for three consecutive times, the base station reports ALM-25899 BFD Session Fault.
	IP layer link	ICMP ping	By default, the base station performs the detection once every 1000 ms. The detection interval is configurable within the range of 1000 ms to 10000 ms.
	IP layer link	Trace route	None

Protocol Layer	Object	Maintenance and Detection Mechanism	Detection Interval
	S1/X2/Xn/Abis/Iub user-plane path, eX2 control-plane link and user-plane path, uX2 control-plane link and user-plane path, and eXn user-plane path	IP link QoS detection	IP link QoS detection includes IP PM, TWAMP, and UDP packet injection. ● For details on IP PM, see *IP Performance Monitor*. ● For details on TWAMP, see *IP Active Performance Measurement*. ● By default, the base station performs detection by UDP packet injection once every 1000 ms. The detection interval is configurable within the range of 1 ms to 1000 ms.
Data link layer	Ethernet trunk port	LACP	The detection is performed once every second. ● Upon detecting a faulty Ethernet trunk link, the base station reports ALM-25887 Ethernet Trunk Link Fault and the BSC6900/BSC6910 reports ALM-21350 Ethernet Trunk Link Fault. ● Upon detecting a faulty Ethernet trunk, the base station reports ALM-25895 Ethernet Trunk Group Fault and the BSC6900/BSC6910 reports ALM-21349 Ethernet Trunk Group Fault.
	Single-hop Ethernet link	IEEE 802.3ah protocol	The detection is performed once every second. Upon detecting the loopback at the local end, the base station reports ALM-25894 ETHOAM 3AH Remote Loopback, the BSC6960 reports ALM-21657 ETHOAM 3AH Remote Loopback, and the BSC6900/BSC6910 reports ALM-21374 ETHOAM 3AH Remote Loopback.

Protocol Layer	Object	Maintenance and Detection Mechanism	Detection Interval
	Multi-hop Ethernet link	IEEE 802.1ag protocol	By default, the detection is performed once every second. For details, see *Ethernet OAM*.
	Ethernet link	ITU-T Y.1731 protocol	By default, the detection is performed once every second. For details, see *Ethernet OAM*.
	Ethernet link	LLDP	By default, the detection is performed once every 30s. This function is configured using the **LLDPDU Tx Interval** parameter, which can be set to a value in the range of 5s to 32768s.
	Ethernet link	ARP probe	The default ARP packet timeout duration is 3s for a GBTS and 2s for a base station controller. The default number of ARP packet retransmission times is 3 for a GBTS and 2 for a base station controller. The ARP packet timeout duration ranges from 300 ms to 60s for a GBTS and from 1s to 10s for a base station controller. The number of ARP packet retransmission times ranges from 3 to 10 for a GBTS and from 2 to 10 for a base station controller.

7.2 ARP Probe (GSM/UMTS)

The GBTS, BSC, and RNC support ARP probe.

ARP probe uses the ARP mechanism, where, after sending an ARP request to the peer end, the local end determines link connectivity according to the ARP response from the peer end. The base station controller/base station periodically sends an ARP request to the network. The destination IP address of the request is the peer IP address to be detected. The base station controller/base station determines the link connectivity based on whether it receives an ARP response from the destination. If the base station controller/base station receives this ARP response, the link is normal. If the base station controller/base station does not receive an ARP response from the destination for certain consecutive periods, the link is faulty.

ARP probe applies only to a direct connection with the IP addresses of both ends on the same network segment.

NOTE

ARP messages are broadcast messages. If the base station controller/base station and Layer 3 devices are directly connected, broadcast storms will not occur. If the base station controller/base station and the Layer 3 devices are not directly connected, broadcast storms are likely to occur.

The characteristics of ARP probe are as follows:

- ARP probe does not depend on peer equipment. A single end can start the ARP probe.
- The port status is associated with the detection status. Such association needs to be configured. If a fault is detected, port switchover is triggered and the routes whose next-hop address is the detected address are deleted. The upper-layer services then select other available routes.
- Independent port detection is supported. When boards work in active/standby mode, ARP probe can be performed only on the active port or on the active and standby ports simultaneously.

ARP probe is used to:

- Detect the connectivity of the link between the base station controller and the gateway that is a router.
- Detect the connectivity of the link between the base station controller and the peer equipment when the base station controller and the peer equipment are directly connected. Generally, the peer equipment is core network (CN) equipment.

Parameters related to ARP probe on the GBTS are as follows:

- ***DSTIP*** specifies the destination IP address for ARP probe.
- ***WHETHERAFFECTSWAP*** specifies whether the ARP probe result is associated with a route. If this parameter is set to **NO**, the route remains valid if a fault is detected. If this parameter is set to **YES**, the route becomes invalid if a fault is detected.

The characteristics of BSC6900/BSC6910 detection parameters are as follows:

The base station controller supports ARP probe on the Abis, Iub, Iu, A, Gb, Iur, and Ater interfaces.

Parameters related to ARP probe are as follows:

- ***ARPTIMEOUT*** specifies the interval at which the base station controller sends ARP probe packets. This must be set.
- ***ARPRETRY*** specifies the number of consecutive response packets that are not received by the base station controller before the ARP link fault is reported. This must be set.
- ***PEERIP*** specifies the destination IP address for ARP probe.
- ***BAKIP*** specifies the IP address of the standby Ethernet port. The standby Ethernet port can be used to detect faults at the physical layer and to start ARP probe to detect faults at upper layers. In the case of ARP probe, the IP

addresses of the active and standby ports must be on the same network segment.

- ***BAKMASK*** specifies the subnet mask of the standby port where ARP probe is performed.
- ***CHKTYPE*** specifies the detection type. It must be set to **ARP**.
- ***MODE*** specifies the detection mode. This parameter can be set to:
  - **CHECK_ON_PRIMARY_PORT(Check on Active Port)**
  - **CHECK_ON_STANDBY_PORT(Check on Standby Port)**
  - **CHECK_ON_INDEPENDENT_PORT(Check on Independent Port)**
  - **CHECK_ON_PRIMARY_TRUNKLINK(Check on Active Sublink in Trunk Group)**
  - **CHECK_ON_STANDBY_TRUNKLINK(Check on Standby Sublink in Trunk Group)**

The BSC6960 supports ARP probe, which is specified by the **ARPCHECKSW** parameter. When this parameter is set to **ENABLE**, each port selects a next-hop gateway to start the ARP connectivity check. If the check fails, ALM-21668 ARP Check Fault is reported.

7.3 ICMP Ping Detection

Both the base station and base station controller support ICMP ping.

ICMP ping detection requires that peer devices support ICMP ping and the bearer network should not filter out the response packets. ICMP ping detection is an end-to-end link fault detection. However, peer devices may regard ICMP ping packets as attack packets and discard them. As a result, the local end considers that the transmission link is faulty when it fails to receive ICMP packet response packets.

ICMP ping detection checks whether the peer end is reachable as well as the MTU of the peer end. ICMP ping detection detects whether a peer device is reachable by sending an ICMP ping request message from the local device. If the peer device receives the message, it sends a response message.

A base station supports continuous ping detection. That is, a base station continuously sends ICMP ping request messages to the peer device so that the connectivity with the destination IP address can be continuously checked. This continuous ping detection function is controlled by the **CONTPING** parameter. Users can press **CTRL+Q** on the keyboard or run the **STP PATHCHK** command to stop the continuous ping detection.

IP packets are transmitted using either of the following methods:

- Transmission along routes defined in the routing table. After the destination IP address of a packet is determined, the packet can be transmitted hop by hop along the path to the destination node.
- Direct transmission without searching for routes. After the egress port and next hop of a packet are determined, the packet is directly transmitted.

Accordingly, ICMP ping detection works as follows:

- The local device sends the peer device an ICMP ping request message along the chosen path.

- The local device sends an ICMP ping request message to the configured egress port and next hop.

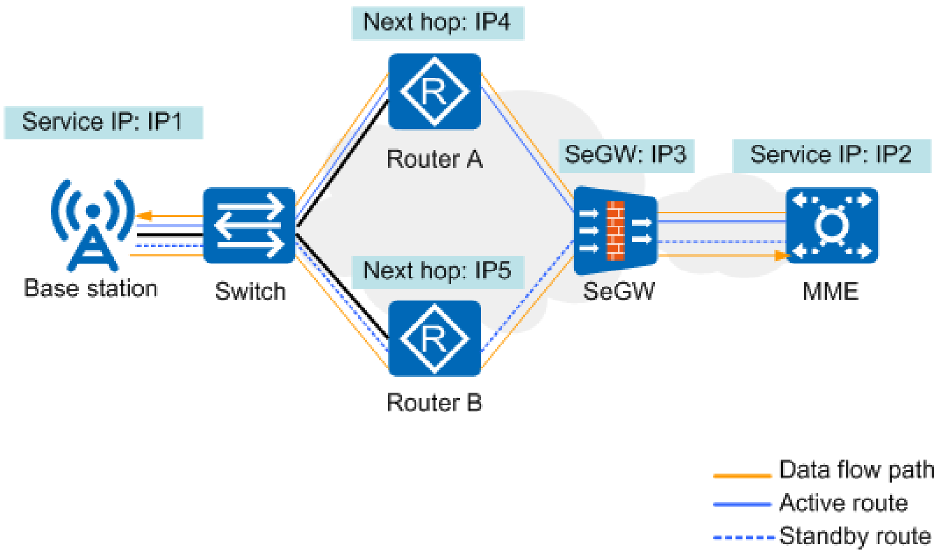
  This method can be used to check continuity of the standby route or routes affected by gateway reconstruction.

  – For PPP, MP, and tunnel ports, specify the source address, destination address, egress port type, and egress port number.

  – For Ethernet links, specify the source address, destination address, and next-hop address.

    On a network with a SeGW, the next hop refers to the hop to the SeGW. In the example network shown in **Figure 7-2**, the next hop is IP5 for continuity check over the standby route destined to IP2.

Figure 7-2 Active and standby routes on a secure network



7.4 BFD

BFD is a type of high-speed independent *Hello* protocol and supports link fault detection in the millisecond range.

BFD is categorized into single-hop BFD (SBFD) and multi-hop BFD (MBFD) and can be linked with upper-layer protocols to trigger fault isolation, which helps minimize service loss and improve system reliability.

7.4.1 Principle

BFD requires the setup of a BFD session with which the BFD detects the connection between two ends.

A BFD point plays either an active role or a passive role during session initialization.

- A BFD end playing the active role can send BFD control packets for a particular session, regardless of whether it has received any BFD packets for that session.

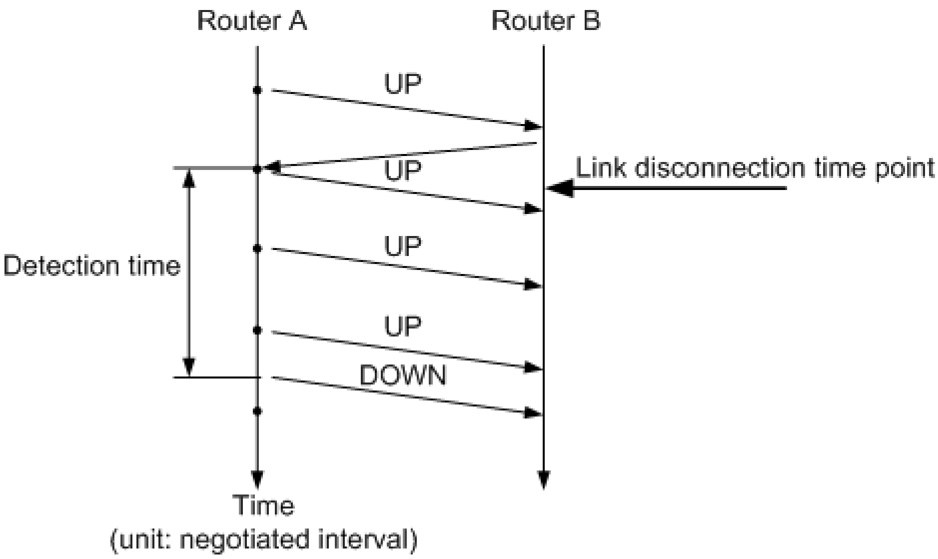
- A BFD end playing the passive role can start sending BFD control packets for a particular session only after it receives a BFD control packet for that session.
- To set up a BFD session, at least one BFD end must take the active role.

Both base stations and base station controllers play the active role of a BFD session.

BFD uses BFD messages as heartbeat messages. If a BFD end transmits several BFD messages over a link but does not receive any response messages, the link is considered faulty.

Figure 7-3 illustrates BFD messages exchanged when a link is disconnected.

Figure 7-3 BFD message exchange when the link is disconnected



In asynchronous BFD mode, both BFD ends periodically send BFD control packets to each other. If the BFD control packets are received within the detection time shown in Figure 7-3, the session is considered **UP**. If BFD control packets are not received within the detection time, the BFD session is considered **DOWN**.

The interval for transmitting BFD control packets is proportional to Max(Minimum transmit interval, Minimum receive interval). The proportion is determined by the detection multiplier. The detection time can be calculated at the BFD transmit end by using the following formula: Detection time = Detection multiplier of the transmit end x Max(Minimum transmit interval, Minimum receive interval).

During BFD session negotiation, packets from a base station carry two fields: my discriminator and your discriminator. My discriminator is generated by the base station, and your discriminator is generated by the router. After parsing my discriminator in the packet from the router, the base station sends it as your discriminator in its own packet to the router. My discriminator of the base station is specified by the **BFDSESSION**.***MYDSCRMGENMODE (LTE eNodeB, 5G gNodeB)*** (old model)/**BFD**.***MYDSCRMGENMODE (LTE eNodeB, 5G gNodeB)*** (new model) parameter.

- If the parameter is set to **STATIC_CONFIG**, my discriminator is manually set to a fixed value.

- If the parameter is set to **DYNAMIC_GEN** or **DYNAMIC_GEN_ENHANCE**, my discriminator is randomly generated. In this case, ensure that the router does not limit the discriminators on the base station side.

7.4.2 Technical Description

SBFD applies to end-to-end direct connections in layer 2 networking.

MBFD applies to end-to-end indirect connections in layer 3 networking. Multiple intermediate nodes can exist between the two ends of an MBFD session.

BFD has the following technical characteristics:

- Both ends of a BFD session must support BFD.
- BFD must be started on both ends of a BFD session, and the detection time must be equal or nearly equal.
- SBFD requires a physical port on the end that sends BFD control packets to be bound to the source and destination IP addresses of a connection to which the SBFD applies.
- MBFD requires the source IP address to be bound to the destination IP address of a connection at the end that sends BFD control packets.

  Either device IP addresses or interface IP addresses of an interface board can be used as the local IP address of an MBFD session. The peer IP address cannot be on the same network segment as any of the local IP addresses.
- Detection results can be linked with upper-layer protocols.

  When BFD detection results are linked with upper-layer protocols:
  - If a fault is detected, the active and standby ports can be switched and the route with the detection IP address of the BFD session as the next-hop IP address will be removed.
  - If a fault is detected, the active and standby routes can be switched and the route bound to the BFD session will be removed.

    Upper-layer services will be carried over other available routes.
- BFD in asynchronous mode is supported and BFD in demand mode is not supported.
- The base station controller supports only BFD on the active port. It does not support simultaneous BFD on the active and standby ports.

If BFD control packets are transmitted in plaintext, security risks such as bogus packets and replay attacks exist. Therefore, BFD authentication is introduced to support SBFD on base stations and base station controllers.

NOTE

BFD authentication complies with the RFC5880 protocol. The protocol version for BFD sessions must be standard and cannot be draft 4.

- To enable BFD authentication for a base station, the peer router must be checked to ensure that the peer router also supports BFD authentication. If the peer router does not, the authentication will fail.

  In addition, the number of configured keys and key values must be consistent between the local and peer ends. Otherwise, the authentication will fail.

- To enable BFD authentication for a base station controller, the number of configured keys, key indexes, and key values must be consistent between the local and peer ends. Otherwise, the authentication will fail.

  In addition, if the base station establishes a BFD session with a Cisco router, it is recommended that the local key length do not exceed 15 characters. Otherwise, the BFD session negotiation may fail.
- If static BFD is enabled on the router and my and your discriminators are specified, the base station's generation mode of my discriminator must be static configuration, and my discriminator on the base station side must be the same as your discriminator configured on the router to avoid negotiation failures.

The parameters used to configure BFD on the base station and base station controller sides are as follows:

- On the base station side (in the old model)
  - **BFDSESSION**.***HT (5G gNodeB, LTE eNodeB)***: used to configure the hop type.
  - **BFDSESSION**.***DM (5G gNodeB, LTE eNodeB)***: used to configure the detection multiplier of a BFD session.
  - **BFDSESSION**.***MINTI (5G gNodeB, LTE eNodeB)***: used to configure the effective minimum TX interval.
  - **BFDSESSION**.***MINRI (5G gNodeB, LTE eNodeB)***: used to configure the effective minimum RX interval.
  - **BFDSESSION**.***CATLOG (5G gNodeB, LTE eNodeB)***: used to configure whether to allow port switchovers.
  - **BFDSESSION**.***MYDSCRMGENMODE (LTE eNodeB, 5G gNodeB)***: used to configure the generation mode of my discriminator.
  - **BFDSESSION**.***BFDSN (LTE eNodeB, 5G gNodeB)*** plus one: used to generate my discriminator.
- On the base station side (in the new model)
  - **BFD**.***HT (5G gNodeB, LTE eNodeB)***: used to configure the hop type.
  - **BFD**.***DM (5G gNodeB, LTE eNodeB)***: used to configure the detection multiplier of a BFD session.
  - **BFD**.***MINTI (5G gNodeB, LTE eNodeB)***: used to configure the effective minimum TX interval.
  - **BFD**.***MINRI (5G gNodeB, LTE eNodeB)***: used to configure the effective minimum RX interval.
  - **BFD**.***CATLOG (5G gNodeB, LTE eNodeB)***: used to configure whether to allow port switchovers.
  - **BFD**.***MYDSCRMGENMODE (LTE eNodeB, 5G gNodeB)***: used to configure the generation mode of my discriminator.
  - **BFD**.***MYDISCREAMINATOR (LTE eNodeB, 5G gNodeB)***: used to configure my discriminator.
- On the GBTS side

NOTE

The GBTS data is configured on the BSC. When a GBTS is reset, the router uses the BFD function to detect the route to the GBTS and finds it unreachable because the GBTS does not have BFD data. In this case, the router may not forward the BSC configuration data to the GBTS. As a result, the GBTS reset may fail. Therefore, BFD is not recommended for the GBTS.

- **BTSBFD**.***HT***: used to configure the hop type.
- **BTSBFD**.***BTSWTR***: used to configure the time for which the GBTS waits to restore a BFD session.
- **BTSBFD**.***DETECTMULT***: used to configure the maximum number of detection periods before the GBTS reports a BFD session failure.
- **BTSBFD**.***MINTXINTERVAL***: used to configure the minimum interval at which the GBTS sends BFD packets.
- **BTSBFD**.***MINRXINTERVAL***: used to configure the minimum interval at which the GBTS receives BFD packets.

- On the base station controller side (BSC6900/BSC6910)
  - **IPCHK**.***BfdAuthSw***: used to control the BFD authentication function.
  - **IPCHK**.***CHKTYPE***: used to configure the BFD type.
  - **IPCHK**.***BFDDETECTCOUNT***: used to configure the detection multiplier of a BFD session.
  - **IPCHK**.***MINTXINT***: used to configure the effective minimum TX interval.
  - **IPCHK**.***MINRXINT***: used to configure the effective minimum RX interval.
  - **IPCHK**.***WHETHERAFFECTSWAP***: used to configure whether to trigger port switchover.
  - **IPCHK**.***ROUTEASSOCIATEDFLAG***: used to configure whether to trigger route switchover.
- On the base station controller side (BSC6960)
  - **BFD**.***HT***: used to configure the hop type.
  - **BFD**.***DM***: used to configure the detection multiplier of a BFD session.
  - **BFD**.***MINTI***: used to configure the effective minimum TX interval.
  - **BFD**.***MINRI***: used to configure the effective minimum RX interval.
  - **BFD**.***CATLOG***: used to configure whether to trigger port switchover.

7.4.3 Binding Relationship Between SBFD and IP Route

To implement SBFD, an SBFD session must be bound to an IP route (defined by **IPRT** in the old model/**IPROUTE4** in the new model).

After BFD authentication is enabled on a base station, associated route switchover is performed after BFD. Once BFD authentication fails or a link fault occurs, link switchover is triggered. The switchover procedure is the same as that after the existing BFD.

After BFD authentication is enabled on a base station controller, if BFD is associated with the route, the system automatically isolates the route with the destination IP address as the next-hop IP address once the BFD authentication fails or a link fault occurs.

The parameters used to configure a binding relationship between an SBFD session and an IP route are different between the base station and base station controller sides.

- On the base station side, the **BFDSESSION.*CATLOG (5G gNodeB, LTE eNodeB)*** (in the old model)/**BFD**.***CATLOG (5G gNodeB, LTE eNodeB)*** (in the new model) parameter is used:
  - When this parameter is set to **RELIABILITY**:

    If an SBFD detects a fault, the base station will automatically isolate the route of which the peer IP address is the next hop.

    If active and standby routes have been configured, a route switchover will be performed after the fault is detected. After the faulty route goes **UP**, a route switchback will take place when the time specified by **GTRANSPARA.*SBTIME (5G gNodeB, LTE eNodeB)*** elapses.
  - When this parameter is set to **MAINTENANCE**:

    If an SBFD detects a fault, the base station will not automatically isolate the route of which the peer IP address is the next hop.

    A route switchover will not be performed even if active and standby routes have been configured.
- On the base station controller side, the **IPCHK**.***ROUTEASSOCIATEDFLAG*** parameter is used:
  - When this parameter is set to **YES**:

    If an SBFD detects a fault, the base station controller will automatically isolate the route of which the peer IP address is the next hop.

    If active and standby routes have been configured, a route switchover will be performed after the fault is detected. After the faulty route goes **UP**, a route switchback will take place when the time specified by **IPCHK**.***RouteSwitchoverDelay*** elapses.
  - When this parameter is set to **NO**:

    If an SBFD detects a fault, the base station controller will not automatically isolate the route of which the peer IP address is the next hop.

    A route switchover will not be performed even if active and standby routes have been configured.

  NOTE

  The active/standby routing networking is not recommended for the BSC6960. This mechanism does not involve the BSC6960.

7.4.4 Binding Relationship Between MBFD and IP Route

If an MBFD detects a fault, the base station controller considers the route bound to the MBFD session unavailable, without associating this fault with the status of other routes or triggering a port switchover. In this case, packets can be forwarded through other routes to implement IP re-routing.

MBFD is not recommended for the base station controller.

On the base station side, if an MBFD of the reliability type is not activated, routes bound to the MBFD session are disabled in descending order of priority. If the detection on a lower-priority route is successful, the MBFD session turns activated. If the lowest-priority route is unreachable, all the disabled routes bound to the MBFD session will be enabled.

After the binding relationship between an MBFD session and an IP route (defined by **IPRT** in the old model/**IPROUTE4** in the new model) or an IP path (defined by **IPPATH**) is specified, the bound IP route or IP path is considered faulty if the MBFD session detects a fault.

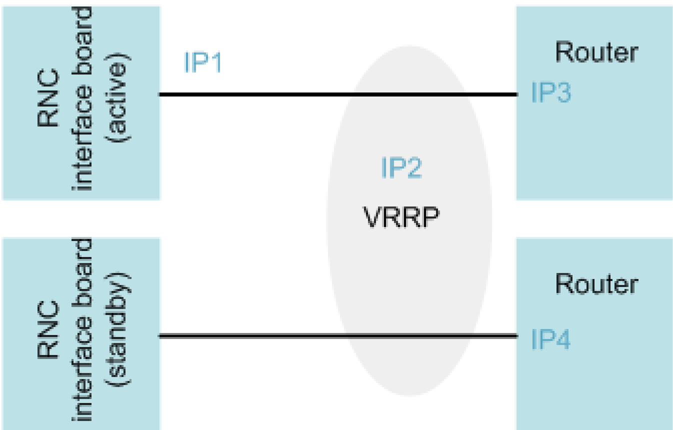
7.4.5 Application of SBFD or ARP in VRRP Networking Mode

In VRRP networking, a router functions as the active router to forward data packets and uses a virtual MAC address as the SBFD or ARP source address to respond to the host. The other router on the VRRP network functions as a standby router. The active router periodically sends VRRP multicast packets to the standby router to notify the standby outer of its status.

If the active router works properly, the standby router, rather than directly processing the received packets, forwards them to the active router. If the active router is faulty and the standby router does not receive VRRP messages at a specified interval, the standby router takes over the services handled by the active router. In this manner, the continuity and reliability of communications are ensured.

Figure 7-4 illustrates an example networking between the RNC (6900/BSC6910) and routers in VRRP mode.

Figure 7-4 VRRP networking mode



The virtual IP address in a VRRP networking mode does not support BFD. In the network shown in Figure 7-4, the RNC uses a virtual IP address (IP2) as the next-hop IP address to communicate with routers that work in VRRP mode. As a common practice, BFD does not use a virtual IP address in a VRRP network. For example, IP2 cannot be designated as the source or destination IP address of an SBFD session.

To detect connectivity between IP1 and IP2, only key ARP probe can be implemented. On the RNC side, SBFD not linked with upper-layer protocol can be started from IP1 to IP3 and from IP1 to IP4 to detect the status of the respective links (IP1 is a port IP address of the RNC, and IP3 and IP4 are port IP addresses of the VRRP routers). On the router side, links connected to the RNC can also be detected by starting SBFD. Therefore, the router can perform a route switchover once a link becomes faulty. The RNC can detect faults on links connected to IP2 as well and then determine whether to switch over the port that is used for connecting the router.

7.5 IP Path Ping Detection

7.5.1 Introduction

Path ping detection provides the end-to-end fault detection function for links over the Abis, A, Iub, Iu, and Iur interfaces. Path ping detection is enabled when the ***PATHCHK*** parameter is set to **ENABLED**. Path ping detection can be classified into two types: UDP ping detection and ICMP ping detection. Users can specify the type by setting the ***CHECKT*** parameter.

UDP ping detection is supported only by the Abis and Iub interfaces.

Path ping is supported only by the BSC6900/BSC6910.

7.5.2 ICMP Ping Detection

Unlike **7.3 ICMP Ping Detection**, ICMP ping in path ping detection is automatically performed.

ICMP ping detection requires that peer devices support ICMP ping response packets and the bearer network should not filter out the response packets. However, peer devices may regard ICMP ping packets as attack packets and discard them. As a result, the base station controller considers that the transmission link is faulty when it fails to receive ICMP packet response packets. The ICMP ping detection process is as follows:

- The base station controller periodically sends ICMP request packets to the peer device. If the peer device returns a response packet to the base station controller, the base station controller considers that the transmission link is working properly.
- If the base station controller fails to receive any response packets from the peer device for N consecutive periods (N is specified by the ***CHECKCOUNT*** parameter), it considers that the transmission link is faulty. Then, if the base station controller receives any correct ICMP response packet, it determines that the transmission link is recovered.

On a transmission link that is working properly, the ICMP ping detection packets may be discarded by a device as attack packets during transmission. Therefore, a path ping detection failure does not necessarily indicate that the transmission link is faulty. When path ping detection fails on the RNC side:

- If the path is over the Iu-CS or Iu-PS interface, the RNC decreases the admission priority of the IP path. The RNC does not isolate it or release the accessed calls carried on it. If there are no other IP paths available, the IP path is still used for admission.
- If the IP path is over the Iub interface, calls will not be admitted on this IP path. When there are no other IP paths available, the RNC rejects new calls.

On the BSC side, the ***AbisPathFaultPolicy*** and ***ApathFaultPolicy*** parameters are used to set the processing policies to be followed after a fault is detected on an IP path over the Abis or A interface. Each of these parameters has the following values:

- **PRIORLOWED**: If a fault is detected on an IP path, the path is not isolated and ongoing call services carried on the path are not released. New services will be preferentially carried on other IP paths. If no other IP path is available, new services will still be carried on the path where the fault is detected.
- **ISOLATED**: If a fault is detected on an IP path, new call services will not be carried on this path. If no other IP path is available, the BSC rejects admission of new calls.

7.5.3 UDP Ping Detection

The BSC/RNC periodically sends UDP ping detection packets to the peer device. On receiving the UDP ping detection packets, the peer device processes the packets by switching the source IP address with the destination IP address, and returns the processed UDP packets to the BSC/RNC. The BSC/RNC receives the UDP response packets and determines that the transmission link is normal. If the BSC/RNC does not receive any UDP response packet for N consecutive periods (N is specified by the ***CHECKCOUNT*** parameter), the BSC/RNC determines that the transmission link is faulty. Then, if the RNC receives any correct ICMP response packet, the BSC/RNC determines that the transmission link is recovered.

ICMP ping detection packets may be filtered by the bearer network. However, this never happens to UDP ping detection packets.

NOTE

UDP ping detection requires that peer devices support UDP ping response packets.

7.6 Trace Route

Trace route is used to determine the reachability, path maximum transmission unit (PMTU) of a network, and whether the DSCP value in the packet has been changed.

The base station, base station controller, and the intermediate devices support trace route.

NOTE

Trace route can be used only when the transport network can send ICMP packets to the transmit end and the firewall in the transport network does not block ICMP packets or trace route packets.

7.6.1 Route Reachability Detection

Trace route IP packets pass through all routers on the path. If a router is faulty, no ICMP response packet will be returned. The transmit end then knows which IP address is not reachable.

Trace route detects the reachability of a network based on the TTL value of the trace route IP packet: The transmit end sends a trace route IP packet with a TTL value. Each router decreases the TTL value of the IP packet by 1 when routing the packet. If the TTL value becomes 0 at a router, the router discards this packet and returns an ICMP error message "ICMP Time Exceeded" to the transmit end. The transmit end sends a trace route IP packet with a TTL value of 1 to the network and increases the TTL value by 1 after it receives the ICMP response packet. In this way, route reachability can be determined. The maximum TTL value is specified by the **TTLMAX** parameter.

Continuous route reachability detection can be performed to detect whether routers are continuously reachable on a network. That is, after completing a tracert detection, a base station resends a trace route IP packet with a TTL value of 1 and proceeds with the preceding steps cyclically. Users can press **CTRL+Q** on the keyboard or run the **STP PATHCHK** command to stop the continuous route reachability detection. This function is controlled by the **CONTRACERT** parameter.

7.6.2 PMTU Detection

The process of PMTU detection is as follows:

1. The RAN sends trace route packets to the network. The MTU for the trace route packets is the same as that configured for the RAN device port. The **DF** bit is set to **1**, indicating that routers are not allowed to segment these packets.
2. If the MTU for a router is smaller than the size of the trace route packet, the router sends an ICMP packet to the RAN device. This notifies the RAN device that the trace route packet is excessively large.
3. The RAN device then decreases the packet size and retransmits the packet. If the packet can be transmitted, the RAN device repeatedly adjusts the packet size and retransmits the packet until the PMTU is determined.

For the base station controller, PMTU detection can also be used to detect whether large packets are discarded unidirectionally from the base station controller to the peer end. The length of detection packets ranges from 46 bytes to 1500 bytes.

For example, the following command output shows that only the packets of less than or equal to 300 bytes can pass through the path to the IP address 20.20.20.152, and the packets of more than 300 bytes are discarded.

```
traceroute to 20.20.20.152(20.20.20.152) 3 hops max,outgoing MTU = 1500
1 165.143.0.254 6 ms  4 ms  4 ms
2 165.143.0.254 7 ms  5 ms  3 ms
3 165.143.0.254 8 ms
Fragmentation required and DF set,next hop MTU = 300.
3 199.199.10.10 70 ms  64 ms  68 ms
3 20.20.20.152 75 ms  65 ms  69 ms
Path MTU detection success. The path to 20.20.20.152 is 300 Bytes.
```

To enable PMTU detection:

- On the base station side

  Run the **TRACERT** command with ***DF*** set to **ON**.
- On the base station controller side

  When the BSC6900/BSC6910 is used:

  Run the **TRC IPADDR** command with ***DETECTMTU*** set to **ENABLE**.

  When the BSC6960 is used:

Run the **TRACERT** command with ***DF*** set to **ON**.

7.6.3 DSCP Change Detection

The process of DSCP change detection is similar to that of route reachability detection:

1. The RAN device sends a trace route IP packet with an initial TTL value of 1 to the network. Each time the IP packet passes through a router, the TTL value of this packet decreases by 1. When the value of TTL decreases to 0, the router discards the IP packet and sends an ICMP message that contains the header of the received IP packet to notify the transmit end that TTL expires. In this way, the RAN device learns the DSCP value received by the router. The RAN device repeatedly sends the trace route IP packets
2. The RAN device repeatedly sends the trace route IP packets with an increasing TTL value until all routers have returned a DSCP value.
3. By checking for inconsistencies in all received DSCP values, users can learn whether the DSCP value has been changed, and if so, which router has changed the DSCP value.

To enable DSCP change detection:

- On the base station side

  Run the **TRACERT** command with ***PATHDSCPSW*** set to **ON**.

- On the base station controller side

  When the BSC6900/BSC6910 is used:

  Run the **TRC IPADDR** command with ***DSCPDETECTSW*** set to **ON**.

  When the BSC6960 is used:

  Run the **TRACERT** command with ***PATHDSCPSW*** set to **ON**.

7.7 IP Link QoS Detection

IP link QoS detection is an end-to-end quality measurement mechanism provided by Huawei devices to detect the quality of a transmission link. The measurement result can be used as a reference for network planning, deployment, operation, and upgrades.

IP link QoS detection can be performed on the Abis/Iub and S1/X2/eX2/NG/Xn/eXn interfaces. IP link QoS detection methods include UDP packet injection, IP PM, and TWAMP.

This document describes only UDP packet injection. For information about IP PM, see *IP Performance Monitor*. For information about TWAMP, see *IP Active Performance Measurement*.

For the Abis and Iub interfaces, UDP packet injection and loopback are performed on the base station controller and base station, respectively. For the S1/X2/eX2/NG/Xn/eXn interfaces, loopback is performed on the peer end, and packet injection on the local end is remotely started on the MAE.

UDP Packet Injection

The base station controller/eNodeB/gNodeB supports UDP packet injection from the local end to the peer NE. Through loopback at the peer NE, the intermediate link quality indicators, such as round-trip time, jitter, and packet loss rate, are obtained. The peer NE must support UDP loopback.

By means of UDP packet injection, counters such as the packet loss rate, received bytes, and delay can be measured. UDP packet injection has the following functions and requirements:

- Packet injection

  When there is no traffic, one end can transmit simulated packets and then they are processed in the same way as real packets. Based on measurement results, the link quality can be estimated.

- Loopback

  Loopback is performed on received packets that meet the predefined conditions.

  – When loopback is enabled at a port on a transmission board, the attributes of packets to be looped back must be specified. The attributes include the source and destination addresses, source and destination port numbers, DSCP value, and protocol type.
  – After receiving a packet with the specified attributes, the port exchanges the source and destination addresses, modifies the packet header (such as the TTL field), and returns the packet. The loopback packet is processed only by the port, not by the application layer.
  – You can evaluate the link QoS based on the statistics about transmitted and received packets. For example, the port can obtain the packet loss rate based on the number of transmitted packets and the number of received packets.

  Loopback is configured on the peer NE. If the peer NE is an eNodeB/gNodeB, run the **SET UDPLOOP** command to configure loopback.

- Statistics collection

  During link quality measurement, both ends measure the number of received packets, number of transmitted packets, throughput, and duration. Currently, statistics are collected on the side that injects packets. The statistical results are viewed on the MAE.

Comparison Among Three Detection Methods

Table 7-2 describes the application scenarios, advantages, and disadvantages of the three detection methods.

Table 7-2 Comparison among three detection methods

Method	Application Scenario	Comparison
IP PM	Routine O&M	• IP PM uses Huawei-proprietary protocols and requires Huawei equipment to be used on both ends of a measurement. This detection method is implemented by collecting traffic data on the live network and therefore requires no packet injection during a measurement, which does not affect the ongoing services and can be used for performance monitoring for a long period of time. • IP PM is recommended if detection is conducted between Huawei equipment, for example when the detection is performed between two base stations, between a base station and a base station controller, or between a base station and core network equipment.
TWAMP	• Site deployment • Routine O&M	• TWAMP uses RFC protocols and therefore supports measurement conducted between multi-vendor devices. • TWAMP requires packet injection during a measurement and therefore affects ongoing services though it is independent of services. • TWAMP measures the packet loss rate of measurement packets and IP PM measures the packet loss rate of service packets. Therefore, IP PM offers a more accurate packet loss result. • TWAMP measures the delay and jitter of measurement packets and IP PM measures the delay and jitter of control packets.
UDP packet injection	Site deployment	• UDP packet injection uses Huawei-proprietary protocols and requires Huawei equipment to be used on both ends of a measurement. This detection method is implemented by injecting UDP packets into the live network, which affects ongoing services. This detection method allows UDP packets of a larger size to be injected compared with TWAMP. • The detection result is not as accurate as IP PM and TWAMP.

7.8 GPM Congestion Detection

7.8.1 Introduction

General Performance Management (GPM) congestion detection over the Iub interface is used to discover congestion over end-to-end IP links.

This feature offers operators reference for transmission bandwidth expansion and transmission troubleshooting.

It also provides the basis for end-to-end dynamic flow control. This prevents bit-error caused packet loss from falsely triggering flow control and rate reduction, ensuring the high transmission resource usage.

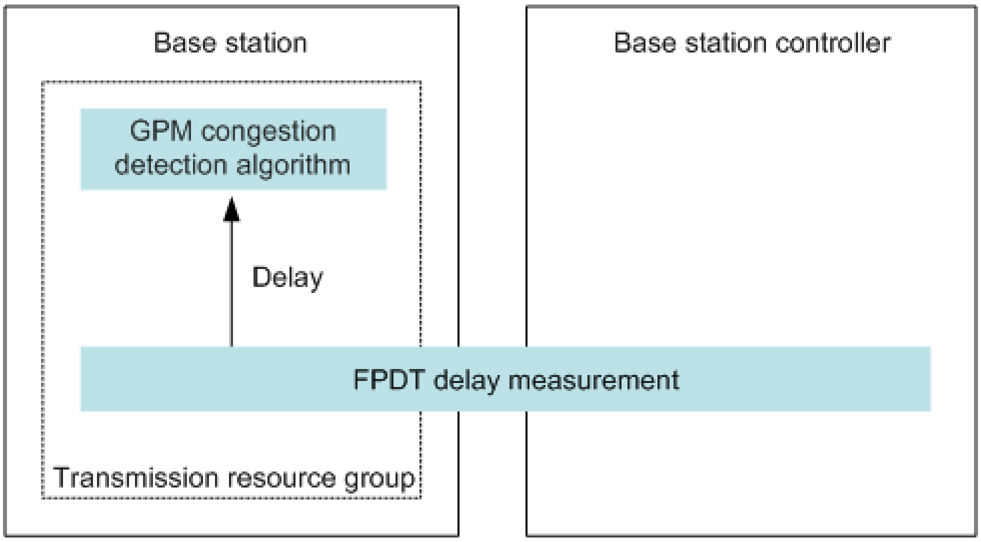
7.8.2 Principle

GPM congestion detection works on a basis of transmission resource group to detect congestion over end-to-end IP transmission networks, regardless of whether bit errors occur during data transmission.

GPM congestion detection is implemented on the base station side. The GPM congestion detection algorithm is introduced to measure the delay variations over the IP transmission links. This algorithm performs congestion detection separately on uplink and downlink IP transmission links. The GPM congestion detection algorithm applies only to HSUPA and HSDPA services.

Figure 7-5 illustrates the architecture of GPM congestion detection.

Figure 7-5 Architecture of GPM congestion detection



The GPM congestion detection algorithm uses the delay analysis results as its input to detect congestion. The NodeB uses frame protocol detection (FPDT) to perform GPM congestion detection. Specifically, the NodeB uses time information in frame protocol (FP) frames for HSDPA or HSUPA services to analyze the delay over the IP links that carry those services. To implement GPM congestion detection, the Iub transmission network must carry the following HSUPA or HSDPA services:

- Interactive or background services for uplink GPM congestion detection
- Interactive, background, or streaming services for downlink GPM congestion detection

If no such services are carried on the network, the working state of the GPM congestion detection is **NO_MEASUREMENT**.

NOTE

The working state of the GPM congestion detection is occasionally NO_MEASUREMENT due to discontinuous HSUPA or HSDPA service traffic over the Iub transmission network, which may occur in the following two scenarios:

- The IP link selected for GPM congestion detection is lightly loaded.
- The packet loss on the IP link selected for GPM congestion detection is high to the extent that the TCP decreasing algorithm is triggered, causing light traffic over the IP link.

The GPM congestion detection configurations are as follows:

- **Detect Object** is set to **RSCGRP** (in the old model)/**IPRSCGRP** (in the new model).
- **Detect Direction** is set to **UP** or **DOWN**.
- **Delay Detect Method** is set to **FPDT**.

7.9 GTP-U Echo

The RNC, eNodeB, and gNodeB support GTP-U echo.

GTP-U echo for the eNodeB and gNodeB uses GTP-U control packets to monitor S1/X2/NG/Xn/eXn user-plane paths at the application layer. GTP-U echo for the RNC detects faults on Iu-PS paths.

GTP-U Echo (RNC)

When GTP-U echo is enabled on the RNC (**Echo Detection Switch** is **ON** in the **LST UGTPU** command output), the Iu-PS interface board sends a GTP-U echo request message once every minute. If no response is returned within a period of time (**Path Timeout Threshold** in the **LST UGTPU** command output), the transmit end sends the message again. If the number of message retransmissions exceeds the maximum number (**Maximum Path Timeouts** in the **LST UGTPU** command output), the RNC considers the path to the SGSN faulty. If all channels for PS user data transmission on the Iu-PS interface boards are detected as faulty for two consecutive times, the BSC6900/BSC6910 generates ALM-22510 GTPU Faulty and the BSC6960 generates ALM-21697 GTPU Faulty, and reports this fault to the service layer. Then the service layer releases services.

NOTE

For UMTS BSC6900, if the **PTT_RAB_SETUP_ATT_CNT_CORRECT_SWITCH** option of the **URRCTRLSWITCH**.***ImprovementSwitch*** parameter is selected, the service layer does not release services. It is recommended that this option be selected when the Iu-PS interface is configured with load sharing routes.

GTP-U Echo (eNodeB and gNodeB)

GTP-U echo can be static or dynamic for the eNodeB and gNodeB.

- When static GTP-U echo is enabled, it is performed regardless of whether there are services. In endpoint configuration mode, the base station reports ALM-25954 User Plane Fault after detecting a fault, and reports the fault to the service layer.
  - In link configuration mode, static GTP-U echo has a global switch and a link-level switch.
    - The global switch is configured using the **GTPU**.***STATICCHK (LTE eNodeB, 5G gNodeB)*** parameter.
    - The link-level switch is configured using the **IPPATH**.***STATICCHK (LTE eNodeB, 5G gNodeB)*** (IPv4) parameter.

    If the link-level switch is not set to **FOLLOW_GLOBAL**, the link-level switch setting is used. If the link-level switch is set to **FOLLOW_GLOBAL**, the global switch setting is used. By default, the link-level switch setting is the same as the global switch setting.
  - In endpoint configuration mode, static GTP-U echo can be configured using the global switch, link-level switch, or endpoint group switch.
    - The global switch is configured using the **GTPU**.***STATICCHK (LTE eNodeB, 5G gNodeB)*** parameter.

      The link-level switch is configured using the **USERPLANEPEER**.***STATICCHK (LTE eNodeB, 5G gNodeB)*** parameter.
    - The endpoint group switch is configured using the **EPGROUP**.***STATICCHK (LTE eNodeB, 5G gNodeB)*** parameter.

    When **GEPMODELPARA**.***STATICCHKMODE (LTE eNodeB, 5G gNodeB)*** is set to **UPPEERSTATICCHK**, the following are true: If the link-level switch is not set to **FOLLOW_GLOBAL**, the link-level switch setting is used. If the link-level switch is set to **FOLLOW_GLOBAL**, the global switch setting is used. By default, the link-level switch setting is the same as the global switch setting.

    When **GEPMODELPARA**.***STATICCHKMODE (LTE eNodeB, 5G gNodeB)*** is set to **EPSTATICCHK**, the endpoint group switch setting is used. By default, the endpoint group switch is turned off.

  NOTE

  - If the **USERPLANEPEER** MO has been configured, the system automatically updates the configuration of this MO when static GTP-U echo is enabled, without changing the value of the **USERPLANEPEER**.***STATICCHK (LTE eNodeB, 5G gNodeB)*** parameter.
  - ALM-25954 User Plane Fault of the major severity is reported when the total number of user-plane path faults is greater than or equal to 1 for the same service type. ALM-25954 User Plane Fault of the critical severity is reported when all the IP paths and user-plane paths are faulty for S1/NG services.
  - In the simplified endpoint mode, as the **EPGROUP** MO cannot be manually configured, static GTP-U echo does not support the endpoint group configuration mode.
- Dynamic GTP-U echo takes effect only when there are services and static GTP-U echo is disabled. When a fault occurs, dynamic GTP-U echo does not

report ALM-25954 User Plane Fault (in endpoint configuration mode) but only reports the fault to the service layer.

It is recommended that static GTP-U echo be enabled. This prevents UEs from being released without any prompt information when a transmission link becomes faulty. Users can specify the timeout duration of echo frames and the number of echo frame timeouts by setting the **GTPU**.***TIMEOUTTH (LTE eNodeB, 5G gNodeB)*** and **GTPU**.***TIMEOUTCNT (LTE eNodeB, 5G gNodeB)*** parameters, respectively. The two parameters are globally configured and cannot be set to different values for individual links.

In link configuration mode, the transmission of GTP-U echo packets depends on the address pair of the IP path. That is, the transmission of GTP-U echo packets is determined by the address pair of the IP path and the DSCP value of GTP-U control packets. GTP-U echo automatically sends control packets based on the IP address pair of each IP path. If multiple IP paths have the same IP address pair but different DSCP values, GTP-U echo control packets are separately transmitted on these IP paths.

- If the type of an IP path is **FIXED**, GTP-U control packets with the same DSCP value as that of the IP path are transmitted.
- If the type of an IP path is **ANY**, the value of the **GTPU**.***DSCP*** parameter for the GTP-U control packets must be planned uniformly within the network.

In endpoint configuration mode, GTP-U echo uses the local IP address of the **USERPLANEHOST** MO and the peer IP address of the **USERPLANEPEER** MO in the same **EPGROUP** MO as the source and destination IP addresses, respectively. In this case, the DSCP values of the GTP-U control packets are uniformly configured for the entire network.

When static GTP-U echo is enabled, the mechanism of reporting X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn user-plane fault alarms is optimized to improve the accuracy of user-plane fault alarms. If a user-plane path fault is detected and there are X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn services triggering a bearer setup attempt, an X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn service alarm will be reported, enabling precise reporting of user-plane fault alarms. Precise reporting of X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn user-plane fault alarms is enabled when the **INTER_UP_ALM_PRECISE_SW** option of the **TRANSALGEXTPARA**.***TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)*** parameter is selected.

NOTE

- Precise reporting of eX2/gNBCUX2/gNBCUXn/gNBDUeXn user-plane fault alarms only works in the endpoint configuration mode, while precise reporting of X2 user-plane fault alarms works in both the link and endpoint configuration modes.
- Precise reporting of eX2 user-plane fault alarms applies only to eX2 interfaces in non-ideal backhaul mode (with **EX2**.***BackhaulType*** set to **NON_IDEAL**).

7.10 Active and Backup Link ARP Probe (BSC6960)

The BSC/RNC (BSC6960) supports active and backup link ARP probe.

The procedure for active and backup link ARP probe is as follows:

1. The function checks whether the transmit link of the standby port is normal by periodically sending ARP request packets with unicast addresses on the board where the standby port is located and receiving the packets on the board where the active port is located.
   – The source MAC address of the ARP packets is the MAC address of the standby port.
   – The destination MAC address is the MAC address of the active port.
   – The source and destination IP addresses are the local maintenance IP address of the board (set by running the **SET LOCALIP** command).
2. The function checks whether the receive link of the standby port is normal by sending ARP request packets with unicast addresses on the board where the active port is located at every 02:40 in the morning and receiving the packets on the board where the standby port is located.
   – The source MAC address of the ARP request packets is the MAC address of the active port.
   – The destination MAC address is the MAC address of the standby port.
   – The source and destination IP addresses are the local maintenance IP address of the board (set by running the **SET LOCALIP** command).
3. The base station controller detects normal link connectivity by checking whether the ARP packets have been received on the port. If the ARP packets are not received within a configured period, the VRRP networking is considered faulty.

Active and backup link ARP probe is recommended only when the interconnection networking of active/backup trunk in manual mode is VRRP. If the networking is not VRRP, this function is enabled, and ALM-21406 ARP Detection Failure is reported, you do not need to handle the alarm.

To enable active and backup link ARP probe, run the **SET GTRANSPARA** command with the following settings:

- Set ***STANDBYPORTCHKSW*** to **ON**.
- Set **ARPTIMEOUT** to **2** in the unit of second.
- Set **ARPRETRY** to **2**. If the base station controller does not receive ARP packets from the peer end for two consecutive times, ALM-21406 ARP Probe Failure is reported.

7.11 Quick Transmission Congestion Detection

7.11.1 Introduction

Quick transmission congestion detection is used to detect the end-to-end IP link status (such as congestion-caused delay and available bandwidth) over the eX2/X2 interface, and promptly and accurately detect and predict the congestion on the transmission link. The detection result can serve as a reference for bandwidth addition and fault location.

When the transmission bandwidth over the eX2/X2 interface is insufficient, the detection result provided by this function can act as the basis for dynamic flow control on coordination services, which ensures that these coordination services are distributed according to the transmission bandwidth to enhance service gains.

Quick transmission congestion detection applies only to the eX2/X2 interface in non-ideal transmission scenarios. The delay and bandwidth of the eX2/X2 interface in ideal transmission scenarios are guaranteed and do not require quick transmission congestion detection.

Quick transmission congestion detection applies only to the transmit direction of the eX2/X2 interface and the detection in the receive direction is not supported currently.

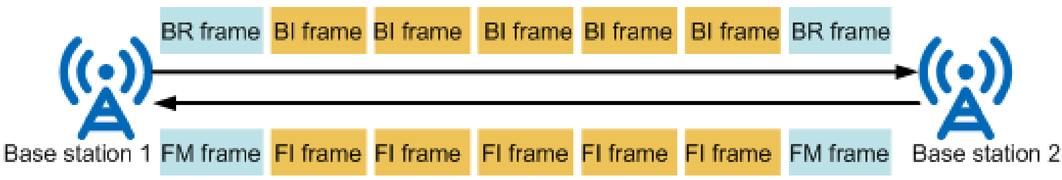
Quick transmission congestion detection depends on the IP performance monitoring (PM) feature.

Quick transmission congestion detection requires the UMPTb, UMPTe, UMPTg, UMPTga, UMPTha, UMPThb, or UMDU board. Other main control boards do not support this function.

7.11.2 Principle

Quick transmission congestion detection is an enhancement of Huawei proprietary IP PM frames. **Figure 7-6** shows the IP PM frame for quick transmission congestion detection. Without changing the original IP PM frame (including the frame sending interval and frame format definition) defined in the IP PM protocol, the forward indicator (FI) frame and backward indicator (BI) frame are added for quick transmission congestion detection. For details on the ACT, ACT-ACK, forward monitoring (FM), backward reporting (BR), DEA, and DEA-ACK frames defined in the IP PM protocol, see *IP Performance Monitor Feature Parameter Description*.

Figure 7-6 IP PM frame for quick transmission congestion detection



Quick transmission congestion detection consists of the following two phases:

- Negotiation phase

  If a local base station receives a BI frame from the peer end within 30s after sending an FI frame, it is determined that the peer end supports quick transmission congestion detection and the negotiation succeeds. If the local base station does not receive a BI frame from the peer end within 30s, it is determined that the peer end does not support quick transmission congestion detection and the negotiation fails. Then the local base station stops sending the FI frame. The cause of a failed negotiation may be that the versions do not match or the peer board does not support the detection.

- Normal detection phase

  After the negotiation succeeds, quick transmission congestion detection enters the normal detection phase. The local base station sends FI frames every 10 ms and the peer base station responds with the BI frames upon receiving the FI frames. Both the FI and BI frames contain time information. After receiving the BI frames, the local base station calculates the delay between the local and peer ends based on the time information, and performs congestion detection based on the delay.

Before the quick transmission congestion detection, IP PM must have been activated on the link to be detected because quick transmission congestion detection requires IP PM.

Quick transmission congestion detection over the eX2/X2 interface is controlled by the **TransportCongDetectSw** option of the **ENodeBAlgoSwitch**.***OverBBUsSwitch*** parameter. When this switch is turned on and the IP PM status specified by the **IPPMSESSION**.***IPPMUPACTSTATE*** parameter changes, the status of quick transmission congestion detection may be affected because it depends on IP PM. The status of quick transmission congestion detection changes with the IP PM status as follows:

- When the IP PM status is **IP PM UP**:

  The local base station performs negotiation. If the negotiation succeeds, the local base station then proceeds with the normal detection. If the negotiation fails, quick transmission congestion fails because it is not supported by the peer base station.

- When the IP PM status changes from **IP PM UP** to **IDLE**:

  If quick transmission congestion detection is in the negotiation phase, the local base station stops the detection and quick transmission congestion detection fails. That is because IP PM has not been activated.

  If quick transmission congestion detection is in the normal detection phase, the local base station stops the detection and quick transmission congestion detection fails. That is because IP PM has not been activated.

- When the IP PM status changes from **IDLE** to **IP PM UP**:

  If quick transmission congestion detection is in the negotiation phase, the local base station continues with the negotiation.

  If quick transmission congestion detection is in the failed state, the local base station restarts the negotiation.

- When the IP PM status is **IDLE**:

  The local base station does not start quick transmission congestion detection and the detection fails. That is because IP PM has not been activated.

7.12 LLDP

Link Layer Discovery Protocol (LLDP) is a layer 2 discovery protocol defined in the IEEE 802.1ab standard. It allows network devices to advertise local information in the local subnet, including the system name, system description, port ID, and MAC address. The OSS can use LLDP to quickly obtain the layer 2 network topology information and topology changes and to further obtain the topology relationship between local and peer devices. With this information, transmission faults can be quickly located.

Both the base station controller and base station support LLDP.

Application on the Base Station Controller (GSM/UMTS)

When the BSC6900/BSC6910 is used:

The **ADD LLDPPORT** command is used for adding the local end information of LLDP ports.

- If the peer device is configured with a VLAN, the **LLDPPORT**.***BNDVLAN*** parameter that specifies the VLAN information bound to the LLDP local port must be set to **YES**. The **LLDPPORT**.***VLANID*** parameter that specifies the VLAN ID and the **LLDPPORT**.***VLANPRI*** parameter that specifies the VLAN priority must be set based on the configuration on the peer device.
- If the peer device is not configured with a VLAN, **LLDPPORT**.***BNDVLAN*** must be set to **NO**.

The **SET LLDPGLOBALINFO** command is used to set the LLDP global configuration information, including the interval for transmitting LLDP packets (**LLDPGLOBALINFO**.***TXINTVAL***), LLDP TTL multiplier (**LLDPGLOBALINFO**.***HOLDMULTI***), reinitialization delay (**LLDPGLOBALINFO**.***REINITDELAY***), and LLDP packet transmission delay (**LLDPGLOBALINFO**.***DELAY***).

Users can run the **ADD LLDP** command to add the local end information of an LLDP port.

- If the peer device is configured with a VLAN, the **LLDP**.***BNDVLAN*** parameter that specifies the VLAN information bound to the LLDP local port must be set to **YES**. The **LLDP**.***VLANID*** parameter that specifies the VLAN ID and the **LLDP**.***VLANPRI*** parameter that specifies the VLAN priority must be set based on the configuration on the peer device.
- If the peer device is not configured with a VLAN, **LLDP**.***BNDVLAN*** must be set to **NO**.

The **SET LLDPGLOBALINFO** command is used to set the LLDP global configuration information, including the interval for transmitting LLDP packets (**LLDPGLOBAL**.***TXINTVAL***), LLDP TTL multiplier (**LLDPGLOBAL**.***HOLDMULTI***), reinitialization delay (**LLDPGLOBAL**.***REINITDELAY***), and LLDP packet transmission delay (**LLDPGLOBAL**.***DELAY***).

When the **ETHTRUNK**.***INTERBOARDFLAG*** parameter is set to **YES**, the Ethernet trunk supports inter-board connections. The BSC6960 does not support LLDP.

Application on the Base Station (GSM/UMTS/LTE/NR)

The base station supports manual and automatic LLDP configurations. The LLDP configuration mode can be specified by setting **LLDPGLOBAL**.***PORTCFGMODE (LTE eNodeB, 5G gNodeB)*** to **MANUAL** or **AUTO**.

- Manual LLDP configuration

  Users can run the **ADD LLDPPORT** (old model)/**ADD LLDP** (new model) command to add the local end information of an LLDP port.

  – If the peer device is configured with a VLAN, **LLDPLOCAL**.***BNDVLAN (LTE eNodeB, 5G gNodeB)*** (old model)/**LLDP**.***BNDVLAN (LTE eNodeB, 5G gNodeB)*** (new model) must be set to **YES**, and **LLDPLOCAL**.***VLANID (LTE eNodeB, 5G gNodeB)*** (old model)/**LLDP**.***VLANID (LTE eNodeB, 5G***

gNodeB) (new model) as well as **LLDPLOCAL**.***VLANPRI (LTE eNodeB, 5G gNodeB)*** (old model)/**LLDP**.***VLANPRI (LTE eNodeB, 5G gNodeB)*** (new model) must be set based on the configuration on the peer device.

  – If the peer device is not configured with a VLAN, **LLDPLOCAL**.***BNDVLAN (LTE eNodeB, 5G gNodeB)*** (old model)/**LLDP**.***BNDVLAN (LTE eNodeB, 5G gNodeB)*** (new model) must be set to **NO**.

- Automatic LLDP configuration

  The base station automatically configures LLDP port information based on the configured Ethernet port. The automatically configured LLDP port is not bound to any VLAN.

The **SET LLDPGLOBALINFO** command is used to set the LLDP global configuration information, including the interval for transmitting LLDP packets (**LLDPGLOBAL**.***TXINTVAL (LTE eNodeB, 5G gNodeB)***), LLDP TTL multiplier (**LLDPGLOBAL**.***HOLDMULTI (LTE eNodeB, 5G gNodeB)***), reinitialization delay (**LLDPGLOBAL**.***REINITDELAY (LTE eNodeB, 5G gNodeB)***), LLDP packet transmission delay (**LLDPGLOBAL**.***DELAY (LTE eNodeB, 5G gNodeB)***), and base station and microwave collaboration switch (**LLDPGLOBAL**.***BTSMWCOSW (LTE eNodeB, 5G gNodeB)***). The **BKP_PSAV_EVALUATE_DATA_SW** option of the **LLDPGLOBAL**.***BTSMWCOSW (LTE eNodeB, 5G gNodeB)*** parameter specifies whether the base station transmits its backup power saving data to the microwave device. If this option is selected and the microwave device's backup power subscription switch is turned on, the base station sends the obtained backup power saving data to the microwave device through LLDP packets.

8 Related Features

8.1 GBFD-118601 Abis over IP

Prerequisite Features

None

Mutually Exclusive Features

RAT	Feature ID	Feature Name
GSM	GBFD-117801	Ring Topology
GSM	GBFD-117301	Flex Abis
GSM	GBFD-116601	Abis Bypass
GSM	GBFD-116701	16Kbit RSL and OML on A-bis Interface
GSM UMTS	MRFD-210206	Tree Topology
GSM UMTS	MRFD-210205	Chain Topology
GSM	GBFD-113729	Adaptive Transmission Link Blocking
GSM	GBFD-119301	Voice Fault Diagnosis

Impacted Features

None

8.2 GBFD-118611 Abis IP over E1/T1

Prerequisite Features

None

Mutually Exclusive Features

RAT	Feature ID	Feature Name
GSM	GBFD-117801	Ring Topology
GSM	GBFD-117301	Flex Abis
GSM	GBFD-116701	16Kbit RSL and OML on A-bis Interface
GSM	GBFD-119301	Voice Fault Diagnosis

Impacted Features

None

8.3 GBFD-118602 A over IP

Prerequisite Features

None

Mutually Exclusive Features

RAT	Feature ID	Feature Name
GSM	GBFD-115601	Automatic Level Control (ALC)
GSM	GBFD-115602	Acoustic Echo Cancellation (AEC)
GSM	GBFD-115603	Automatic Noise Restraint (ANR)
GSM	GBFD-115703	Automatic Noise Compensation (ANC)
GSM	GBFD-115704	Enhancement Packet Loss Concealment (EPLC)
GSM	GBFD-115701	TFO
GSM	GBFD-117701	BSC Local Switch
GSM	GBFD-117702	BTS Local Switch
GSM	GBFD-115711	EVAD

Impacted Features

None

8.4 GBFD-118622 A IP over E1/T1

Prerequisite Features

None

Mutually Exclusive Features

RAT	Feature ID	Feature Name
GSM	GBFD-115601	Automatic Level Control (ALC)
GSM	GBFD-115602	Acoustic Echo Cancellation (AEC)
GSM	GBFD-115603	Automatic Noise Restraint (ANR)
GSM	GBFD-115703	Automatic Noise Compensation (ANC)
GSM	GBFD-115704	Enhancement Packet Loss Concealment (EPLC)
GSM	GBFD-115701	TFO

Impacted Features

None

8.5 GBFD-118603 Gb over IP

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.6 WRFD-050402 IP Transmission Introduction on Iub Interface

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.7 WRFD-050409 IP Transmission Introduction on Iu Interface

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.8 WRFD-050410 IP Transmission Introduction on Iur Interface

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.9 WRFD-050411 Fractional IP Function on Iub Interface

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.10 LBFD-003003/MLBFD-003003/TDLBFD-003003 VLAN Support (IEEE 802.1p/q)

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.11 LBFD-003007/TDLBFD-003007/MLBFD-12000308 IP Route Backup

Prerequisite Features

RAT	Feature ID	Feature Name
LTE FDD	LOFD-003007	Bidirectional Forwarding Detection
LTE TDD	TDLOFD-003007	Bidirectional Forwarding Detection
NB-IoT	MLOFD-003007	Bidirectional Forwarding Detection

Mutually Exclusive Features

None

Impacted Features

None

8.12 MRFD-210103 Link Aggregation

Prerequisite Features

RAT	Feature ID	Feature Name
GSM	GBFD-118601	Abis over IP
GSM	GBFD-118602	A over IP
GSM	GBFD-118603	Gb over IP
UMTS	WRFD-050402	IP Transmission Introduction on Iub Interface
UMTS	WRFD-050409	IP Transmission Introduction on Iu Interface
UMTS	WRFD-050410	IP Transmission Introduction on Iur Interface

Mutually Exclusive Features

None

Impacted Features

None

8.13 LOFD-003008/MLOFD-003008/TDLOFD-003008 Ethernet Link Aggregation (IEEE 802.3ad)

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.14 LBFD-00202103/MLBFD-00202103/TDLBFD-00202103 SCTP Multi-homing

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.15 LOFD-003005/TDLOFD-003005/MLOFD-003005 OM Channel Backup

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.16 LOFD-003007/TDLOFD-003007/MLOFD-003007 Bidirectional Forwarding Detection

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.17 Bidirectional Forwarding Detection (GSM/UMTS)

Prerequisite Features

RAT	Feature ID	Feature Name	Description
GSM	GBFD-118601	Abis over IP	Applied to the eGBTS/ GBTS
UMTS	WRFD-050402	IP Transmission Introduction on Iub Interface	Applied to the NodeB
GSM	GBFD-118601	Abis over IP	Applied to the Abis interface of the BSC6900
GSM	GBFD-118602	A over IP	Applied to the A interface of the BSC6900
GSM	GBFD-118603	Gb over IP	Applied to the Gb interface of the BSC6900
UMTS	WRFD-050402	IP Transmission Introduction on Iub Interface	Applied to the Iub interface of the BSC6900
UMTS	WRFD-050409	IP Transmission Introduction on Iu Interface	Applied to the Iu interface of the BSC6900
UMTS	WRFD-050410	IP Transmission Introduction on Iur Interface	Applied to the Iur interface of the BSC6900
UMTS	WRFD-140207	Iu/Iur Transmission Resource Pool in RNC	Applied to the BSC6900 in IP pool mode
UMTS	WRFD-140208	Iub Transmission Resource Pool in RNC	
GSM	GBFD-150201	A over IP Based on Dynamic Load Balancing	Applied to the A interface of the BSC6910/BSC6960
UMTS	WRFD-150243	Iub IP Transmission Based on Dynamic Load Balancing	Applied to the Iub interface of the BSC6910/ BSC6960

RAT	Feature ID	Feature Name	Description
UMTS	WRFD-150244	Iu/Iur IP Transmission Based on Dynamic Load Balancing	Applied to the Iu/Iur interface of the BSC6910/ BSC6960

Mutually Exclusive Features

None

Impacted Features

Function Name	Impact
Single-Hop BFD	If an SBFD session is shut down while it is running, the base station or base station controller automatically disables the routes whose next-hop IP address is the peer IP address of the failed SBFD session. If SBFD is configured to be applicable only to the active port on the base station controller side and the ***WHETHERAFFECTSWAP*** parameter is set to **YES(YES)**, an SBFD link fault triggers a port switchover. Otherwise, an SBFD link fault does not trigger a port switchover, but the availability of the related routes is affected. After the binding relationship between an MBFD session and an IP route (defined by **IPRT** in the old model/**IPROUTE4** in the new model) or an IP path (defined by **IPPATH**) is specified, the bound IP route or IP path is considered faulty if the MBFD session detects a fault.

8.18 ARP Probe

Prerequisite Features

When this feature is used on the eGBTS/GBTS, the GBFD-118601 Abis over IP feature is required.

When this feature is used on the NodeB, the WRFD-050402 Iub over IP feature is required.

When this feature is used on the eNodeB, no features are required.

When this feature is used on the BSC6900, the following features are required:

- GBFD-118601 Abis over IP

- GBFD-118602 A over IP
- GBFD-118603 Gb over IP
- WRFD-050402 IP Transmission Introduction on Iub Interface
- WRFD-050409 IP Transmission Introduction on Iu Interface
- WRFD-050410 IP Transmission Introduction on Iur Interface

When this feature is used on the BSC6900 in IP pool mode, the features WRFD-140207 Iu/Iur Transmission Resource Pool of RNC and WRFD-140208 Iub Transmission Resource Pool of RNC are also required.

When this feature is used on the BSC6910/BSC6960, the following features are required:

- GBFD-150201 A over IP Based on Dynamic Load Balancing
- WRFD-150243 Iub IP Transmission Based on Dynamic Load Balancing
- WRFD-150244 Iu/Iur IP Transmission Based on Dynamic Load Balancing

Mutually Exclusive Features

None

Impacted Features

If the ARP probe fails, the system automatically isolates the route whose next hop is the peer IP address. If load-balancing routes or alternative routes are configured, route switchover is performed.

8.19 ICMP

Prerequisite Features

When this feature is used on the eGBTS/GBTS, the GBFD-118601 Abis over IP feature is required.

When this feature is used on the NodeB, the WRFD-050402 Iub over IP feature is required.

When this feature is used on the eNodeB, no features are required.

When this feature is used on the BSC6900, the following features are required:

- GBFD-118601 Abis over IP
- GBFD-118602 A over IP
- GBFD-118603 Gb over IP
- WRFD-050402 IP Transmission Introduction on Iub Interface
- WRFD-050409 IP Transmission Introduction on Iu Interface
- WRFD-050410 IP Transmission Introduction on Iur Interface

When this feature is used on the BSC6900 in IP pool mode, the features WRFD-140207 Iu/Iur Transmission Resource Pool of RNC and WRFD-140208 Iub Transmission Resource Pool of RNC are also required.

When this feature is used on the BSC6910/BSC6960, the following features are required:

- GBFD-150201 A over IP Based on Dynamic Load Balancing
- WRFD-150243 Iub IP Transmission Based on Dynamic Load Balancing
- WRFD-150244 Iu/Iur IP Transmission Based on Dynamic Load Balancing

Mutually Exclusive Features

None

Impacted Features

None

8.20 IP Path Ping Detection

Prerequisite Features

- GBFD-118601 Abis over IP
- GBFD-118602 A over IP
- WRFD-050402 IP Transmission Introduction on Iub Interface
- WRFD-050409 IP Transmission Introduction on Iu Interface
- WRFD-050410 IP Transmission Introduction on Iur Interface

Mutually Exclusive Features

None

Impacted Features

None

8.21 Trace Route

Prerequisite Features

When this feature is used on the eGBTS/GBTS, the GBFD-118601 Abis over IP feature is required.

When this feature is used on the NodeB, the WRFD-050402 Iub over IP feature is required.

When this feature is used on the eNodeB, no features are required.

When this feature is used on the BSC6900, the following features are required:

- GBFD-118601 Abis over IP
- GBFD-118602 A over IP

- GBFD-118603 Gb over IP
- WRFD-050402 IP Transmission Introduction on Iub Interface
- WRFD-050409 IP Transmission Introduction on Iu Interface
- WRFD-050410 IP Transmission Introduction on Iur Interface

When this feature is used on the BSC6900 in IP pool mode, the features WRFD-140207 Iu/Iur Transmission Resource Pool of RNC and WRFD-140208 Iub Transmission Resource Pool of RNC are also required.

When this feature is used on the BSC6910/BSC6960, the following features are required:

- GBFD-150201 A over IP Based on Dynamic Load Balancing
- WRFD-150243 Iub IP Transmission Based on Dynamic Load Balancing
- WRFD-150244 Iu/Iur IP Transmission Based on Dynamic Load Balancing

Mutually Exclusive Features

This function cannot be used together with the **PING**, **CFMTRACE**, or **CFMPING** command.

Impacted Features

None

8.22 IP Link QoS Detection

Prerequisite Features

When this feature is used on the eNodeB, no features are required.

When this feature is applied to the Abis interface, GBFD-118601 Abis over IP is required.

When this feature is applied to the Iub interface, WRFD-050402 IP Transmission Introduction on Iub Interface is required.

When this feature is used on the BSC6900 in IP pool mode, the WRFD-140208 Iub Transmission Resource Pool of RNC feature is required. When this feature is used on the BSC6910/BSC6960 in IP pool mode, the WRFD-150243 Iub IP Transmission Based on Dynamic Load Balancing feature is required.

Mutually Exclusive Features

This function cannot be used together with the **PING**, **CFMTRACE**, or **CFMPING** command.

Impacted Features

None

8.23 GPM Congestion Detection

Prerequisite Features

- WRFD-050402 IP Transmission Introduction on Iub Interface
- Downlink GPM congestion detection: WRFD-01061008 Interactive and Background Traffic Class on HSDPA or WRFD-010630 Streaming Traffic Class on HSDPA
- Uplink GPM congestion detection: WRFD-01061206 Interactive and Background Traffic Class on HSUPA

Mutually Exclusive Features

- WRFD-050420 FP MUX for IP Transmission
- PPP MUX

NOTE

When this feature is used with FP MUX for IP Transmission, the length of the timer for FP MUX must be less than or equal to 1 ms.

This feature cannot be used together with PPP MUX.

Impacted Features

None

8.24 GTP-U Echo

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8.25 Source-based Routing on the Base Station Side

Prerequisite Features

When this feature is used on the eGBTS, the GBFD-118601 Abis over IP feature is required.

When this feature is used on the NodeB, the following features are required: WRFD-050402 Iub over IP, WRFD-140208 Iub Transmission Resource Pool of RNC, and WRFD-150243 Iub IP Transmission Based on Dynamic Load Balancing.

When this feature is used on the eNodeB, no features are required.

Mutually Exclusive Features

- DHCP sub-feature of the WRFD-031101 NodeB Self-discovery Based on IP Mode feature
- E1 Backup
- WRFD-050301 ATM Transmission Introduction Package
- WRFD-050404 ATM/IP Dual Stack NodeB
- IEEE802.1ag binding route, multi-hop binding route, OM channel binding route, and MTU on the base station side

Impacted Features

On a separate-MPT base station, data is exchanged between two RATs through the backplane tunnel.

If the main control board not providing the co-transmission port obtains its IP address through DHCP, DHCP relay must be configured on the main control board providing the co-transmission port. If the main control board providing the co-transmission port uses the panel port IP address as its DHCP relay IP address and the source-based routing is configured on the base station side, the DHCP relay IP address must be manually configured.

8.26 Active and Backup Link ARP Probe

When this feature is used on the BSC6900, the following features are required:

- GBFD-118601 Abis over IP
- GBFD-118602 A over IP
- GBFD-118603 Gb over IP
- WRFD-050402 IP Transmission Introduction on Iub Interface
- WRFD-050409 IP Transmission Introduction on Iu Interface
- WRFD-050410 IP Transmission Introduction on Iur Interface

When this feature is used on the BSC6910/BSC6960, the following features are required:

- GBFD-150201 A over IP Based on Dynamic Load Balancing
- WRFD-150243 Iub IP Transmission Based on Dynamic Load Balancing
- WRFD-150244 Iu/Iur IP Transmission Based on Dynamic Load Balancing

8.27 Quick Transmission Congestion Detection

Prerequisite Features

RAT	Feature ID	Feature Name	Description
LTE FDD	LBFD-00301201	IP Performance Monitor	The IP PM frame is required by quick delay measurement that depends on IP performance monitoring.
LTE TDD	TDLBFD-00301201	IP Performance Monitoring	
NB-IoT	MLBFD-12100309	IP Performance Monitoring	

Mutually Exclusive Features

RAT	Feature ID	Feature Name	Description
LTE FDD	LBFD-00300402	PPP MUX	PPP MUX introduces the multiplexing packet waiting delay and the waiting delays of different packets may vary. As a result, the packet delay variation increases. However, the basis of quick transmission congestion detection is to detect packet delay changes. Therefore, PPP MUX may affect the accuracy of quick transmission congestion detection results. The maximum multiplexing packet waiting delay is 10 ms by default, introducing a maximum packet delay variation of 10 ms. This greatly affects quick transmission congestion detection. Therefore, this function is not recommended when PPP MUX is enabled.
LTE TDD	TDLBFD-003004	PPP MUX	

Impacted Features

None

8.28 LLDP

Prerequisite Features

When this feature is used on the BSC6900, the following features are required:

- GBFD-118601 Abis over IP
- GBFD-118602 A over IP
- GBFD-118603 Gb over IP
- WRFD-050402 IP Transmission Introduction on Iub Interface
- WRFD-050409 IP Transmission Introduction on Iu Interface
- WRFD-050410 IP Transmission Introduction on Iur Interface

When this feature is used on the BSC6910/BSC6960, the following features are required:

- GBFD-118601 Abis over IP
- GBFD-150201 A over IP Based on Dynamic Load Balancing
- GBFD-118603 Gb over IP
- WRFD-150243 Iub IP Transmission Based on Dynamic Load Balancing
- WRFD-150244 Iu/Iur IP Transmission Based on Dynamic Load Balancing

Mutually Exclusive Features

None

Impacted Features

None

8.29 VLAN Priority Mapping

Prerequisite Features

None

Mutually Exclusive Features

RAT	Feature ID	Feature Name	Description
LTE TDD	TDLBFD-001134	VRF	A non-default VRF instance (with the **VRF**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter not set to **0**) cannot coexist with a VLAN priority mapping (configured using the **VLANCLASS** MO).

Impacted Features

None

8.30 PPP Link

Prerequisite Features

None

Mutually Exclusive Features

RAT	Feature ID	Feature Name	Description
LTE TDD	TDLBFD-001134	VRF	A non-default VRF instance (with the **VRF**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter not set to **0**) cannot coexist with a PPP link (configured using the **PPPLNK** MO).

Impacted Features

None

8.31 MLPPP Group

Prerequisite Features

None

Mutually Exclusive Features

RAT	Feature ID	Feature Name	Description
LTE TDD	TDLBFD-001134	VRF	A non-default VRF instance (with the **VRF**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter not set to **0**) cannot coexist with an MLPPP group (configured using the **MPGRP** MO).

Impacted Features

None

9 Network Impact

9.1 GBFD-118601 Abis over IP

System Capacity

No impact.

Network Performance

This feature uses as many bandwidth resources as TDM transmission, because it allows encapsulation at the physical layer, data link layer, network layer, and transport layer. This feature provides statistical multiplexing and packet multiplexing, thereby improving transmission efficiency.

9.2 GBFD-118611 Abis IP over E1/T1

System Capacity

No impact.

Network Performance

This feature uses as many bandwidth resources as TDM transmission, because it allows encapsulation at the physical layer, data link layer, network layer, and transport layer. This feature provides statistical multiplexing, voice activity detection (VAD), and packet multiplexing, thereby improving transmission efficiency.

In certain scenarios (such as sites with a relatively high percentage of half-rate services), service KPIs may deteriorate after TDM transmission is replaced by IP over E1/T1 transmission, because there are cases where the latter mode requires more bandwidth than the former one during transmission. Therefore, evaluate the transmission bandwidth required before the reconstruction to avoid service KPI deterioration.

9.3 GBFD-118602 A over IP

System Capacity

No impact.

Network Performance

Intra-BSC decoding is not required when this feature is used with the TrFO feature, improving voice quality. This feature also facilitates MSC pool networking, enhancing network reliability.

9.4 GBFD-118622 A IP over E1/T1

System Capacity

No impact.

Network Performance

This feature allows the MGW to provide TC functions. In addition, intra-BSC decoding is not required when this feature is used with the TrFO feature, improving voice quality.

9.5 GBFD-118603 Gb over IP

System Capacity

No impact.

Network Performance

No impact.

9.6 WRFD-050402 IP Transmission Introduction on Iub Interface

System Capacity

No impact.

Network Performance

No impact.

9.7 WRFD-050409 IP Transmission Introduction on Iu Interface

System Capacity

No impact.

Network Performance

No impact.

9.8 WRFD-050410 IP Transmission Introduction on Iur Interface

System Capacity

No impact.

Network Performance

No impact.

9.9 WRFD-050411 Fractional IP Function on Iub Interface

System Capacity

No impact.

Network Performance

No impact.

9.10 LBFD-003003/MLBFD-003003/TDLBFD-003003 VLAN Support (IEEE 802.1p/q)

System Capacity

No impact.

Network Performance

No impact.

9.11 LBFD-003007/TDLBFD-003007/MLBFD-12000308 IP Route Backup

System Capacity

No impact.

Network Performance

No impact.

9.12 LOFD-003008/MLOFD-003008/TDLOFD-003008 Ethernet Link Aggregation (IEEE 802.3ad)

System Capacity

No impact.

Network Performance

Link aggregation in load sharing mode improves bandwidth efficiency.

9.13 LBFD-00202103/MLBFD-00202103/ TDLBFD-00202103 SCTP Multi-homing

System Capacity

No impact.

Network Performance

No impact.

9.14 LOFD-003005/TDLOFD-003005/MLOFD-003005 OM Channel Backup

System Capacity

No impact.

Network Performance

No impact.

9.15 ARP Probe

System Capacity

No impact.

Network Performance

No impact.

9.16 ICMP

System Capacity

No impact.

Network Performance

No impact.

9.17 LOFD-003007/TDLOFD-003007/MLOFD-003007 Bidirectional Forwarding Detection

System Capacity

No impact.

Network Performance

No impact.

9.18 IP Path Ping Detection

System Capacity

No impact.

Network Performance

No impact.

9.19 Trace Route

System Capacity

No impact.

Network Performance

No impact.

9.20 IP Link QoS Detection

System Capacity

No impact.

Network Performance

UDP packet injection consumes network bandwidth. Bandwidth usage is related to the interval for sending injected packets as well as the packet size. For example, transmitting 80-byte-sized packets at an interval of 100 ms requires 6.4 kbit/s of bandwidth.

9.21 GPM Congestion Detection

System Capacity

No impact.

Network Performance

The NodeB uses FPDT to measure the delay over uplink IP transmission links, which requires the RNC to send private frames that carries uplink FP frame to the NodeB over the Iub interface at an interval of 20 ms. This process increases downlink transmission bandwidth overhead over the Iub interface. If a single baseband processing unit is configured for a single NodeB, GPM congestion detection for one IP link increases the overhead by 17.6 kbit/s if Type 1 FP frames are sent. If Type 2 FP frames are sent, the overhead increases by 35.2 kbit/s.

9.22 GTP-U Echo

System Capacity

No impact.

Network Performance

Static GTP-U echo have the following impacts on network performance:

- If the physical transmission link is interrupted, there is a possibility that the CPU usage increases.
- If the user-plane path detection result shows that the path is faulty, no new bearer will be set up on the path. As a result, the access success rate of the bearer decreases and the access failure rate of the bearer caused by transport-layer faults increases in some scenarios.

9.23 Source-based Routing on the Base Station Side

System Capacity

No impact.

Network Performance

No impact.

9.24 Active and Backup Link ARP Probe

System Capacity

No impact.

Network Performance

No impact.

9.25 Quick Transmission Congestion Detection

System Capacity

No impact.

Network Performance

Quick transmission congestion detection requires periodical sending (every 10 ms) of FI and BI frames, which improves the transmission bandwidth overhead. Every one quick transmission congestion detection instance increases about 60 kbit/s transmission bandwidth overhead.

9.26 LLDP

System Capacity

No impact.

Network Performance

No impact.

9.27 VLAN Priority Mapping

System Capacity

No impact.

Network Performance

No impact.

9.28 PPP Link

System Capacity

No impact.

Network Performance

No impact.

9.29 MLPPP Group

System Capacity

No impact.

Network Performance

No impact.

10 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

11 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

12 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

13 Reference Documents

- 3GPP TS 29.281, "General Packet Radio System (GPRS) Tunnelling Protocol User Plane (GTPv1-U)"
- 3GPP TS 29.414, "Core network Nb data transport and transport signalling"
- *SRAN Networking and Evolution Overview Feature Parameter Description*
- *IP Performance Monitor Feature Parameter Description*
- *Ethernet OAM Feature Parameter Description*
- *IP Active Performance Measurement Feature Parameter Description*
- *Common Transmission Feature Parameter Description*
- Feature parameter description documents in *GBSS Feature Documentation*:
  - *Transmission Resource Pool in BSC Feature Parameter Description*
  - *IP BSS Engineering Guide Feature Parameter Description*
  - *Abis Transmission Backup Feature Parameter Description*
- Feature parameter description documents in *RAN Feature Documentation*:
  - *Transmission Resource Pool in RNC Feature Parameter Description*
  - *Synchronization Feature Parameter Description*
  - *ATM&IP Dual Stack Feature Parameter Description*
  - *Hybrid Iub IP Transmission Feature Parameter Description*
  - *uX2 Interface Management Feature Parameter Description*
  - *IP RAN Engineering Guide Feature Parameter Description*
- Feature parameter description documents in *eRAN Feature Documentation*:
  - *Synchronization Feature Parameter Description*
  - *S1 and X2 Self-Management*
  - *IP eRAN Engineering Guide Feature Parameter Description*
- Feature parameter description documents in *5G RAN Feature Documentation*:
  - *NG and Xn Self-Management Feature Parameter Description*
  - *IP NR Engineering Guide Feature Parameter Description*
- Documents in *BSC6900 GSM Product Documentation*:
  - *BSC6900 GSM Technical Description*
- Documents in *BSC6910 GSM Product Documentation*:

  - *BSC6910 GSM Technical Description*
- Documents in *BSC6900 GU Product Documentation*:
  - *BSC6900 GU Technical Description*
- Documents in *BSC6910 GU Product Documentation*:
  - *BSC6910 GU Technical Description*

